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Topic	**Contrastive Spectrum–Molecular Pretraining (CSMP): Building a Chemical Search Engine for Matching Tandem Mass Spectra with Molecular Structures**

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Abstract

Tandem mass spectrometry (MS/MS) is a technique that produces molecular fragmentation spectra to identify the structural composition of chemical compounds in experimental samples. However, most existing annotation systems rely on exact reference search in spectral libraries or hand-crafted expert heuristics, which limits their ability to generalize across broader chemical space and reliably determine the structures of rare or previously unobserved molecules. This thesis addresses this limitation by formulating spectrum–molecule matching as a contrastive representation learning problem, where MS/MS spectra and molecular structures are projected into a shared embedding space for scalable retrieval. Following the implementation of the proposed research methodology, **a unified benchmark dataset containing 1,382,425 validated spectrum–molecule pairs and a molecular search database with 872,935 candidate structures** were constructed from heterogeneous public resources. Two modern dual-encoder retrieval models, JESTR and CSU-MS2, were fine-tuned and benchmarked under identical experimental conditions, with **CSU-MS2 achieving a self-similarity mean of 0.51 and retrieval performance ranging from Hit Rate@1 of 0.21 to Hit Rate@100 of 0.91**. The best-performing model was integrated into a **deployable retrieval platform combining spectrum preprocessing, precursor-mass candidate filtering, embedding-based ranking, and indexed molecular search**. The developed retrieval framework and supporting materials are publicly available at https://ivproger.github.io/csmp-search-engine/.

Chapter 1

Introduction

1.1 Challenges in MS/MS Spectrum Annotation

Tandem mass spectrometry (MS/MS) represents a widely used analytical approach applied across metabolomics, the synthesis of novel compounds and materials, as well as environmental and pharmaceutical research [1]. This technique precisдн identifies and structurally elucidates chemical entities ranging from small molecules to complex biomolecules and synthetic materials.

Despite the capabilities of modern spectrometers, annotating assigning experimental spectra to corresponding molecular structures remains challenging. Conventional annotation approaches rely on spectral reference libraries such as NIST [2] or Wiley [3], achieving correct identification through direct fingerprint spectral matching. These approaches perform well for known compounds but are limited by the incompleteness of available databases, which cover only a fraction of chemical space. As a result, their effectiveness diminishes for novel molecules, structural isomers, or complex compounds.

Furthermore, annotation is complicated by variable experimental conditions and instrument configuration. Indeed, differences in selected ionization modes, collision energies, and fragmentation techniques can produce markedly different spectra diagrams for identical molecules [4]. Such variability hinders reproducibility and contributes to a large fraction of unannotated spectra often termed "spectral dark matter", limiting both reliability and throughput in chemical analyses and experiments [5].

In addition to these limitations, annotation itself often remains labor-intensive and requires substantial expert involvement. When spectral library matching fails to provide a reliable identification, chemists must interpret the spectrum manually by examining individual fragment ions, neutral losses, and characteristic fragmentation pathways to infer plausible molecular substructures and candidate compounds. This procedure relies heavily on domain knowledge and typically involves iterative hypothesis generation and validation based on observed spectral patterns [6, 7]. In practice, such interpretation resembles solving a complex puzzle: analysts must integrate multiple sources of evidence, including fragment mass differences, isotope patterns, and known chemical transformation rules, to reconstruct the most probable molecular structure. Consequently, manual spectrum interpretation is often slow, error-prone, and difficult to scale for the rapidly growing volumes of high-resolution MS/MS data generated in modern analytical workflows [8].

These challenges highlight the need for generalizable computational strategies that capture the underlying complex relationships between molecular structure and the fragmentation behavior of experimental mass spectra. Addressing these limitations, Machine Learning (ML) has recently emerged as a promising direction,

1.2 The role of Machine Learning

Machine learning (ML) has recently emerged the inherent complexity of tandem mass spectrometry data analysis [9]. In contrast to traditional rule-based or library-dependent approaches, ML models are able to capture complex non-linear dependencies directly from observed data, making it possible to uncover the patterns that connect molecular structure with the resulting fragmentation behavior. Within this paradigm, the spectrum interpretation problem may be cast either as a predictive task or as aretrieval one, which opens the way to automated and scalable molecular identification with steadily improving accuracy of structural elucidation.

Among the more recent directions, representation learning and contrastive learning have shown notable promise. The core concept is the construction of a shared embedding space in which different data modalities, such as molecular structures and MS/MS spectra, are mapped into a unified representation form. Spectrum–molecule matching can then be performed by measuring similarity between embeddings, effectively transforming the identification problem into a cross-modal retrieval task. Recent studies have shown that such methods can capture meaningful correspondences between spectral patterns and molecular features, improving identification performance in cross-modal matching [10, 11].

Despite these advances, three practical challenges continue to limit the effective application of machine learning methods to the spectrum–molecule matching problem.

First, most publicly available MS/MS data are distributed in heterogeneous and often raw formats. These datasets frequently contain inconsistent metadata, duplicated records, low-quality spectra, and other anomalies that complicate their direct use for machine learning. As a result, substantial preprocessing and curation are required to construct reliable datasets suitable for model training and evaluation.

Second, the availability of standardized and diverse benchmarking datasets remains limited. Benchmarks, such as MassSpecGym, CANOPUS, and CASMI [12–14], typically cover restricted subsets of spectral data and represent only specific experimental conditions, instrument types, or acquisition protocols. Consequently, models trained and evaluated on these datasets may not generalize well to the broader diversity of spectra encountered in real analytical workflows.

Finally, many recently proposed machine learning approaches for spectrum–molecule matching remain at the proof-of-concept stage. Published methods are often evaluated under different experimental settings and datasets, which complicates direct comparison. Moreover, reproducibility and practical adoption are frequently hinderd by incomplete code releases, missing or low-quality pre-trained model weights, and limited documentation.

1.3 Thesis Objective

The above challenges demonstrate that a substantial gap remains between methodological advances and the development of a practical, reproducible, and scalable solution suitable for real-world analytical workflows. Studies primarily focus on isolated model proposals evaluated under heterogeneous datasets, inconsistent preprocessing pipelines, and varying benchmarking conditions, which limits both direct comparison and practical deployment. At the same time, the lack of unified large-scale curated datasets and operational retrieval systems constrains the applicability of these methods in research environments.

Therefore, this thesis primarily aims to design, implement, and evaluate an end-to-end computational framework for spectrum–molecule matching that bridges the above gap by integrating three complementary contributions: (i) construction of a large-scale curated dataset from heterogeneous public MS/MS sources, (ii) unified benchmarking and comparative evaluation of modern cross-modal contrastive learning models under identical experimental conditions, and (iii) deployment of the model achieving the highest retrieval performance under the conducted evaluation protocol within a scalable chemical retrieval system capable of processing real spectral data under practical computational constraints. To achieve this aim, the thesis is guided by the following research questions:

RQ1: Which data engineering, preprocessing, and curation methods is required to consolidate heterogeneous and noisy public MS/MS resources into a consistent dataset appropriate for reliable model training and reproducible evaluation procedure?

RQ2: How effectively do modern cross-modal machine learning architectures address the spectrum–molecule matching task when evaluated under a unified benchmarking protocol based on standardized real-world experimental data?

RQ3: How can the most effective model be integrated into a deployable retrieval system that balances retrieval accuracy, computational efficiency, scalability, and robustness for realistic analytical use?

To address these questions, this thesis adopts a dual scientific and engineering approach. First, heterogeneous MS/MS and molecular data are systematically collected, standardized, and transformed into a structurally disjoint large-scale benchmark dataset. Second, selected state-of-the-art models are adapted and evaluated using a unified training and retrieval evaluation framework. Third, the best-performing model is deployed within a modular, service-oriented retrieval architecture that transforms raw spectral inputs into ranked molecular candidates via a sequence of interoperable system components.

The experimental results confirm the effectiveness of the proposed framework across data engineering, model evaluation, and system deployment. The implemented data processing pipeline produced a large-scale curated benchmark dataset containing 1,382,425 validated spectrum–molecule pairs and a molecular search database comprising 872,935 candidate structures. Under the unified experimental protocol, JESTR and CSU-MS2 were comparatively evaluated under identical training and retrieval conditions. Although both models successfully learned meaningful cross-modal spectrum–molecule representations, CSU-MS2 consistently outperformed JESTR in both embedding-space discrimination and retrieval effectiveness, achieving a KL divergence of 3.4280 versus 1.6815, Hit Rate@1 of 0.2147 versus 0.1137, Hit Rate@100 of 0.9053 versus 0.8660, Recall@1 of 0.1737 versus 0.0988, and Recall@100 of 0.8719 versus 0.8510. These results established CSU-MS2 as the strongest-performing model for downstream integration. Deployment within the developed retrieval system further demonstrated practical feasibility, with an average processing time of 0.0899 seconds per spectrum, stable parallel execution of up to 11 concurrent file requests, support for large-scale spectral inputs under defined operational constraints, and a request-level failure rate of $F = 0$ under malformed or corrupted input conditions.

These findings establish a reproducible foundation for a scalable AI-assisted annotation system by demonstrating that modern contrastive spectrum–molecule learning approaches can be effectively translated into analytical computational mass spectrometry workflows.

Chapter 2

Background and preliminaries

2.1 Fundamentals of MS/MS Mass Spectrometry

Mass spectrometry (MS) was developed to address the need to detect, characterize, and quantify individual chemical compounds within complex molecular mixtures. Biological or chemical samples often contain numerous coexisting molecular species whose composition cannot be resolved through bulk chemical measurements alone in disciplines metabolomics, pharmacology, environmental science, and proteomics. This limitation is addressed by MS through enabling molecule-specific analysis based on the measurable physical properties of ionized compounds [15].

Mass spectrometry operates by converting neutral analyte molecules into gas-phase ions and separating these ions according to their mass-to-charge ratio (m/z). Conventional single-stage mass spectrometry (MS^1) consists of sequential ionization, mass analysis, and ion detection, ultimately generating a mass spectrum that represents the distribution and relative abundance of detected ions (Figure 1). This process provides an initial characterization of molecular mass and ion composition, thereby establishing a basis for molecular analysis [16].

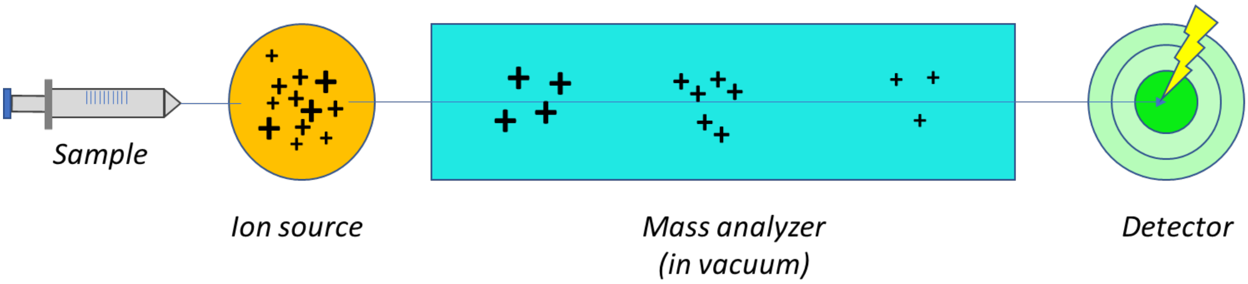

Fig. 1. General principle of mass spectrometry. Adapted from JEOL USA Mass Spectrometry Basics [17].

Although MS^1 is highly effective for detecting molecular ions and estimating candidate molecular masses, molecular mass alone is often insufficient for confident structural identification. Distinct compounds may exhibit similar or identical masses, structural isomers may generate overlapping precursor-ion signals, and complex mixtures may contain co-eluting species with indistinguishable mass profiles. Consequently, MS^1 data alone often cannot uniquely resolve molecular structure, necessitating additional structurally informative evidence for reliable compound identification [18].

Tandem mass spectrometry (MS/MS or MS^2) was developed to overcome this limitation by extending conventional MS with an additional stage of controlled ion fragmentation. Rather than measuring only intact molecular ions, MS/MS isolates selected precursor ions and examines how they dissociate under defined activation conditions. This distinction is fundamental: MS^1 primarily determines which ions are present, whereas MS^2 investigates how a selected ion fragments, thereby revealing structural information that cannot be inferred from molecular mass alone [19].

Fragmentation behavior is governed by molecular structure, bond dissociation energy, charge localization, functional group composition, and ion stability.

Consequently, precursor dissociation commonly follows chemically meaningful and partially predictable pathways rather than random decomposition. Precursor-ion activation may induce preferential bond cleavage, generating product ions, fragment ions, and characteristic neutral losses associated with specific molecular substructures (Figure 2).

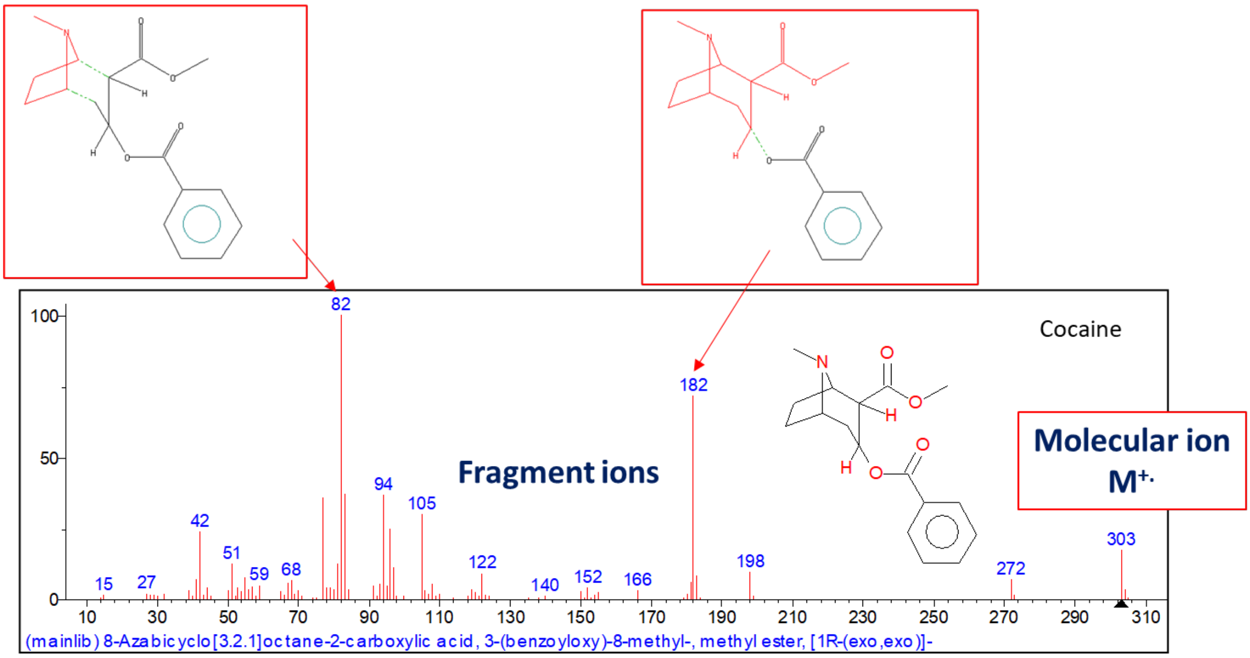

Fig. 2. Example of precursor-ion fragmentation pathways, characteristic product ions, and neutral losses in tandem mass spectrometry. Adapted from JEOL USA Mass Spectrometry Basics [17].

Chemists can infer structural features and progressively reduce the set of plausible molecular candidates by calculating mass differences between precursor and product ions, evaluating plausible fragmentation pathways, and comparing observed spectral patterns against established chemical rules, reference spectral libraries, or computational fragmentation models. Thus, MS/MS transforms mass spectrometry from molecular mass measurement into a structural analysis methodology in which fragmentation patterns function as chemically interpretable molecular fingerprints [18].

The generalized MS/MS workflow is summarized in Pipeline 1.

PIPELINE 1

General Tandem Mass Spectrometry (MS/MS)

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1. Introduce sample into the analytical system
2. Ionize analyte molecules
3. Perform MS^1 analysis
4. Detect and select precursor ion(s)
5. Transfer precursor ion(s) to activation region
6. Apply ion activation (e.g., CID)
7. Fragment precursor ion(s)
8. Perform MS^2 analysis
9. Measure product-ion m/z values and intensities
10. Construct tandem spectrum:

$$S = \{(m/z_i, I_i)\}_{i=1}^{N}$$

Depending on instrument architecture, these stages may be implemented either through physically separate analyzers or through sequential operations within a single instrument. Regardless of implementation, tandem spectra are influenced by ionization conditions, precursor selection strategy, fragmentation method, collision energy, and mass analyzer design. These factors shape spectral quality, fragmentation reproducibility, and structural interpretability. Consequently, while MS/MS provides chemically rich structural evidence, its practical utility in large-scale automated molecular identification depends on effective preprocessing, normalization, and computational standardization [20].

2.2 Computational Representations of MS/MS Spectra and Molecular Structures

Modern cheminformatics and machine learning methods cannot operate directly on molecular structures or tandem mass spectra in their native scientific form. Instead, both modalities must be transformed into standardized computational representations that preserve chemically relevant information while enabling efficient numerical processing. Within the spectrum–molecule matching problem, the design of universal representation forms plays a central role because MS/MS spectra correspond to experimentally acquired signal patterns, while molecular structures are represented as discrete chemical graphs. Consequently, effective computational frameworks require machine-readable encodings that translate both modalities into structured formats suitable for statistical analysis, similarity estimation, and representation learning [18, 21].

Computational Representation of MS/MS Spectra

In analytical chemistry, an MS/MS spectrum is canonically represented as a finite set of fragment peaks:

$$S = \{(m/z_i, I_i)\}_{i=1}^{N}, \tag{1}$$

where m/z_i denotes the mass-to-charge ratio of fragment ion i, I_i denotes its measured intensity, and N is the total number of detected peaks.

This representation reflects the experimental output of tandem mass spectrometry, in which each detected peak corresponds to a product ion generated through precursor fragmentation. Conventional spectral visualization represents fragment mass-to-charge ratio (m/z) along the horizontal axis and relative ion intensity along the vertical axis (Figure 3). Therefore, peak position captures fragment mass characteristics, whereas peak intensity reflects relative fragment abundance.

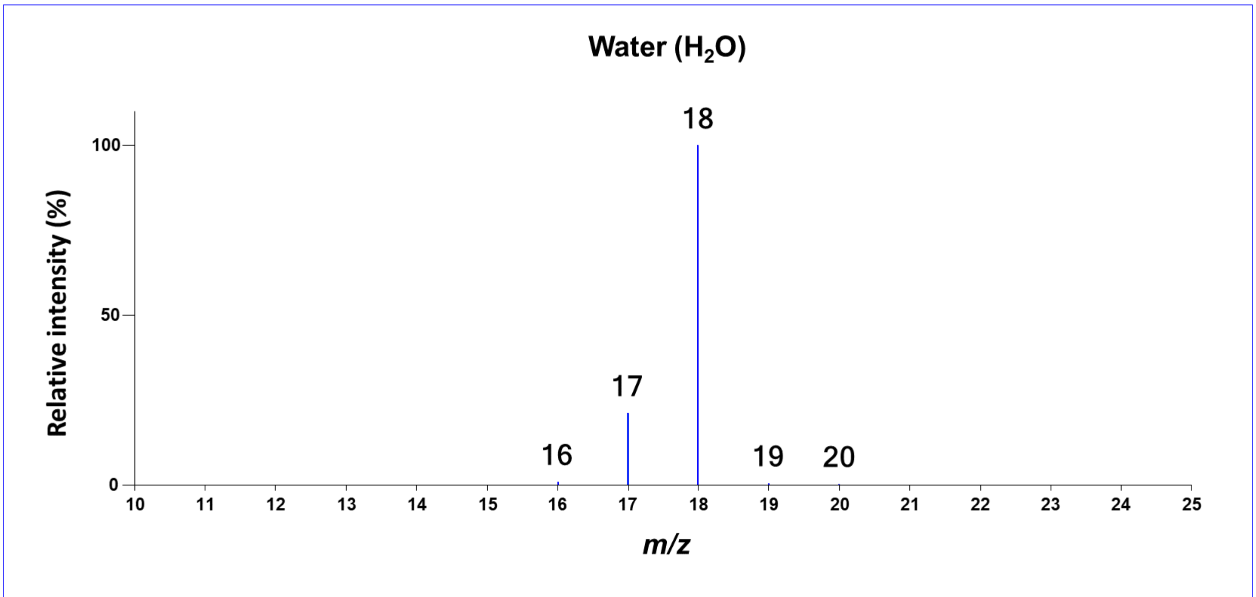


Fig. 3. Example of mass spectrum diagram. Adapted from JEOL USA Mass Spectrometry Basics [17].

Such spectral diagrams are highly interpretable for human experts and support manual structural reasoning through fragment inspection, neutral loss analysis, and diagnostic peak recognition.

However, although the peak-list formulation is chemically meaningful, it presents major challenges for computational modeling. Spectra are inherently variable in length, sparse, and instrument-dependent, which complicates the direct use in conventional machine learning architectures. To address this limitation, modern computational pipelines frequently transform spectra into fixed-dimensional numerical vectors through **spectral binning** [18].

In binned representations, the continuous m/z axis is partitioned into discrete intervals (bins), and fragment intensities are assigned to corresponding positions within a fixed-length vector. Formally, for a predefined mass range $[m_{\min}, m_{\max}]$ divided into K bins of width Δ, each bin corresponds to:

$$b_j = [m_{\min} + (j-1)\Delta,\ m_{\min} + j\Delta), \quad j = 1, 2, \ldots, K \tag{2}$$

where:

$$K = \left\lfloor \frac{m_{\max} - m_{\min}}{\Delta} \right\rfloor \tag{3}$$

The spectrum is then transformed into a fixed-dimensional vector:

$$\mathbf{x} = (x_1, x_2, \ldots, x_K) \tag{4}$$

such that each component x_j stores the aggregated intensity of all peaks whose m/z values fall within bin b_j:

$$x_j = \sum_{i:\, m/z_i \in b_j} I_i \tag{5}$$

Thus, each vector dimension represents a specific mass interval rather than an individual observed peak. This transformation converts variable-length spectral peak lists into standardized numerical vectors suitable for computational modeling.

Despite these advantages, spectral binning introduces reduced mass resolution, sensitivity to bin width selection, and partial information compression. Nevertheless, fixed-dimensional spectrum encoding remains a foundational strategy for large-scale computational mass spectrometry.

Computational Representation of Molecular Structures

Unlike spectra, molecular structures are not experimental signals but chemically defined arrangements of atoms and bonds. For computational analysis, these structures must likewise be transformed into machine-readable representations that preserve structural information while enabling algorithmic comparison.

One of the most widely used textual representations is the **Simplified Molecular Input Line Entry System (SMILES)** [22]. SMILES encodes molecular graph topology as a linear character sequence, representing atoms, bonds, branches, and ring structures through standardized syntax. For example, ethanol may be encoded as CCO. This representation is compact, standardized, and highly compatible with text-based machine learning models, including sequence encoders and language-model-inspired architectures.

However, SMILES also has limitations. The same molecule may possess multiple valid SMILES forms unless canonicalization is enforced, and small syntax errors may invalidate the entire representation. Consequently, although SMILES is highly practical, it is often complemented by more explicitly structural numerical encodings.

A dominant alternative is the use of **molecular fingerprints**, particularly Extended Connectivity Fingerprints (ECFP), commonly referred to as Morgan fingerprints [21]. Morgan fingerprints algorithmically transform molecular graphs into fixed-length binary or count-based vectors by iteratively encoding local atomic neighborhoods (Figure 4). Each vector dimension captures the presence of particular substructural patterns, thereby enabling efficient numerical comparison between molecules.

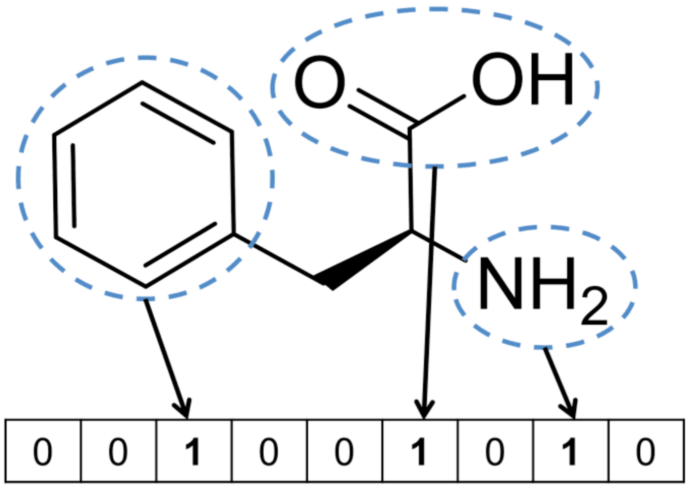


Fig. 4. Overview of Morgan fingerprint generation from molecular graph structure.

These fingerprint vectors form the basis of widely used the Tanimoto coefficient, which quantifies the degree of shared substructural features between molecular representations by comparing the overlap of encoded fingerprint patterns. In cheminformatics, Tanimoto similarity is extensively applied to molecular retrieval, virtual screening, clustering, candidate ranking, and structural benchmarking because it provides a computationally efficient and chemically interpretable approximation of structural relatedness between compounds [23, 24].

2.3 Representation Learning

Representation learning is a central concept in modern deep learning. It refers to learning compact and informative embeddings from raw data that capture essential structure, patterns, or semantics. Unlike handcrafted features, representation learning automatically maps data into a latent space where relationships are easily exploited for classification, clustering, retrieval, or similarity estimation [25].

Formally, given a dataset

$$\mathcal{X} = \{x_i\}_{i=1}^{N}, \tag{6}$$

a representation learning model f_θ learns a mapping

$$f_\theta : \mathcal{X} \to \mathbb{R}^d, \quad z_i = f_\theta(x_i), \tag{7}$$

where $z_i \in \mathbb{R}^d$ encodes the essential characteristics of x_i. The latent space is structured such that distances or similarities between embeddings correspond to meaningful relationships in the input space.

2.4 Contrastive Learning

2.4.1 Fundamental Principles of Contrastive Learning

Contrastive learning explicitly models relationships between samples. Its core idea is to minimize the distance between semantically similar inputs (*positive pairs*) and maximize the distance between dissimilar inputs (*negative pairs*) [26].

Given a dataset $\mathcal{X} = \{x_i\}_{i=1}^{N}$, define positive and negative pairs as:

$$\mathcal{P} = \{(x_i, x_j^+) \mid x_j^+ \text{ is semantically similar to } x_i\}, \tag{8}$$

$$\mathcal{N} = \{(x_i, x_k^-) \mid x_k^- \text{ is semantically dissimilar to } x_i\}. \tag{9}$$

The contrastive objective learns a mapping $f_\theta : \mathcal{X} \to \mathbb{R}^d$ such that embeddings of positive pairs are close, while embeddings of negative pairs are well separated:

$$\theta^* = \arg\min_{\theta} \sum_{(x_i, x_j^+) \in \mathcal{P}} \sum_{(x_i, x_k^-) \in \mathcal{N}} \mathcal{L}(z_i, z_j^+, z_k^-), \tag{10}$$

where $\mathcal{L}(\cdot)$ encodes relational constraints and θ denotes the parameters of the embedding function.

2.4.2 Mathematical Formulations of Contrastive Objectives

Several specific loss functions have been proposed to solve the defined optimization objective. Each of these losses introduces distinct notions about how positive and negative relationships should be enforced in the latent space.

Contrastive Loss [26] requires that positive pairs lie close together in the embedding space, while negative pairs are pushed apart by at least a fixed margin m. Formally, for two embeddings z_i and z_j with similarity label $y \in \{0, 1\}$ the loss is defined as follows:

$$\mathcal{L}_{\text{contrastive}}(z_i, z_j^+, z_k^-) = y\|z_i - z_j^+\|_2^2 + (1-y)\max(0, m - \|z_i - z_k^-\|_2)^2, \tag{11}$$

Here:

- $\|z_i - z_j^+\|_2^2$ pulls positive pairs together under Euclidean distance.
- $\max(0, m - \|z_i - z_k^-\|_2)^2$ enforces that negative pairs are separated by at least the margin m; if they are already farther than m, no penalty is applied.
- The binary label y distinguishes between positive ($y = 1$) and negative ($y = 0$) pairs.

This loss introduces the notion of **explicit distance-based constraints** in the latent space, directly encoding similarity and dissimilarity.

Triplet Loss [27] extends this idea by considering three embeddings simultaneously: an anchor z_a, a positive sample z_p (semantically similar to the anchor), and a negative sample z_n (semantically dissimilar). The loss is:

$$\mathcal{L}_{\text{triplet}}(z_a, z_p, z_n) = \max(0, \|z_a - z_p\|_2^2 - \|z_a - z_n\|_2^2 + \alpha), \tag{12}$$

where $\alpha > 0$ is a margin hyperparameter. Key notions introduced:

- The loss enforces a **relative constraint**: the distance between the anchor and positive must be smaller than the distance between the anchor and negative by at least α.
- Only triplets that violate this relative ordering contribute to the gradient, leading to a sparse but focused learning signal.

Lifted Structured Loss [28] extends the pairwise formulation to the entire batch and exploits the full set of pairwise relations to make better use of the gradient signal.

$$\mathcal{L}_{\text{lifted}} = \frac{1}{2\mid \mathcal{P}\mid} \sum_{(i,j)\in\mathcal{P}} \left[\log \sum_{(i,k)\in\mathcal{N}} e^{m-D_{ik}} + \log \sum_{(j,l)\in\mathcal{N}} e^{m-D_{jl}} + D_{ij}\right]_+^2, \tag{13}$$

Here $D_{ij} = \|z_i - z_j\|_2$ and $[\cdot]_+$ stands for the hinge operator. Several points are worth noting:

- Rather than acting on isolated negative pairs, the loss **combines the contributions of all in-batch negatives**, which naturally puts more weight on hard ones.
- Due to the hinge operator, only margin violations enter the gradient.

- This loss introduces a **batch-level structural constraint**, which leads to a more globally organized latent space.

InfoNCE Loss [29] has become a standard choice in modern self-supervised learning. It recasts contrastive learning as a **probabilistic classification problem**, in which the model is asked to identify the positive sample among K negatives. For a query z, a positive z^+, and and negatives $\{z_k^-\}_{k=1}^K$ it is defined as:

$$\mathcal{L}_{\text{InfoNCE}}(z, z^+, \{z_k^-\}_{k=1}^K) = -\log \frac{\exp(\text{sim}(z, z^+)/\tau)}{\sum_{k=1}^K \exp(\text{sim}(z, z_k^-)/\tau)}, \tag{14}$$

where $\text{sim}(\cdot,\cdot)$ denotes a similarity function such as cosine similarity, and τ is the temperature parameter that controls the sharpness of the resulting softmax. Several observations follow:

- The positive embedding plays the role of the correct class in a categorical distribution defined over the considered embeddings.
- The temperature τ regulates how peaked or smooth the distribution is, which in turn shapes the magnitude of the gradients.
- Because many negatives are taken into account at once, the loss yields a **richer and more informative gradient**, which is especially helpful when working in high-dimensional embedding spaces.

2.4.3 Architectural design and Practical Considerations

After defining contrastive objectives and loss functions, it is important to understand how these objectives are realized in practice. In general, effective contrastive learning representations are exercised through two complementary paradigms each with distinct architectural choices and engineering strategies.

Single-Modality Representation Learning:

In the single-modality setting, the goal is to obtain robust embeddings such that semantically similar inputs lie close together in the latent space, while dissimilar ones are clearly separated. A common solution is the **dual-encoder architecture**, in which two augmented "views" of the same input are processed by identical networks with shared weights. The views are produced by **data augmentation strategies**, for example random cropping, color jittering, masking, or noise injection, designed to encourage invariance to surface-level changes in the input.

Formally, let $x \in \mathcal{X}$ be an input sample, and let $t, t' \sim \mathcal{T}$ be two stochastic augmentations. The augmented views $x_i = t(x)$ and $x_j = t'(x)$ are then fed into a shared encoder f_θ to produce representations:

$$h_i = f_\theta(x_i), \quad h_j = f_\theta(x_j). \tag{15}$$

A learnable **projection head** g_ϕ subsequently maps these representations to embeddings in the contrastive latent space:

$$z_i = g_\phi(h_i), \quad z_j = g_\phi(h_j), \quad \tilde{z}_i = \frac{z_i}{\|z_i\|_2}, \quad \tilde{z}_j = \frac{z_j}{\|z_j\|_2}. \tag{16}$$

This setup encourages the network to recover semantically meaningful embeddings that are invariant to augmentation while still being discriminative across distinct samples.

Three methods have shaped this paradigm:

- **SimCLR** [30]: SimCLR shows that competitive representations can be learned through a combination of straightforward augmentations and normalized embeddings, without resorting to momentum encoders or memory banks. The pipeline consists of a base encoder followed by a projection head that maps encoder outputs into a normalized embedding space. Positive pairs come from two augmented views of the same sample, and negatives are obtained from the remaining batch elements. Training is driven by the InfoNCE objective.

- **MoCo v2** [31]: MoCo v2 was introduced to handle the practical difficulty of keeping a large pool of negative samples available during training. The solution is a **momentum encoder** whose parameters are updated more slowly than those of the main encoder. Embeddings produced by previous batches are held in a queue-based dictionary, so that the model has access to a continuously refreshed set of negatives. Formally, the momentum encoder f_{θ_m} is updated as:

$$\theta_m \leftarrow m\theta_m + (1-m)\theta, \tag{17}$$

where $m \in [0,1)$ is the momentum coefficient. This approach enables efficient scaling to large datasets and provides a stable source of negative samples for contrastive learning.

- **BYOL** [32]: BYOL demonstrates that competitive embeddings can be learned without explicitly sampling negatives. The method uses a **target network** and a predictor network to enforce consistency between two augmented views. Let $z_i = f_\theta(t(x))$ be the online embedding and $z'_j = f_{\theta'}(t'(x))$ be the target embedding, where θ' is an exponential moving average of θ. The predictor q_ϕ then maps z_i to $p_i = q_\phi(z_i)$, and training minimizes the mean squared error between the normalized vectors:

$$\mathcal{L}_{\text{BYOL}} = \|\text{normalize}(p_i) - \text{normalize}(z'_j)\|_2^2. \tag{18}$$

Together, the moving-average target network and the predictor network make training more stable and prevent representation collapse.

Overall, single-modality contrastive learning is built around weight-sharing encoders that process augmented views of the same input, with the contrastive objective and the augmentation strategy playing complementary roles. Together, these elements give a reliable starting point for learning meaningful embeddings that transfer well to a range of downstream tasks.

Cross-Modal Representation Learning

Cross-modal representation learning addresses the problem of building embeddings for inputs of different modalities within a **shared latent space**. Due to inherent differences in structure, distribution, and statistical properties of each modality **separate encoders** must be employed for each modality. This design allows each encoder to specialize in feature extraction appropriate to its input type, while still enabling alignment across modalities in the shared latent space.

Formally, let $x^{(A)} \in \mathcal{X}^{(A)}$ and $x^{(B)} \in \mathcal{X}^{(B)}$ denote inputs from two different modalities. Two separate encoders below,

$$f_\theta : \mathcal{X}^{(A)} \to \mathbb{R}^{d'}, \quad g_\phi : \mathcal{X}^{(B)} \to \mathbb{R}^{d'}, \tag{19}$$

map the inputs to modality-specific embeddings $h^{(A)}$ and $h^{(B)}$:

$$h^{(A)} = f_\theta(x^{(A)}), \quad h^{(B)} = g_\phi(x^{(B)}). \tag{20}$$

To enable comparison and alignment in a shared latent space, each embedding is then passed through a **modality-specific projection head**:

$$z^{(A)} = p_\psi(h^{(A)}), \quad z^{(B)} = q_\xi(h^{(B)}), \tag{21}$$

where p_ψ and q_ξ are learnable projection functions (e.g., MLPs) mapping the modality-specific embeddings $h^{(A)}, h^{(B)} \in \mathbb{R}^{d'}$ into a shared latent space $\mathbb{R}^d$.

CLIP [33] effectively implements this paradigm. It employs separate encoders for images and text, mapping each modality into a shared latent space. Positive pairs consist of semantically aligned image-text examples, whereas negative pairs are drawn from other samples in the batch. The model optimizes a symmetric InfoNCE loss.

In general, cross-modal contrastive learning extends the principles of single-modality contrastive learning to heterogeneous input domains.

Engineering Considerations

In practice, the successful application of contrastive learning depends not only on the choice of loss function and architecture but also on several critical engineering strategies. Such design choices help to keep training stable, raise the quality of the resulting embeddings, and ensure that the framework remains workable on large datasets. A few aspects deserve particular attention:

- **Batch Size:** Working with larger batches gives the model more negative examples per step, which strengthens the contrastive signal and stabilizes optimization. In a cross-modal setting, this helps the negative pairs to reflect the actual semantic spread of the data.
- **Hard Negative Mining:** Selecting negatives that are structurally or semantically close to the anchor sharpens the discriminative power of the embed-

dings. In practice, this can be done by mining nearest-neighbor negatives or by maintaining a memory bank of difficult examples.

- **Normalization and Projection Heads:** Applying L2 normalization to embeddings and inserting a projection head before the contrastive loss both contribute to smoother training and better convergence. Projection heads transform encoder outputs into a space where the contrastive objective is more effective, ultimately improving the quality of the learned representations.
- **Augmentation Strategies:** A diverse set of augmentations, such as cropping, masking, jittering, or noise injection, encourages the network to learn invariances and improves both generalization and robustness. The most suitable choice of augmentation depends on the modality and the task, yet it remains essential for extracting semantically meaningful features.

Considered together, these engineering practices act as a set of **broadly applicable recommendations** that hold regardless of the specific contrastive learning paradigm or modality. They contribute to producing embeddings that are well-structured, discriminative, and robust, while at the same time stabilizing optimization and allowing the framework to scale. Combining loss functions, architectural choices, and these engineering strategies makes it possible for contrastive models to deliver high-quality latent representations that are interpretable and perform well on similarity-based tasks, both within and across modalities.

Chapter 3

Literature Review

This literature review considers the main computational and machine learning methodologies that have been proposed for the spectrum–molecule matching problem. The chapter starts from foundational solutions, covering traditional rule-based and database-driven approaches together with more recent machine learning paradigms aimed at improving molecular identification from spectral evidence. It then analyzes contemporary representation learning frameworks, including advanced similarity modeling strategies that aim to overcome key limitations of classical methods such as restricted scalability, domain specificity, and limited generalization. Finally, the review investigates spectrum–molecule retrieval systems as practical implementations of these methodological advances, with the goal of identifying current scientific, engineering, and system-design limitations that inform the development of a scalable and open retrieval framework.

3.1 Computational Methods

3.1.1 Rule-Based and Knowledge-Driven Methods

Rule-based and knowledge-driven methods represent the earliest computational paradigm for spectrum–molecule matching and established the conceptual foundation for computer-assisted molecular annotation from tandem mass spectra. These approaches encode expert chemical knowledge into deterministic computational procedures that interpret fragmentation spectra through predefined bond dissociation rules, neutral-loss heuristics, molecular formula constraints, and chemically plausible fragmentation pathways. Rather than learning from large-scale datasets, such systems rely on explicit fragmentation logic and candidate evaluation strategies, making them highly interpretable but often limited in scalability and adaptability across heterogeneous chemical spaces [34, 35].

Early expert systems such as DENDRAL demonstrated that molecular structure elucidation could be framed computationally as a constrained generate-and-test problem, where chemically valid candidate structures were iteratively evaluated against observed spectra [36]. Although pioneering, these systems required extensive manual rule engineering and were difficult to generalize beyond narrowly defined compound classes. This limitation led to the emergence of automated in silico fragmentation approaches, including MetFrag, MAGMa, and MS-FINDER, which extended the conventional paradigm by generating chemically plausible fragmentation patterns from candidate molecules, thereby enabling the subsequent ranking of structural hypotheses against observed MS/MS evidence [34, 37, 38]. These methods significantly improved throughput and practical usability by integrating precursor mass filtering with heuristic fragment scoring, but they remained fundamentally dependent on database completeness and often struggled with rearrangements, noisy spectra, and structural isomerism.

A major methodological advancement within this category was the introduction of fragmentation trees through the SIRIUS framework [39]. Fragmentation trees model tandem fragmentation as a rooted directed acyclic graph in which precursor ions are recursively decomposed into chemically valid subfragments connected by neutral losses. This transformed spectrum interpretation from isolated peak matching into a global combinatorial optimization problem, substantially improving molecular formula determination and fragmentation consistency analysis. Later developments, such as SIRIUS 3 and SIRIUS 4, further strengthened this framework through probabilistic scoring and integration with downstream molecular fingerprint prediction [40, 41]. However, fragmentation trees primarily improve formula-level or candidate-space reduction rather than direct molecular structure retrieval.

Competitive Fragmentation Modeling (CFM-ID) extended the knowledge-driven paradigm by introducing probabilistic fragmentation graphs learned from reference spectra [42]. While CFM-ID incorporates data-driven parameter estimation, its core formulation remains chemically explicit because fragmentation events are modeled as interpretable structural transitions rather than latent neural embeddings. This hybridization improved candidate ranking and spectrum prediction but introduced greater computational complexity and reduced robustness for underrepresented compound classes.

Overall, rule-based and knowledge-driven methods provide strong interpretability, explicit chemical reasoning, and effective candidate-space reduction under known fragmentation assumptions. Their principal limitations include dependence on curated candidate databases, restricted scalability across chemically diverse molecular spaces, and reduced robustness for novel or poorly characterized compound classes. These limitations ultimately motivated the emergence of spectral library systems and later machine learning approaches. An overview of the evolutionary progression of major rule-based and knowledge-driven computational paradigms is provided in Table I.

TABLE I
Evolution of major rule-based and knowledge-driven computational paradigms for spectrum–molecule matching.

Period	**Method Class**	**Core Principle**	**Foundational Studies**
1960–1980	Expert Systems	Computer-assisted molecular structure elucidation through explicit expert-defined chemical rules, valence constraints, and generate-and-test reasoning over candidate structures	DENDRAL (Lindsay et al., 1993 historical analysis)
2010–2016	In Silico Fragmentation	Candidate retrieval from molecular databases followed by computational fragment enumeration and heuristic scoring of predicted fragments against observed tandem mass spectra	MetFrag (Wolf et al., 2010); MAGMa (Ridder et al., 2014); MS-FINDER (Tsugawa et al., 2016)
2008–2019	Fragmentation Trees	Global combinatorial optimization of chemically valid fragmentation pathways as directed acyclic graphs for precursor decomposition, molecular formula annotation, and structured fragmentation reasoning	Böcker and Rasche (2008); SIRIUS (Böcker et al., 2009); Fragmentation Trees Reloaded (Dührkop et al., 2015); SIRIUS 4 (Dührkop et al., 2019)
2014–Present	Probabilistic Fragmentation Modeling	Data-informed probabilistic modeling of fragmentation transitions to simulate spectra and improve candidate ranking through learned fragmentation behavior	CFM-ID (Allen et al., 2014); CFM-ID 3.0/4.0

3.1.2 Spectral Library and Reference Database Search

Spectral library and reference database search methods constitute the dominant practical baseline for small-molecule identification in mass spectrometry workflows. Rule-based methods, which interpret spectra through chemical fragmentation logic. In contrast, library-based approaches identify unknown compounds by comparing a query spectrum with curated experimental reference spectra. This paradigm assumes that identical, or closely related, molecular structures produce sufficiently similar fragmentation patterns under compatible acquisition conditions [20, 43].

The development of public spectral repositories established the practical foundation for this paradigm. MassBank and METLIN provided curated collections of reference spectra and enabled routine spectrum-to-spectrum matching for known compounds [44, 45]. In a typical workflow, the query spectrum is first filtered by precursor mass, adduct type, ionization mode, and available metadata. These constraints reduce the search space before detailed spectral comparison and help limit false-positive annotations.

The standard implementation of library search relies on spectral similarity scoring, most commonly cosine similarity. A tandem mass spectrum can be represented as a set of fragment peaks $S = \{(m_i, I_i)\}_{i=1}^{n}$, where m_i denotes the fragment m/z value and I_i denotes its intensity. Before comparison, query and reference spectra are usually preprocessed through peak filtering, intensity normalization, and peak alignment within a defined mass tolerance. After alignment, the spectra can be treated as intensity vectors, where each dimension corresponds to a matched or binned fragment feature. Classical cosine similarity measures the degree of overlap between these aligned vectors. A higher score indicates stronger agreement between fragmentation patterns and, therefore, a higher likelihood that the spectra correspond to the same compound or a closely related structure.

A subsequent development was the use of modified cosine similarity and analogue-oriented spectral comparison. Modified cosine similarity extends classical cosine scoring by allowing fragment alignment not only at identical m/z values but also under precursor mass shifts. This accounts for systematic spectral offsets introduced by chemical modifications and enables limited analogue detection beyond exact spectrum-to-spectrum identity. Early molecular networking approaches demonstrated that precursor-aware spectral alignment could reveal structurally related molecular scaffolds even when no exact reference match was available [46].

GNPS further expanded this direction by formalizing large-scale molecular networking, where spectra are represented as graph nodes and similarity relationships are represented as edges [43, 47]. This transformed spectral library search from isolated pairwise matching into relational chemical-space exploration. In this setting, annotations can be propagated across structurally related molecular families, which is especially useful in natural product discovery and untargeted metabolomics.

Despite its operational maturity, spectral library search remains fundamentally constrained by reference coverage. Even extensive spectral libraries cover only a small portion of the overall chemical space, leaving many biologically significant or novel compounds without experimentally acquired reference spectra. In addition, spectral similarity is sensitive to collision energy, ionization mode, instrument platform, preprocessing choices, and acquisition noise, which can reduce reproducibility across datasets [20]. Structural analogues may also produce weak cosine similarity despite meaningful scaffold similarity, limiting broader molecular generalization.

To mitigate coverage limitations, computationally predicted reference spectral libraries have been introduced. For example, LipidBlast expanded lipid identification by generating in silico tandem mass spectra for lipid classes with relatively regular fragmentation behavior [48]. Such resources enlarge the searchable reference space beyond experimentally measured standards, but they remain dependent on prediction quality, compound-class assumptions, and metadata consistency.

Thus, spectral library search remains a high-precision yet coverage-limited paradigm that provides the practical foundation for modern retrieval systems while motivating the transition toward machine learning-based similarity modeling. A chronological overview of major methodological developments within spectral library and reference database search is provided in Table II.

TABLE II
Evolution of major spectral library and reference database search paradigms for spectrum–molecule matching.

Period	**Method Class**	**Core Principle**	**Foundational Studies**
2006–2010	Classical Spectral Library Search	Direct spectrum-to-spectrum matching through alignment of experimental query spectra with curated reference spectra using cosine-based similarity scoring under precursor mass and metadata constraints	METLIN (Smith et al., 2005); MassBank (Horai et al., 2010)
2010–2016	Modified Cosine and Analog Search	Extension of classical cosine similarity through precursor mass shift-aware fragment alignment to improve analogue detection beyond exact spectrum identity matching	Molecular Networking Prototype (Watrous et al., 2012); Meta-Mass Shift Networking (Yang et al., 2013)
2016–2020	Molecular Networking	Graph-based extension of modified cosine similarity into spectrum similarity networks for large-scale relational annotation, analogue propagation, and chemical family exploration	GNPS (Wang et al., 2016); Feature-Based Molecular Networking (Nothias et al., 2020)
2013–2022	In Silico and Predicted Spectral Libraries	Expansion of searchable reference space through computationally predicted tandem mass spectra for compounds lacking experimental reference measurements	LipidBlast (Kind et al., 2013);

3.1.3 Machine Learning Methods

The limitations of classical computational paradigms motivated the emergence of machine learning approaches designed to improve scalability, generalization, and molecular discovery. Within spectrum–molecule matching, two main directions of methodological development can be distinguished. The first treats molecular identification as a direct end-to-end prediction task, in which tandem mass spectra are mapped to molecular structures, fingerprints, or other structural representations through supervised generative or predictive modeling. The second formulates the task as representation learning, where spectra and molecular structures are projected into shared semantic spaces that support cross-modal similarity, retrieval, and large-scale alignment. Although both paradigms address fundamental limitations of traditional computational methods, joint representation learning has emerged as a particularly significant direction due to its superior scalability for open-world retrieval, broad applicability across heterogeneous chemical spaces, and increasing relevance to modern retrieval-oriented system architectures.

End-to-End Structure Prediction Models

End-to-end structure prediction models emerged as a major machine learning paradigm for spectrum–molecule matching by reframing molecular identification as a supervised task that maps tandem mass spectra directly to structural representations. End-to-end models aim to infer structural information directly from spectral evidence. Depending on the prediction target, this paradigm encompasses a broad spectrum of formulations, ranging from molecular fingerprint and descriptor prediction to direct generation of SMILES strings, molecular graphs, scaffolds, or formula-constrained structures [49].

The earliest practical formulation of end-to-end structure prediction focused on molecular fingerprint inference. Rather than attempting to reconstruct a complete molecular graph from sparse and often ambiguous fragmentation evidence, early systems translated spectra into structural proxy representations that could be matched against candidate molecular databases. FingerID introduced this paradigm by framing tandem mass spectrometry as the prediction of molecular fingerprint properties, thereby transforming spectrum annotation into spectra-to-structure-property prediction [50]. CSI:FingerID substantially advanced this framework by integrating fragmentation trees with kernel-based prediction of molecular fingerprints, allowing predicted structural properties to guide database search over candidate compounds [51]. This transition was conceptually significant because spectra were no longer matched exclusively to observed reference spectra. Instead, they were transformed into abstract structural representations that generalized beyond direct library search.

Subsequent developments refined fingerprint prediction through increasingly sophisticated machine learning architectures. Structured-output approaches such as IOKR improved candidate ranking through kernel regression [52]. SIMPLE introduced sparse peak-interaction modeling to enhance interpretability [53]. ADAPTIVE further shifted the paradigm from predefined binary fingerprints toward learned molecular vectors, signaling an important methodological transition from handcrafted structural descriptors to data-driven latent structural spaces [54]. MetFID, CNN-based fingerprint predictors, IDSL_MINT, and FFNet neural architectures progressively replaced engineered spectral kernels with deep learning over peak lists, binned spectra, fragmentation trees, or transformer-style encoders [55–58]. Across this progression, the central objective remained stable: spectra were translated into intermediate structural representations that served as tractable proxies for full molecular identity.

A second major stage introduced direct de novo sequence generation, where molecular identification was framed as conditional molecular language modeling. Rather than predicting fingerprints alone, these methods attempted to generate explicit molecular strings from spectra. MassGenie represented an early transformer-based effort to generate candidate molecular structures from spectral evidence [59]. MSNovelist became a landmark by combining fingerprint prediction with encoder–decoder generation, thereby bridging structural proxy prediction and full de novo generation [60]. Spec2Mol, MS2SMILES, ctMSNovelist, and later MS-BART expanded this paradigm through increasingly direct spectrum-to-SMILES translation, pretrained molecular decoders, grammar constraints, and unified spectrum–molecule tokenization [61–64]. Collectively, these models reframed molecular identification as spectra-conditioned sequence generation, but they also revealed spectral ambiguity, non-unique SMILES representations, stereochemical uncertainty, and benchmark leakage frequently constrained reported performance [49].

The current frontier has shifted toward graph-native and constrained molecular generation. Recent models such as DiffMS, DiffNovo, FlowMS, MADGEN, and OMG increasingly acknowledge that unconstrained de novo generation from spectra alone is often underdetermined [65–69]. Instead, these systems incorporate molecular formula constraints, scaffold priors, diffusion processes, discrete flow matching, or reinforcement learning to reduce chemical search space while preserving generative flexibility. This evolution reflects a major conceptual transition: contemporary end-to-end models rarely rely on spectra as the sole information source, but instead integrate spectra with chemically meaningful priors to produce structurally valid hypotheses. In this sense, the field has evolved from pure direct prediction toward constrained generative modeling [49].

Despite substantial methodological progress, major limitations remain. Specifically, tandem mass spectra do not uniquely specify complete molecular structures, especially in the presence of isomerism, instrument variability, and incomplete fragmentation. As emphasized by recent benchmark analyses many performance gains are highly sensitive to evaluation design, candidate-set construction, and leakage between training and testing splits [49]. Consequently, end-to-end prediction systems are best interpreted not as autonomous structure elucidation engines, but as structure-hypothesis generators whose practical utility depends on representation design, conditioning strategies, and rigorous evaluation protocols.

Overall, end-to-end structure prediction has progressed through three major stages: fingerprint and descriptor prediction, sequence-based molecular generation, and constrained graph-native generation. This progression reflects a broad trajectory from tractable structural proxies toward increasingly explicit molecular reconstruction. While these systems expanded the scope of spectrum-driven molecular annotation beyond classical retrieval, they also revealed that robust structure prediction depends not only on predictive capacity but on chemically informed constraints, benchmark rigor, and realistic treatment of molecular ambiguity (see Table III below).

TABLE III
Evolution of major end-to-end structure prediction paradigms for spectrum–molecule matching.

Period	**Method Class**	**Core Principle**	**Foundational Studies**
2013–2020	Fingerprint, Descriptor, and Structural Proxy Prediction	Spectra-to-structure prediction is formulated as supervised inference of molecular fingerprints, descriptors, formulae, or learned molecular vectors that function as intermediate structural proxies for candidate ranking, thereby avoiding direct reconstruction of full molecular graphs while generalizing beyond exact spectral library matching	FingerID (Brouard et al., 2013); CSI:FingerID (Dührkop et al., 2015); IOKR (Brouard et al., 2019); SIMPLE (Nguyen et al., 2018); ADAPTIVE (Li et al., 2019); MetFID (Plante et al., 2022)
2021–2024	Sequence-Based De Novo Molecular Generation	Molecular identification is reframed as spectra-conditioned molecular language modeling, where encoder–decoder, transformer, or pretrained generative architectures translate tandem mass spectra into explicit molecular strings such as SMILES through direct sequence generation, often supported by fingerprint priors or grammar constraints	MassGenie (Shrivastava et al., 2021); MSNovelist (Stravs et al., 2022); Spec2Mol (Hamid et al., 2023); MS2SMILES (Foley et al., 2023); ctMSNovelist (Vyas et al., 2024)
2023–2025	Neural Structural Representation Expansion	Transformer, CNN, graph-attention, and multimodal architectures expand spectra-to-structure prediction through higher-capacity neural encoders that improve fingerprint prediction, latent structural representation learning, and scalable metabolite annotation across broader chemical domains	CNN-MS Fingerprint Prediction (Lee et al., 2022); IDSL_MINT (2024); Graph-Attention MS Models (2025); FFNet (2026)
2025–2026	Constrained Graph-Native and Generative Structure Modeling	Formula-aware, scaffold-conditioned, diffusion, discrete flow, and reinforcement-guided systems generate chemically constrained molecular graphs or structures by integrating spectral evidence with explicit chemical priors, reducing combinatorial ambiguity while improving structural validity and search-space tractability	DiffMS (2025); MADGEN (2025); DiffNovo (2025); OMG (2025); MS-BART (2025); FlowMS (2026)

Joint Representation Learning for Spectrum–Molecule Alignment

While end-to-end structure prediction models view molecular identification primarily as a direct spectra-to-structure inference task, an alternative line of work in machine learning recasts spectrum–molecule matching as a representation learning problem. Instead of explicitly predicting fingerprints, molecular graphs, or molecular strings, joint representation learning methods project tandem mass spectra and molecular structures into a shared semantic latent space, where chemically corresponding pairs are placed near each other and unrelated entities are kept apart. This transition signifies a change in methodological perspective, replacing direct structural reconstruction with scalable cross-modal retrieval strategies designed to facilitate large-scale spectrum annotation, molecular search across diverse chemical spaces, and incorporation into retrieval-oriented computational systems.

Early cross-modal retrieval systems established the conceptual foundation for this paradigm through contrastive embedding alignment across heterogeneous spectral and molecular modalities. CReSS introduced one of the first large-scale contrastive learning frameworks for direct spectrum–structure retrieval by aligning encoded spectral representations with molecular embeddings in a shared latent space [70]. Using fixed-length spectral encodings, data augmentation, and dual-modality neural architectures, this framework demonstrated that spectrum–molecule alignment could achieve highly accurate retrieval without explicit fragmentation rules engineering or fingerprint prediction. This innovations represented a significant conceptual departure from earlier machine learning systems by showing that molecular identification could be reformulated as a retrieval problem over learned semantic spaces rather than direct prediction alone.

Subsequent developments generalized early cross-modal contrastive retrieval frameworks beyond a single spectroscopic modality and introduced broader retrieval capabilities. SMEN extended contrastive joint embedding learning to infrared spectrum–molecule retrieval through transformer-based spectral encoders and graph neural molecular representations [71]. Importantly, this framework also incorporated an auxiliary generative decoder, demonstrating that shared latent spaces could simultaneously support retrieval and structure generation. The integration of generative decoding capabilities marked a methodological expansion by revealing that cross-modal embeddings were not solely useful for ranking candidate molecules, but could also function as chemically meaningful intermediate representations for downstream generative tasks.

As joint representation learning matured, architectures became increasingly specialized for tandem mass spectrometry and practical metabolomics workflows. CMSSP refined the paradigm through task-specific dual-encoder design, combining transformer-based spectral encoding with graph-enhanced molecular representation learning for metabolite annotation [72]. Candidate-space reduction through formula and isotopic filtering improved scalability, while contrastive embedding optimization improved retrieval precision on heterogeneous biochemical datasets. These developments emphasized that practical spectrum–molecule retrieval performance depends not only on embedding quality, but also on upstream preprocessing, candidate filtering, and chemically informed search-space restriction.

Further studies demonstrated that joint embedding spaces could directly align spectra with structural proxy spaces such as molecular fingerprints without requiring intermediate fingerprint prediction pipelines. Multimodal deep learning approaches based on dual-tower architectures showed that spectra and molecular descriptors could be embedded directly into comparable latent representations optimized through contrastive similarity objectives [73].The development of dual-tower multimodal embedding architectures broadened the paradigm beyond metabolite annotation and reinforced the idea that joint embedding systems constitute a general multimodal learning framework applicable to retrieval, classification, and property prediction.

A major conceptual refinement emerged through multiview contrastive learning formulations, where spectra and molecular structures were explicitly treated as complementary views of the same underlying chemical entity. JESTR advanced this perspective by applying Contrastive Multiview Coding with InfoNCE optimization to align molecular graphs and spectra without requiring intermediate structural proxy prediction [10]. Additional regularization through unlabeled candidate molecules improved generalization and retrieval robustness across heterogeneous datasets. This framework demonstrated that direct semantic alignment of molecular and spectral views can outperform both traditional retrieval systems and predictive intermediates, reinforcing multiview contrastive learning as a major architectural direction.

The current frontier is represented by scalable unified retrieval infrastructures that combine large-scale pretraining, advanced embedding alignment, and deployment-oriented molecular search. CSU-MS2 extended joint representation learning into a system-level framework by integrating transformer-based spectral encoders, graph isomorphism molecular encoders, and dynamic embedding space alignment optimized through large-scale contrastive pretraining [11]. By enabling retrieval across molecular library containing over a million compounds, this framework illustrates how joint representation learning has evolved from a methodological concept into a deployable solution for large-scale molecular retrieval. Such systems increasingly emphasize scalability, modularity, and integration into practical annotation pipelines while maintaining strong retrieval performance.

An overview of the major developmental stages in joint representation learning for spectrum–molecule alignment is provided in Table IV.

TABLE IV
Evolution of major joint representation learning paradigms for spectrum–molecule alignment.

Period	Method Class	Core Principle	Foundational Studies
2021–2023	Early Cross-Modal Contrastive Retrieval	First-generation contrastive dual-encoder systems established shared latent spaces for direct spectrum–molecule retrieval, demonstrating that spectra and molecular structures could be aligned without explicit rule-based fragmentation or fingerprint prediction pipelines	CReSS (2021); SMEN (2024)
2024	Task-Specialized Spectrum–Molecule Alignment	Domain-focused architectures introduced modality-specific encoder specialization, candidate filtering, and metabolomics-oriented optimization, improving retrieval precision and practical annotation performance on real biochemical datasets	CMSSP (2024); MMDL (2025)
2025	Multiview Contrastive Representation Learning	Spectrum and molecular structure were reformulated as complementary views of the same chemical entity, enabling multiview InfoNCE optimization, stronger generalization, and retrieval without intermediate structural prediction stages	JESTR (2025)
2025	Scalable Unified Retrieval Frameworks	Integration of embedding-space alignment, retrieval over million-scale molecular libraries, and deployment-oriented system infrastructure for scalable molecular search	CSU-MS2 (2025)

Collectively, joint representation learning frameworks illustrate a clear methodological progression from early proof-of-concept contrastive retrieval systems to highly specialized, scalable, and system-oriented cross-modal architectures.

Across this evolution, the field has shifted from establishing the feasibility of shared spectrum–molecule latent spaces toward optimizing retrieval robustness through modality-specific encoders, multiview learning, candidate filtering, and large-scale pretraining. Unlike end-to-end structure prediction, which prioritizes direct molecular reconstruction, joint representation learning increasingly prioritizes semantic alignment and retrieval scalability, making it particularly suitable for open-world molecular discovery and large database search. However, direct comparison across studies remains challenging because datasets, candidate spaces, evaluation protocols, and retrieval constraints vary substantially.

3.2 Spectrum–Molecule Retrieval Systems

Within the scope of this thesis, spectrum–molecule retrieval systems are considered as complete, deployable software solutions intended for practical molecular identification in laboratory environments. Accordingly, this section focuses not on isolated computational methods, machine learning models, software libraries, or search databases as previous sections, but on end-user platforms that combine analytical functionality with practical usability through accessible interfaces, structured MS/MS analysis workflows, and routine laboratory applicability. Under this definition, currently available spectrum–molecule retrieval systems are broadly classified into two principal categories: commercial vendor ecosystems and open-source or academically developed platforms. The subsections below examine each category separately to compare their practical capabilities, retrieval strategies, system-level performance, and current limitations.

3.2.1 Commercial Retrieval Systems

The main systems reviewed in this category are summarized in Table V.

TABLE V
Main commercial spectrum–molecule retrieval systems and their dominant analytical roles in practical laboratory workflows.

Name	**Specialization and Functionality**
Thermo Fisher Scientific ecosystem (Compound Discoverer, mzCloud, mzVault, Mass Frontier, mzLogic) [74, 75]	Integrated proprietary analytical environment combining spectral-library search, spectral-tree analysis, fragmentation interpretation, and candidate reduction workflows within Thermo-centered laboratory infrastructures.
Agilent MassHunter with PCDL [76]	Vendor-specific analytical platform focused on accurate-mass screening, formula prediction, and Personal Compound Database and Library (PCDL)-assisted molecular identification.
Bruker MetaboScape [77]	Metabolomics-oriented software suite supporting feature extraction, molecular annotation, spectral matching, and collision cross section (CCS)-assisted candidate refinement.
Waters Progenesis QI and UNIFI [78]	HRMS-focused analytical environments designed for compound discovery, feature alignment, and integrated library-assisted molecular identification.
SCIEX OS with LibraryView [79]	LC–MS/MS-oriented vendor platform emphasizing spectral-library matching, local library management, and standardized analytical deployment.
Shimadzu LabSolutions [80]	Proprietary analytical ecosystem supporting instrument-integrated molecular analysis, workflow standardization, and laboratory compatibility.
NIST MS Search with the NIST Tandem Mass Spectral Library [2, 81]	Specialized commercial reference platform centered on curated tandem mass spectral-library search and standardized compound identification.
METLIN [82]	Large-scale commercial and web-accessible experimental MS/MS database emphasizing spectral matching, fragment-ion search, and neutral-loss analysis.
Wiley MSforID [83]	Tandem mass spectral-library platform with strong specialization in forensic and toxicological compound identification.

Commercial spectrum–molecule retrieval systems currently represent the most operationally mature class of laboratory-ready molecular identification platforms. These systems are typically developed as proprietary analytical ecosystems by instrument manufacturers or specialized database providers and are designed to integrate spectrum interpretation directly into routine laboratory workflows. In contrast to standalone computational tools, commercial platforms consolidate the complete spectrum analysis and molecular identification workflow within unified software ecosystems designed for vendor-specific instrumentation and routine laboratory workflows.

Collectively, these systems illustrate the breadth of the commercial retrieval ecosystem, spanning tightly integrated vendor software infrastructures and specialized reference-library platforms. Across this category, retrieval workflows are primarily centered on spectral-library matching, accurate-mass constraints, metadata filtering, formula prediction, and curated reference databases.

From a retrieval-quality perspective, commercial systems remain highly effective when target compounds are represented within curated reference libraries. NIST MS Search reports top-1 identification rates exceeding 95% for in-library compounds under suitable spectral quality conditions, while EI-mode identification can exceed 98% at high match-factor thresholds [84]. METLIN currently provides one of the largest experimentally acquired MS/MS databases, containing more than 850,000 spectra and more than 900,000 compounds, while X-Rank scoring has been reported to reduce false-positive identifications by approximately 30% relative to simpler dot-product methods [82, 85]. Thermo mzCloud contains more than four million high-resolution MS/MS and MS^n spectra and extends conventional spectral matching through spectral-tree search capabilities [75]. Bruker MetaboScape additionally integrates CCS evidence, with published studies reporting false-positive reductions of approximately 20–30% relative to MS/MS-only workflows [86]. Together, these systems demonstrate strong practical performance for high-confidence in-library identification and routine workflow efficiency.

Despite these strengths, commercial ecosystems exhibit several structural limitations. Independent benchmark studies are often limited because performance claims are frequently derived from vendor application notes or internal evaluations rather than standardized peer-reviewed comparisons. Many systems are tightly coupled to proprietary instruments, file formats, licensing models, and database infrastructures, which constrain reproducibility, interoperability, and methodological extensibility. Furthermore, reviewed platforms incorporate sophisticated scoring systems or expert-rule frameworks, they generally do not provide advanced library-independent retrieval paradigms, such as representation-learning-based molecular search, contrastive embedding retrieval, or scalable open-database structure matching frameworks.

Overall, commercial retrieval platforms currently provide the strongest examples of deployable spectrum–molecule identification systems for practical laboratory environments. Their principal advantages lie in workflow standardization, technical support, curated reference resources, and instrument-level integration. However, these strengths are frequently balanced by limited methodological transparency, dependence on proprietary infrastructures, and comparatively weak support for modern open machine learning retrieval paradigms. Consequently, commercial systems define the current standard for laboratory usability, but they do not fully address the scientific need for scalable, reproducible, and representation-driven retrieval frameworks.

3.2.2 Open-Source and Academic Retrieval Systems

The main systems reviewed in this category are summarized in Table VI.

TABLE VI
Main open-source and academic spectrum–molecule retrieval systems and their dominant analytical roles in practical laboratory workflows.

Name	Specialization and Functionality
SIRIUS with CSI:FingerID, CANOPUS, and COSMIC [39, 51, 87]	Desktop-based spectrum-to-structure platform combining molecular formula annotation, fragmentation-tree analysis, fingerprint prediction, candidate ranking, compound-class prediction, and confidence estimation for database-assisted molecular identification.
MS-DIAL with MS-FINDER [38, 88]	End-to-end graphical metabolomics and lipidomics environment supporting raw data processing, feature detection, spectral-library search, formula prediction, fragment annotation, and candidate structure ranking.
MZmine 3 [89]	Open-source GUI platform for mass spectrometry data preprocessing, feature detection, spectral-library matching, GNPS integration, and reproducible workflow construction.
GNPS / MASST [43, 90]	Public web-based repository-scale spectral search and molecular networking platform focused on dereplication, analog search, and community-scale MS/MS data exploration.
MetFrag Web [34]	Browser-accessible candidate retrieval and ranking platform based on in silico fragmentation, metadata-assisted scoring, and auxiliary evidence integration.
CFM-ID 4.0 Web Server [42]	Web-accessible compound identification system using competitive fragmentation modeling for spectral prediction, candidate ranking, and peak annotation.

Open-source and academically developed platforms represent the principal noncommercial class of deployable spectrum–molecule retrieval systems. In contrast to proprietary vendor ecosystems, these platforms generally emphasize methodological transparency, reproducibility, scientific flexibility, and compatibility with open data standards. They commonly provide accessible desktop or web-based interfaces without mandatory dependence on vendor-specific analytical infrastructures, making them particularly relevant for academic laboratories, reproducible research, and flexible analytical deployment.

Among currently available systems, the SIRIUS ecosystem represents one of the most advanced integrated open retrieval environments [39, 41]. Rather than functioning as a single standalone tool, SIRIUS operates as a modular platform for spectrum-to-structure analysis. Its foundational workflow performs isotope-informed molecular formula annotation and fragmentation-tree analysis directly from MS/MS spectra. CSI:FingerID extends this framework through machine-learning-based molecular fingerprint prediction and candidate structure ranking against molecular databases [51]. Additional modules further expand analytical functionality: CANOPUS supports compound-class prediction [13], while COSMIC introduces confidence estimation and false-discovery control for structural annotations [87]. Together, these components form a unified desktop-based ecosystem spanning formula determination, candidate ranking, structural classification, and annotation-confidence assessment. Within this framework, CSI:FingerID achieved the strongest performance in CASMI 2016 category 2 for database-assisted molecular structure identification [14], with reported top-10 structure-identification accuracy above 70% and molecular formula accuracy exceeding 98% under suitable precursor-mass and charge-state conditions [41]. This combination of modular breadth, benchmark strength, and user accessibility makes SIRIUS one of the most complete noncommercial retrieval systems currently available, although dependence on remote services and candidate-database availability can constrain fully offline deployment.

MS-DIAL with MS-FINDER represents one of the most practical open platforms for routine untargeted metabolomics workflows. MS-DIAL provides the primary end-to-end processing environment, including raw data import, feature detection, deconvolution, alignment, and spectral-library search [88], while MS-FINDER extends this workflow through formula prediction, fragment annotation, and candidate structure ranking for unknown compounds [38]. MS-DIAL 5 reports lipid species annotation rates approaching 90% in reference mixtures with false annotation rates below 5% under benchmark conditions, and Flash Entropy Search substantially improves retrieval scalability for very large spectral libraries [91, 92]. MS-FINDER also demonstrated competitive CASMI 2016 category 3 performance, including approximately 47% top-1 identification for natural products and approximately 66% for lipid subsets [38]. Despite this strong practical utility, these systems remain primarily rule-, library-, and score-driven rather than based on modern representation-learning retrieval architectures.

MZmine 3 provides one of the most reproducible open-source graphical environments for mass spectrometry preprocessing and spectral-library workflows [89]. Its strengths lie in feature detection quality, workflow modularity, and interoperability with GNPS, MassBank, and MoNA ecosystems. Published benchmarks report feature-detection F1 scores of approximately 0.85–0.92 under suitable conditions. However, MZmine functions primarily as a robust data-processing and library-search platform rather than as a dedicated spectrum-to-structure retrieval engine. GNPS/MASST similarly offers powerful web-scale spectral dereplication, analog search, and molecular networking, but its strongest practical role lies in repository-scale public-data exploration rather than local deployable molecular retrieval. Annotation rates through GNPS workflows typically remain constrained by reference-library coverage.

MetFrag and CFM-ID represent specialized open academic platforms centered on candidate-centric molecular structure annotation rather than complete deployable retrieval ecosystems. MetFrag emphasizes precursor-mass-constrained candidate retrieval followed by in silico fragmentation and flexible multi-evidence ranking, with later versions incorporating metadata, retention time, and user-defined scoring beyond fragmentation alone [34, 93]. In contrast, CFM-ID focuses on Competitive Fragmentation Modeling through probabilistic spectral prediction, compound identification, and peak annotation, with CFM-ID 4.0 reporting approximately 63% top-1 and 83% top-10 identification accuracy on MoNA holdout benchmarks [94, 95]. Although both platforms substantially reduce programming barriers for molecular annotation, they are more appropriately characterized as specialized candidate-ranking systems for focused structure identification than as scalable, production-grade laboratory retrieval platforms.

Collectively, open-source and academic systems provide substantially greater methodological openness, reproducibility, and scientific flexibility than commercial ecosystems. Several platforms, particularly SIRIUS and MS-DIAL, demonstrate analytical performance that approaches or exceeds many proprietary systems under benchmark conditions. However, important limitations remain. Most current platforms continue to rely primarily on spectral-library matching, fragmentation heuristics, candidate databases, or fingerprint intermediates rather than on fully deployable representation-driven molecular retrieval paradigms. In addition, system-level scalability, enterprise deployment simplicity, and laboratory-grade operational integration often remain weaker than in commercial vendor ecosystems. Consequently, although open-source systems currently offer the strongest balance of scientific transparency and practical accessibility outside proprietary infrastructures, substantial gaps remain in the scalable deployment of modern machine-learning-centered, library-independent retrieval systems.

3.3 Conclusions from Literature Review

The current spectrum–molecule matching landscape is defined not by methodological scarcity, but by fragmentation across computational paradigms, deployment maturity, and practical accessibility. Collectively, the reviewed evidence supports several major analytical conclusions that directly shape the scientific and engineering positioning of the this thesis.

1. **The field has achieved methodological diversity, but not methodological convergence.** Spectrum–molecule matching has evolved into a broad ecosystem comprising rule-based systems, spectral-library platforms, end-to-end predictive models, and representation learning frameworks. Each paradigm addresses important aspects of molecular identification, yet no currently dominant approach simultaneously satisfies the combined requirements of:

   - scalable open-world molecular retrieval,
   - methodological modernity,
   - reproducible benchmarking,
   - practical laboratory deployment.

   The coexistence of multiple partially specialized paradigms indicates that scientific progress has been substantial, but fragmented rather than unified.

2. **Classical retrieval paradigms remain operationally dominant because of reliability, but remain fundamentally constrained in universality and open-world scalability.** Rule-based systems and spectral-library search continue to define the practical baseline for real-world laboratory molecular identification due to the following:

   - strong precision within curated candidate spaces,

- chemical interpretability,
- mature software ecosystems,
- deployment readiness.

However, these strengths are primarily realized within known analytical domains. Their dependence on predefined candidate databases, experimentally measured reference spectra, and bounded search spaces substantially limits their universality across chemical space and reduces effectiveness in blind molecular identification scenarios, where unknown compounds lack direct reference standards. As a result, while classical systems remain highly reliable for established compounds, they do not fully address the scientific challenge of scalable unknown-compound discovery, thereby directly motivating the emergence of AI-driven paradigms capable of hypothesis generation, semantic generalization, and open-world retrieval.

3. **End-to-end structure prediction expands scientific ambition, but currently provides limited retrieval infrastructure value.** Direct spectra-to-structure systems significantly broadened the conceptual scope of the field by enabling the following:

   - fingerprint prediction,
   - molecular string generation,
   - graph-native generation,
   - chemically constrained hypothesis construction.

   Nevertheless, current literature consistently indicates that these approaches remain constrained by spectral ambiguity, structural underdetermination, benchmark instability, and limited deployment practicality. As a result, they currently function more effectively as candidate-generation systems than as scalable retrieval infrastructures.

4. **Representation learning provides the strongest strategic foundation for next-generation retrieval systems.** Joint spectrum–molecule embedding frameworks most directly align with the practical requirements of modern retrieval because they do as follows:

   - support large-scale molecular database search,
   - enable semantic similarity retrieval,
   - generalize beyond exact spectral identity,
   - remain compatible with modular deployment architectures.

   Unlike direct generation approaches, representation learning naturally integrates machine learning scalability with retrieval-oriented system design.

5. **The principal gap is now translational rather than conceptual.** The literature strongly suggests that representation learning is scientifically promising, but current implementation remains limited by the following:

   - inconsistent datasets,
   - heterogeneous benchmarking,
   - weak standardization,
   - lack of full laboratory-ready deployment ecosystems.

   Accordingly, the primary limitation no longer lies in demonstrating methodological feasibility, but rather in the limited incorporation of modern AI approaches into reproducible, scalable, and practically accessible retrieval systems.

Chapter 4

Design and Methodology

This thesis presents a structured methodological design comprising two complementary components that addressed both the scientific and engineering aspects of the spectrum–molecule matching problem.

The first component focused on the research methodology and encompasses the collection and preprocessing of molecular and tandem mass spectrometry data, as well as the development of a unified protocol for training and evaluating candidate models. This protocol is was designed to learn general cross-modal semantic representations effectively, enabling accurate retrieval and ranking of candidate molecular structures from input spectra.

The second component focused on the system-level design and addressed the integration of the developed models and curated data into an end-to-end retrieval system, enabling practical deployment for spectrum-driven molecular identification.

Such an organization ensures a coherent transition from methodological foundations to the implementation of a functional solution.

4.1 Research Methodology

4.1.1 MS/MS Data Collection and Processing

To address RQ1 and construct a representative high-quality dataset of paired MS/MS spectra and molecular structures, the data acquisition and preprocessing workflow was organized as a staged pipeline. This design enabled the systematic transformation of heterogeneous raw data into a consistent and validated dataset suitable for model training and evaluation.

Stage 1: Data Source Identification

The first stage aimed to identify a sufficiently broad set of candidate data sources to maximize coverage of publicly available MS/MS spectra. Since no single comprehensive repository exists, multiple complementary discovery strategies were employed to balance the diversity and relevance of the collected data.

Candidate sources were identified through three primary channels: (i) references extracted from the literature review, (ii) recommendations from domain experts and the chemistry community, and (iii) targeted web search for publicly available annotated MS/MS datasets. This multi-channel strategy reduced selection bias and ensures coverage of both widely used benchmark datasets and less standardized but informative resources.

Stage 2: Source Selection Criteria

Following the discovery stage, the identified sources were filtered using a defined set of criteria. The selected sources were required to be openly accessible for research use, provide paired MS/MS spectra and molecular structures with essential attributes such as m/z, intensity, precursor mass, adduct, and SMILES, and be distributed in standard machine-readable formats.

These criteria ensured that the resulting dataset is both consistent and suitable for downstream preprocessing, model training, and reproducible evaluation.

Stage 3: Selected Data Sources

Applying the above selection criteria resulted in a curated set of the following data sources that jointly provided sufficient coverage, diversity, and annotation quality.

- **Global Natural Products Social Molecular Networking Library** [43]: A community-maintained platform for sharing and analyzing tandem mass spectrometry data. Beyond its own contributions, GNPS aggregates content from MONA, HMDB, and MassBank [44, 96, 97], so that a single access point gives access to experimentally acquired MS/MS spectra spanning many compound classes.
- **CANOPUS (NPLIB1)** [13]: A curated set of natural products with annotated molecular structures and matching spectra, originally assembled for structure-informed learning.
- **MassSpecGym** [12]: A benchmark-oriented dataset. Unlike the two previous sources, MassSpecGym is not so much a spectral repository as a ready-made evaluation setup, with fixed splits and an explicit protocol intended for reproducible comparison of retrieval and identification methods.

Together, these three resources covered both the experimental breadth of GNPS and the controlled benchmarking conditions of MassSpecGym, with CANOPUS contributing the natural-product portion of chemical space.

Stage 4: Data Standardization and Schema Design

Since the chosen sources differed in how they store spectra and in how complete their metadata is, the next step had to bring them onto common ground. A unified schema was critical for enabling joint processing, model training, and fair evaluation.

At the spectral level, every MS/MS record is held as a set of $(m/z, \text{intensity})$ pairs. Molecular structures are kept as canonical SMILES, which removes the ambiguity that arises when the same molecule can be written in multiple syntactic forms.

The schema contained the following fields:

$$\text{peaks\_array, precursor\_mz, adduct, instrument, smiles,}$$
$$\text{inchikey, molecular\_formula, polarity}$$

Some of these fields were not always available in the source data. When inchikey or molecular_formula was missing, it was reconstructed deterministically from the SMILES string. Polarity was, where necessary, inferred from the adduct.

The result of this stage is that spectra and molecules are described in the same way regardless of the source they came from, which is the precondition for the preprocessing and validation procedures applied next.

Stage 5: Preprocessing, Validation, and Dataset Construction

With the schema fixed, the collected data was put through a single curation pipeline that turns the raw records into a structured dataset.

Within this pipeline, molecular and spectral records were checked for validity, incomplete or malformed entries were dropped, and the structural and spectral representations were brought into a uniform form. Metadata fields coming from different sources were aligned, and the values were normalized so that differences in formatting or in annotation conventions did not propagate further.

After preprocessing, splitting into training and test subsets was performed independently for each source. Doing the split at the source level meant that the original distributional properties of every dataset were preserved inside the corresponding split.

The per-source training subsets were then concatenated into a single global training set, and the test subsets into a global test set. To remove duplicates across sources, a hash-based identification was applied: each spectrum–molecule pair was hashed using its InChIKey together with a discretized version of the spectrum, and pairs sharing the same hash were treated as the same record.

To prevent information leakage and ensure an unbiased evaluation setting, structural overlap between the global training and test sets was eliminated. This was achieved by comparing molecular structures using standardized SMILES representations and regrouping intersecting entries across splits.

As a result, the final dataset consistsed of two curated subsets of training and test, with strict separation between molecules used for model training and evaluation, while preserving internal consistency and uniqueness of spectrum–molecule pairs.

4.1.2 Molecular Data Collection and Processing

Although the dataset in Subsection 4.1.1 provided paired spectrum–molecule samples for supervised learning and benchmarking, its coverage of chemical space is inherently limited by the availability of experimentally annotated MS/MS data. In contrast, molecular structure data were substantially more abundant. Therefore, a separate staged data collection process was required to construct a large-scale candidate database for a retrieval system.

To ensure scalability while minimizing the effort required for data collection and processing, this thesis built upon the molecular database introduced by Xie et al. [11]. This resource provided a ready-to-use foundation for large-scale molecular search. It was further extended with molecules derived from the curated MS/MS dataset in Subsection 4.1.1, ensuring alignment between the candidate search space and the compounds observed during model training.

Stage 1: Molecular Data Source Selection

As stated above, the primary source of candidate molecules was the CSU-MS2-DB[1], which integrated multiple established molecular databases. This aggregation provided broad structural diversity and served as a realistic approximation of the real chemical space.

Stage 2: Schema Definition and Data Integration

To enable consistent storage and retrieval, all molecular entries were mapped to a unified data schema:

$$\text{formula TEXT, smiles TEXT, inchikey TEXT (UNIQUE),}$$
$$\text{monoisotopic\_mass REAL}$$

This schema ensured a compact and standardized representation of molecular attributes while supporting efficient indexing and lookup operations. All records were subsequently merged into a single integrated dataset.

[1] https://huggingface.co/datasets/Tingxie/CSU-MS2-DB/tree/main

Stage 3: Molecular Standardization and Validation

To ensure chemical consistency, SMILES representations were canonicalized, invalid or ambiguous structures were removed, and missing attributes were deterministically derived from available metadata information. Only structurally valid and uniquely identifiable molecules were retained.

Stage 4: Integration of MS/MS-Derived Molecules

To improve the search chemical space consistency of the molecular database, unique structures extracted from the MS/MS dataset in Subsection 4.1.1 were integrated into the existing collection. For these molecules, monoisotopic masses were computed using precursor mass, adduct information, and molecular formula. This integration step ensured that compounds observed in the curated spectral dataset were explicitly represented in the candidate search space.

Stage 5: Database Design and Retrieval Strategy

The resulting molecular collection was structured as a database designed to support both constraint-driven filtering and retrieval based on learned embeddings. Molecular embeddings were produced in advance by the trained encoder and stored alongside structured attributes, which makes similarity search efficient.

Three complementary indexing strategies were chosen to make retrieval scalable; they are described in Chapter 5.

By coupling physicochemical constraints with representation learning techniques, this design provided a scalable and flexible foundation for candidate retrieval in the spectrum–molecule matching problem.

4.1.3 Model Identification and Adaptation

To address RQ2, the present subsection sets out the experimental framework for selecting, training, and evaluating state-of-the-art models on the spectrum–molecule matching problem. The objective is to put in place a fair and reproducible benchmarking procedure under a unified setting. As discussed in Chapter 3, existing methods are commonly evaluated on heterogeneous benchmarks and under inconsistent protocols, which complicates direct model comparison. The proposed framework addresses this issue by enforcing consistency across all stages of the experimental and evaluation process.

Identification of Candidate Models

Based on the literature review, recent cross-modal spectrum–molecule matching methods were systematically analyzed. Candidate models were selected using a defined set of criteria to balance methodological relevance and practical applicability.

Preference was given to modern approaches from 2024 to 2026 to reflect the current state of the field. The architectural scope was restricted to dual-encoder models. This paradigm enabled independent modality encoding and efficient similarity-based retrieval in a shared latent space. In addition, preference was given to models based on transformers and graph architectures, which demonstrated effectiveness in representation learning across diverse chemical domains.

From a practical perspective, only methods with publicly available implementations, pretrained weights, and sufficient technical documentation were considered. This constraint ensures reproducibility and enables direct integration into the retrieval system proposed in Subsection 4.2.1.

Based on these criteria, two models were selected for benchmarking. JESTR [10] was chosen as a lightweight baseline based on a dual-encoder architecture, combining a multilayer perceptron (MLP) for spectral encoding with a graph neural network (GNN) molecular encoder. In contrast, CSU-MS2 [11] was selected as an advanced and task-specialized approach. Its architecture combined a transformer-based spectral encoder and a GIN-based molecular encoder with an External Space Attention Aggregation (ESA) module for cross-modal alignment.

The comparison between these models reflects a deliberate trade-off between architectural simplicity and representational capacity, while remaining directly aligned with the retrieval setting considered in this study.

Unified Training Protocol

To ensure comparability, both models were trained under a unified protocol with minimal modifications to their original implementations. The guiding principle of this stage was reproducibility while preserving the original technical implementation of each method.

Official source code, pretrained weights, and training scripts were used whenever available, with modifications limited to adapting the data processing pipeline to the unified dataset in Subsection 4.1.1. This approach avoided introducing implementation bias while enabling consistent evaluation.

The models were trained on a single NVIDIA A100 GPU with a batch size of 1024 samples. The learning rate was set to 1×10^{-4} to keep optimization stable. Rather than fixing the number of training epochs in advance, the procedure relied on a convergence-driven schedule: training continued until the validation performance stabilized, with early stopping triggered as soon as the validation loss began to grow. This ensured that each model is evaluated at its best operating point under identical training conditions.

Throughout training, the behavior of the model was monitored using the similarity metric computed on correct spectrum–molecule pairs. A unified experiment-tracking pipeline was used to log metrics, examine training dynamics, and store intermediate checkpoints, ensuring consistency and reproducibility across runs.

Evaluation Protocol

The evaluation protocol covered two complementary aspects: the quality of the learned representation space and the practical effectiveness of each model when used for retrieval. All models were evaluated based on the test set defined in Subsection 4.1.1, ensuring unbiased comparison.

Embedding Space Evaluation

The first stage evaluated the quality of the learned latent space by assessing how well the model separates matched and mismatched spectrum–molecule pairs. In particular, the objective was to verify that embeddings of true pairs exhibit consistently higher cosine similarity than randomly paired samples.

Using cosine similarity as the similarity measure, two empirical distributions were constructed: P_{pos}, corresponding to similarities of correct spectrum–molecule pairs, and P_{neg}, corresponding to similarities of randomly paired samples. The degree of separation between these distributions was quantified using the Kullback–Leibler divergence formula:

$$D_{\mathrm{KL}}\left(P_{\mathrm{pos}} \parallel P_{\mathrm{neg}}\right) = \sum_{i} P_{\mathrm{pos}}(i) \log \frac{P_{\mathrm{pos}}(i)}{P_{\mathrm{neg}}(i)}. \tag{22}$$

Higher divergence indicated better separation and a more discriminative embedding space. In addition, descriptive statistics of mean, standard deviation, minimum, maximum, and median supported quantitative and qualitative evaluation of similarity values.

Retrieval Evaluation

The second stage evaluated model performance in a realistic retrieval scenario. A query spectrum was used to rank candidate molecular structures from the search database. The evaluation followed a three-step procedure: (i) candidate generation, (ii) embedding-based ranking, and (iii) structural similarity assessment using the Tanimoto coefficient.

First, a candidate pool was constructed for each query using mass filtering, based on the neutral mass computed from precursor and adduct information. In each case, the ground-truth molecule was explicitly included in the candidate set.

Second, all candidates in the pool were encoded into the shared latent space and ranked according to cosine similarity between the spectrum and molecular embeddings. This ranking defined the ordered list used for structural evaluation.

Let N denote the number of queries, $\mathcal{R}_i^{(K)}$ the set of top-K retrieved candidates for query i, and $\mathcal{G}_i$ the set of relevant (ground-truth) molecules for query i. $\mathbb{I}(\cdot)$ denoted the indicator function, which equals 1 if the condition is true and 0 otherwise. Retrieval performance was evaluated using standard top-K metrics:

Hit Rate@K

Quantifies the proportion of queries for which at least one structurally correct candidate is present among the top-K retrieved results.

$$\mathrm{HitRate@K} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\left(g_i \in \mathcal{R}_i^{(K)}\right). \tag{23}$$

Precision@K

Quantifies the proportion of relevant candidates among the top-K retrieved results, reflecting the quality of the ranked list.

$$\text{Precision@K} = \frac{1}{N} \sum_{i=1}^{N} \frac{\left| \mathcal{R}_i^{(K)} \cap \mathcal{G}_i \right|}{K}, \tag{24}$$

Recall@K

Quantifies the proportion of ground-truth candidates retrieved within the top-K results, reflecting the coverage of the complete set of valid candidates.

$$\text{Recall@K} = \frac{1}{N} \sum_{i=1}^{N} \frac{\left| \mathcal{R}_i^{(K)} \cap \mathcal{G}_i \right|}{\mid \mathcal{G}_i \mid}. \tag{25}$$

Finally, to assess structural similarity beyond exact matches, retrieved candidates were compared with the ground-truth molecule using the Tanimoto coefficient computed over molecular fingerprints. For a predicted fingerprint f_p and a ground-truth fingerprint f_g, the Tanimoto similarity was defined as:

$$T(f_p, f_g) = \frac{f_p \cdot f_g}{\mid f_p \mid_2^2 + \mid f_g \mid_2^2 - f_p \cdot f_g}. \tag{26}$$

Since retrieval produces a ranked list of candidates, structural similarity was evaluated not only for the top-ranked prediction but also over the top-K candidates. Let $\{f_p^{(1)}, \ldots, f_p^{(K)}\}$ denoted the fingerprints of the top-K retrieved molecules. The average structural similarity at cutoff K is defined as:

$$\text{Tanimoto@K} = \frac{1}{K} \sum_{i=1}^{K} T\left(f_p^{(i)}, f_g\right). \tag{27}$$

A retrieved molecule was considered structurally correct if $T(f_p, f_g) \geq \tau$, where τ was a predefined similarity threshold. In addition, a summary of Tanimoto scores statistics of mean, standard deviation, minimum, and maximum values were reported to characterize the distribution of structural similarity among the retrieved candidates.

Best Model Selection Criteria

The final model selection integrated the results of latent space analysis and retrieval evaluation to identify the model that achieves the balance between quantitative performance and efficient deployment within the retrieval system. The selected model was subsequently integrated into the end-to-end architecture described in section below.

4.2 System Design

4.2.1 System Architecture

To address RQ3, the proposed system was designed as an end-to-end retrieval pipeline that transformed raw MS/MS spectra into a ranked set of molecular structure candidates. Unlike traditional reference library matching, which depends on direct comparison with pre-annotated spectra, the system followed a representation learning paradigm in which spectra and molecular structures are projected into a shared latent space, allowing flexible and data-driven retrieval.

The overall architecture followed a modular service-oriented design built from five main components Fig. 5 as follows:

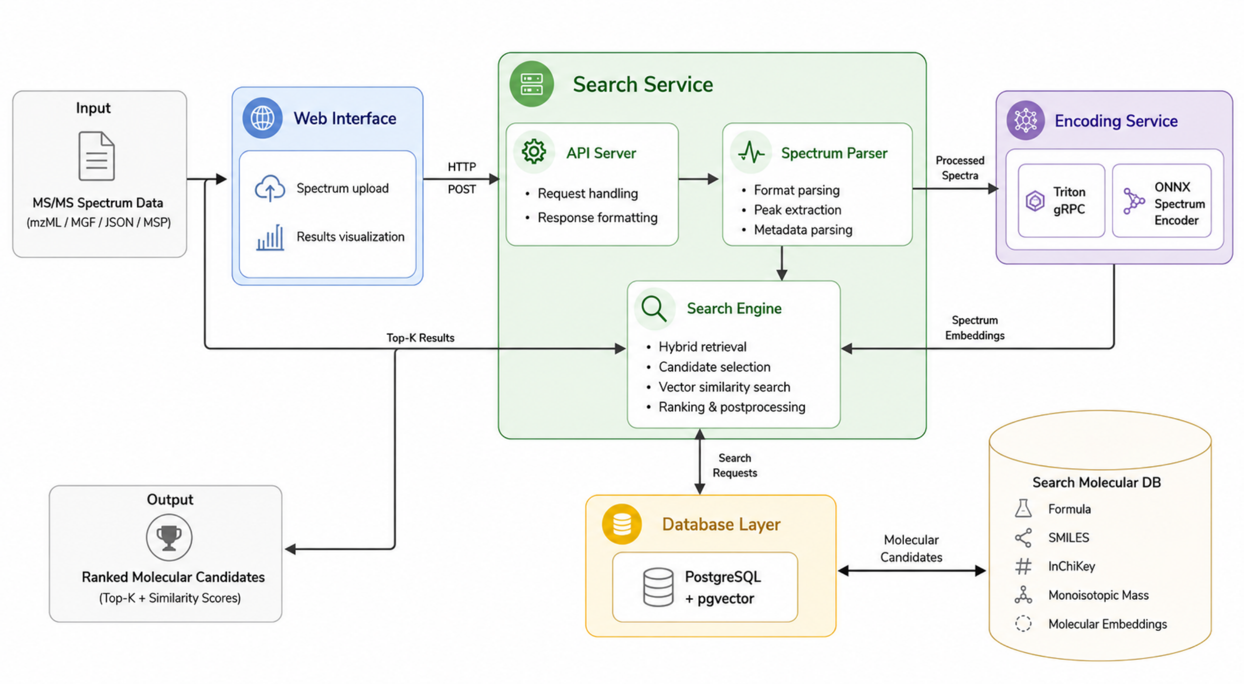


Fig. 5. System architecture of the proposed spectrum–molecule retrieval system.

(1) Client Interface: A lightweight web interface lets the user upload MS/MS spectra in several formats and inspect the retrieved candidates. This layer hides the underlying complexity of the system and provides an accessible interaction point.

(2) API Layer: A FastAPI-based service plays the role of central orchestration component, accepting incoming requests, managing batching, and coordinating the work of the downstream modules.

(3) Spectrum Processing Module: This component normalizes heterogeneous input formats, extracts peak lists, and parses the metadata. It ensures consistent data representation for the spectrum encoder inference.

(4) Inference Service: A dedicated embedding generation service encodes spectra into a shared latent space using a pretrained cross-modal model deployed via an optimized inference backend.

(5) Retrieval Engine and Molecular Search Database: The retrieval subsystem combines constraint-based filtering and vector similarity search over a large-scale molecular database, where each molecule is represented by both structured attributes and precomputed embeddings.

Together, these components defined a modular and extensible architecture that supported efficient processing, scalable candidate retrieval, and practical deployment in real-world analytical settings.

4.2.2 System Efficiency and Robustness Evaluation

The evaluation of the proposed system focused on its performance as a deployed retrieval service, with particular emphasis on efficiency, robustness, and stable operation under realistic workload conditions. Since retrieval quality was determined by the pretrained cross-modal model used within the system, it was not reassessed at the system level. The performance of the underlying model was evaluated separately in Subsection 4.1.3. In contrast, the purpose of system-level evaluation was to assess how efficiently and reliably the deployed pipeline processes input spectral files in practice.

In the proposed workflow, the system accepted an input file containing multiple MS/MS spectra, where each spectrum was processed as an individual query. Therefore, the system evaluation was designed to characterize four key aspects of its operational performance.

Per-spectrum processing time

First, the average processing time per spectrum was measured. This metric quantified the time required to handle a single record and provided a normalized estimate of system efficiency independent of input file size.

Parallel file processing

Second, the system was evaluated under concurrent usage by processing multiple input files in parallel. This assessment determined the level of concurrency that could be sustained while maintaining stable execution, acceptable response times, and correct result generation.

Maximum supported file size

Third, the system was tested on input files containing an increasing number of spectra to determine the maximum capacity that can be processed without failures or significant performance degradation. This evaluation provided an estimate of the practical upper bound on the number of records per file.

Robustness to invalid input

Fourth, the system was evaluated on incorrect, incomplete, or corrupted spectral files to verify reliable handling of invalid inputs. The evaluation included detection of parsing errors, prevention of system failures, and verification of proper error reporting.

Based on these scenarios, the following metrics were recorded:

- **Average processing time per spectrum record:**

$$T_{\text{avg}} = \frac{T_{\text{file}}}{N}, \tag{28}$$

  where T_{file} is the total time required to process the input file, and N is the number of spectrum records contained in the file.

- **Parallel processing capacity:**

$$F_{\text{parallel}} = \max\{F \mid \text{all } F \text{ files are processed concurrently without failure}\}, \tag{29}$$

  where F denotes the number of input files processed in parallel.

- **Maximum number of spectrum records per file:**

$$N_{\text{max}} = \max\{N \mid \text{a file with } N \text{ records is processed without error}\}. \quad (30)$$

- **Request-level failure rate:**

$$F = \frac{N_{\text{failed}}}{N_{\text{requests}}}, \quad (31)$$

where N_{failed} is the number of requests that result in controlled error responses (e.g., HTTP error codes), and N_{requests} is the total number of processed requests.

- **Record-level degradation rate:**

$$D = \frac{N_{\text{degraded}}}{N_{\text{processed}}}, \quad (32)$$

where N_{degraded} is the number of spectrum records that could not be fully processed due to missing or invalid metadata, and $N_{\text{processed}}$ is the total number of parsed records.

This evaluation protocol provided a practical characterization of the deployed system as a retrieval service.

4.2.3 Design Decisions and Justification

The design of the proposed system was determined by the practical constraints of its intended deployment environment. Specifically, the system was aimed at research and private laboratories, where mass spectrum was typically analyzed by small teams under limited computational resources and non-continuous workloads. These operating conditions required a solution that ensured **system deployability and maintainability**, **computational efficiency without losing retrieval accuracy**, and **stable behavior under variable demand**.

Deployability and maintainability

The system was designed as a modular retrieval service that could be deployed on commonly available infrastructure, with no need for specialized computing environments. In particular, it was optimized for CPU-based execution and not relied on dedicated GPU resources, which are typically not present in a standard laboratory. This decision followed directly from the intended usage scenario, in which the annotation tool had to be easy to install, maintain, and integrate into existing workflows. A lightweight deployment model lowered the operational overhead and improved accessibility in environments with limited technical support.

Computational efficiency with accurate retrieval

To preserve computational efficiency while maintaining reliable performance, the system implemented a retrieval approach in which input spectra are projected into a shared embedding space aligned with molecular representations from search database. This approach enabled it possible to compare spectra directly with candidate molecules without relying on reference spectral libraries, opened the way to retrieval over large molecular databases with precomputed embeddings, and improved coverage of chemical space.

To control computational cost, the candidate search space was first reduced through mass-based filtering, which provided a fast and chemically grounded constraint on plausible molecules. The remaining candidates were then ranked by similarity in the learned embedding space. This two-stage design reduced the number of expensive similarity computations while keeping the ability to surface structurally relevant molecules, which produced a workable balance between efficiency and retrieval accuracy.

Robust operation under variable demand

The system was built to behave reliably under the variable workloads typical of laboratory environments. Instead of assuming continuous high-throughput processing, it handled requests independently, which keeps performance stable under both low and high load.

The architecture supported efficient handling of single queries as well as batch inputs without requiring persistent resource allocation. As a result, the system scaled with the number of processed spectra while keeping its execution behavior and response times predictable. Therefore, the system stayed responsive when demand fluctuates and avoided unnecessary computational overhead when idle.

Taken together, these design decisions translated the practical requirements of the target deployment setting into a retrieval framework that was deployable, computationally efficient, operationally reliable, and well-suited for accurate spectrum-based molecular annotation.

4.2.4 System Assumptions and Constraints

The proposed system introduced several assumptions and constraints that defined its operational scope

The first assumption is that input spectra must contain a valid precursor m/z value. This information is required for mass-based candidate filtering, which forms the first stage of the retrieval. Spectra without a precursor m/z cannot be processed reliably by the current pipeline.

A second assumption is that the learned embedding space provided a meaningful alignment between spectra and molecular structures. While this feature was supported by empirical evaluation, the alignment may not hold uniformly across all chemical classes, especially for structurally similar isomers.

Third, the retrieval process relied on the completeness of the molecular database. The database did not encompass the entire chemical space, which could limit recall for novel or rare compounds.

A fourth point related to computational efficiency on resource-constrained hardware, which required several simplifications. In particular, candidate selection was restricted to a fixed mass tolerance window, and vector retrieval relies on approximate nearest-neighbor search. These approximations improved performance, although they could introduce minor deviations from exact ranking.

Finally, the system was conceived as a candidate generation and ranking tool rather than as a definitive identification solution. Its main role was to narrow the search space and offer high-quality candidate suggestions, which could then be verified by domain experts.

Future work may address these limitations by adding fallback strategies for missing precursor information, increasing embedding robustness, and integrating additional computational methods for reliable spectrum–molecule matching.

Chapter 5

Implementation and Results

Building on the methodological framework set out in Chapter 4, this chapter presents the practical implementation of the proposed approach together with the results of its experimental evaluation.

Specifically, the chapter describes how the MS/MS spectrum and molecular data processing pipelines were implemented, how the unified spectrum–molecule dataset and the molecular search database were constructed, and how the candidate models for the resulting semantic embeddings were trained and evaluated. The chapter also covers the integration of these components into an end-to-end retrieval system and discusses its performance in terms of computational cost and scalability.

5.1 Data Preparation

The data preparation stage was implemented within a unified codebase[1], which provided all components required for processing MS/MS spectra and molecular structure data.

The pipeline was developed in Python 3.11 and relied on libraries for tabular

[1] https://github.com/IVproger/csmp-data-pipeline

data processing (pandas, numpy, scikit-learn), cheminformatics (RDKit, pubchempy, libchebipy), and mass spectrometry data handling (pymzML, matchms, pyteomics, psims).

The codebase followed a modular architecture that separated reusable processing logic from execution workflows and is organized into functional components as follows:

- src/ **package** — core processing modules for MS/MS data preprocessing and molecular database construction;
- **notebooks** — reproducible scripts implementing the main stages of the pipeline, including:
  - ms_data_processing.ipynb — MS/MS ingestion, preprocessing, and hash-based deduplication;
  - smiles_deduplication.ipynb — molecule-level train/test split construction;
  - ms_dataset_eda.ipynb — statistical analysis of the MS/MS dataset;
  - molecular_database_preprocessing.ipynb — molecular data aggregation and database construction;
  - molecular_database_eda.ipynb — statistical analysis of the molecular database.

A configuration-driven approach was used to ensure consistent integration of heterogeneous sources. Dataset definitions and processing parameters were specified in config/data_sources.yaml and config/data_processing.yaml.

The subsections below describe the implementation of the MS/MS and molecular data processing stages, together with the resulting artifacts, including the curated spectrum–molecule dataset and the molecular search database.

5.1.1 MS/MS Data Processing and Dataset Construction

The MS/MS data processing pipeline followed the procedure defined in Section 4.1.1 and was executed through the ms_data_processing.ipynb workflow.

The pipeline was configured via external YAML files, which specified data sources and processing parameters, enabling reproducible and flexible integration of heterogeneous datasets. All input sources were ingested and processed within a unified execution flow that standardizes spectral and molecular representations.

The pipeline processed each record through a consecutive set of stages involving raw MS/MS data parsing, spectral peak extraction, metadata standardization, and transformation of molecular annotations into a unified representation format. The processing logic was encapsulated in reusable modules, while the notebook orchestrates their execution and keeps the intermediate results traceable.

Duplicate spectrum–molecule pairs were removed by a deterministic hashing procedure. Each pair was encoded through the combination of the molecular identifier (InChIKey) and a discretized spectral representation, which maked deduplication consistent and reproducible across data sources.

Dataset split construction was performed as a separate stage using the smiles_deduplication.ipynb workflow. Records were grouped by standardized SMILES representations, and the split into training and test subsets was enforced at the molecular level. This approach guaranteed structural disjointness between subsets while preserving the overall distribution of the data.

The output of this stage was a curated dataset of validated spectrum–molecule pairs stored in a unified tabular format. This dataset served as the direct input for model training and evaluation in subsequent stages.

The overall data processing workflow was summarized in Pipeline 2.

PIPELINE 2
MS/MS Data Processing and Dataset Construction

Load configuration files data_sources.yaml and data_processing.yaml
Initialize empty dataset $\mathcal{D}$
for each data source S in configuration **do**
    Ingest raw MS/MS records from S
    for each record r in S **do**
        Parse spectrum into $(m/z, \text{intensity})$ pairs
        Extract and normalize metadata fields
        Standardize molecular annotations (SMILES, InChIKey, formula)
        Construct spectrum–molecule pair p
        Add p to $\mathcal{D}$
    end for
end for
Remove duplicates from $\mathcal{D}$ using $\text{hash}(\text{InChIKey}, \text{discretized spectrum})$
Group pairs in $\mathcal{D}$ by standardized SMILES
Split $\mathcal{D}$ into training set $\mathcal{D}_{train}$ and test set $\mathcal{D}_{test}$ with molecule-level separation
Export $\mathcal{D}_{train}$ and $\mathcal{D}_{test}$ in unified tabular format

5.1.2 Constructed Dataset Description

This section describes and formally characterizes curated dataset produced by the data processing stage. The dataset was organized into two disjoint train and test splits, constructed to support consistent model training and unbiased evaluation. The statistical metrics and distributions were computed using the ms_dataset_eda.ipynb script.

Both splits followed an identical schema and consist of validated spectrum–molecule pairs. Each record combined spectral attributes (peaks_json, precursor_mz, instrument, adduct, polarity) with molecular annotations (smiles, inchikey, molecular_formula), along with a unique identifier (pair_id) assigned during deduplication stage. From a chemical perspective, this representation captured both the fragmentation pattern of a molecule (MS/MS peaks) and its underlying structural identity.

The statistics below structurally described of the constructed dataset, reflecting the outcome of the applied curation and integration procedures.

Training Set Properties

The training split contained 1,111,380 pairs corresponding to 64,234 unique molecular structures. Across these records, 83 adduct types were observed, with $[M+H]+$, [M-H]-, and $[M+Na]+$ being most frequent, followed by $[M+HCOO]$- and $[M+Cl]$-. These adducts corresponded to common ionization pathways in mass spectrometry and influence the observed precursor ions and fragmentation behavior.

On the spectral side, the data showed an average of 52.45 peaks per spectrum (median: 16.00), an average TIC of 292.48 (median: 207.63), and a normalized maximum intensity with mean and median both equal to 100.00. The average m/z range was 220.02 (median: 203.07), and the average base peak m/z was 246.03 (median: 199.03). Precursor m/z values had a mean of 404.50 and a median of 363.16, ranging from 32.05 up to 3355.60. This range reflected the diversity of molecular masses and chemical classes that presented in the dataset.

The distributions of functional groups and instrument types are shown in Fig. 6 and Fig. 7.

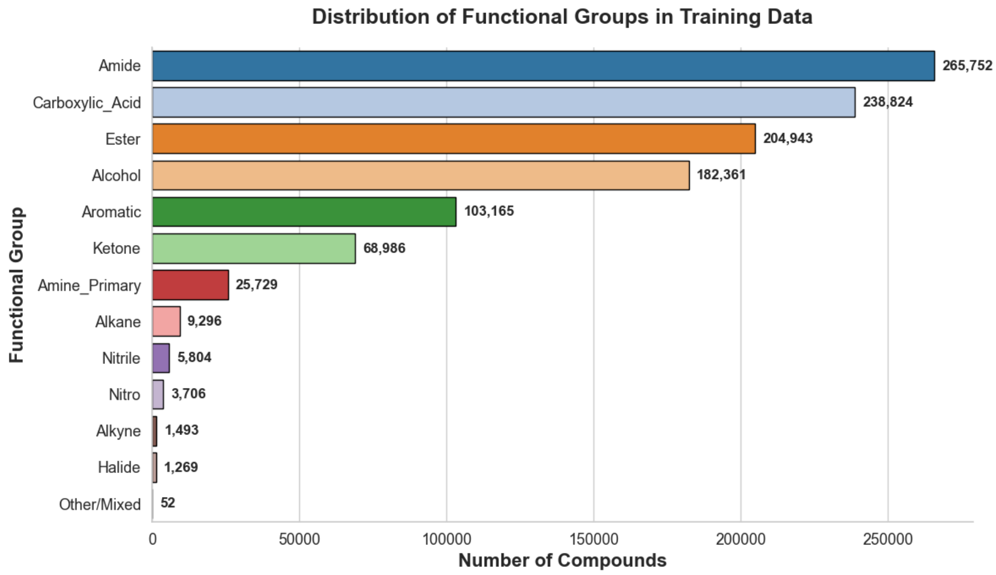


Fig. 6. Distribution of functional groups in the training dataset.

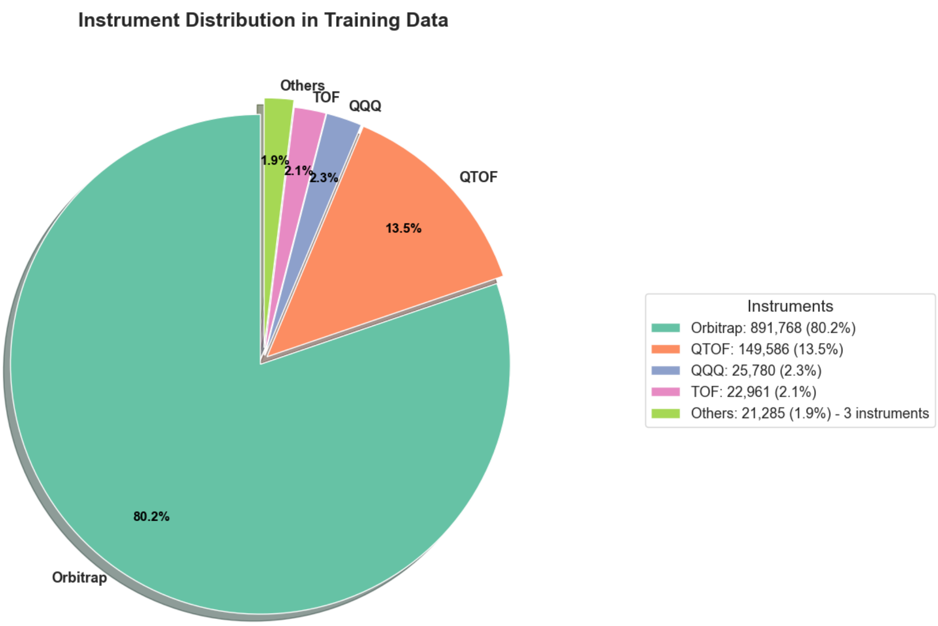


Fig. 7. Distribution of instruments in the training dataset.

Test Set Properties

The test split contained 271,045 pairs and 13,523 unique molecular structures. A total of 45 adduct types were present, dominated by $[M+H]+$ and

[M+Na]+, followed by [M-H]-, [M+HCOO]-, and [M+H-H_2O]+. The reduced diversity of adducts reflected the controlled composition of the evaluation set.

Spectral statistics were comparable to the training split, with an average of 53.31 peaks (median: 10.00), average TIC of 307.14 (median: 209.20), and normalized maximum intensity (mean and median: 100.00). The average m/z range was 175.67 (median: 147.03), and the average base peak m/z was 274.95 (median: 239.11). Precursor m/z values had a mean of 417.68 and a median of 370.20, ranging from 69.04 to 2679.48.

The corresponding distributions of functional groups and instruments are presented in Fig. 8 and Fig. 9.

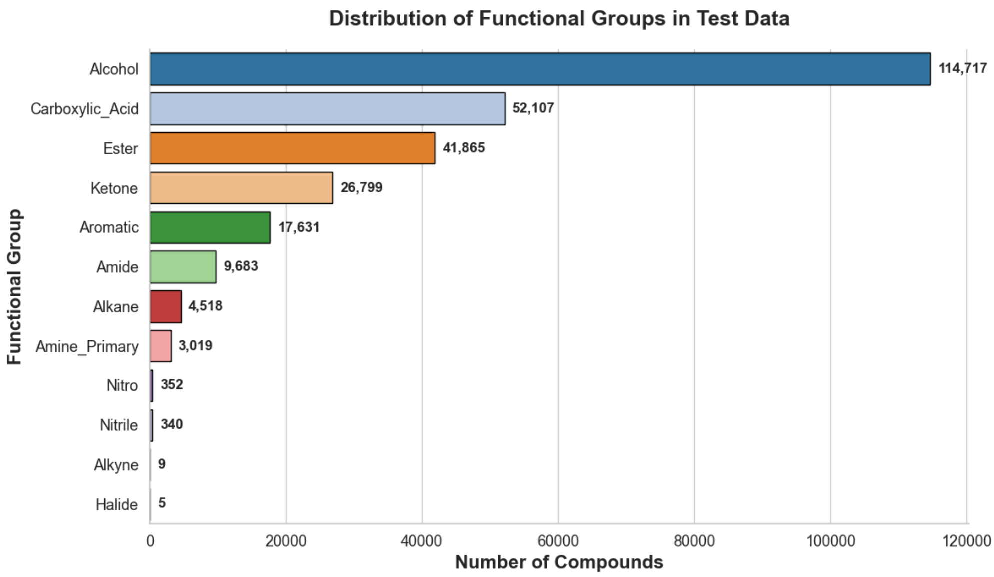


Fig. 8. Distribution of functional groups in the test dataset.

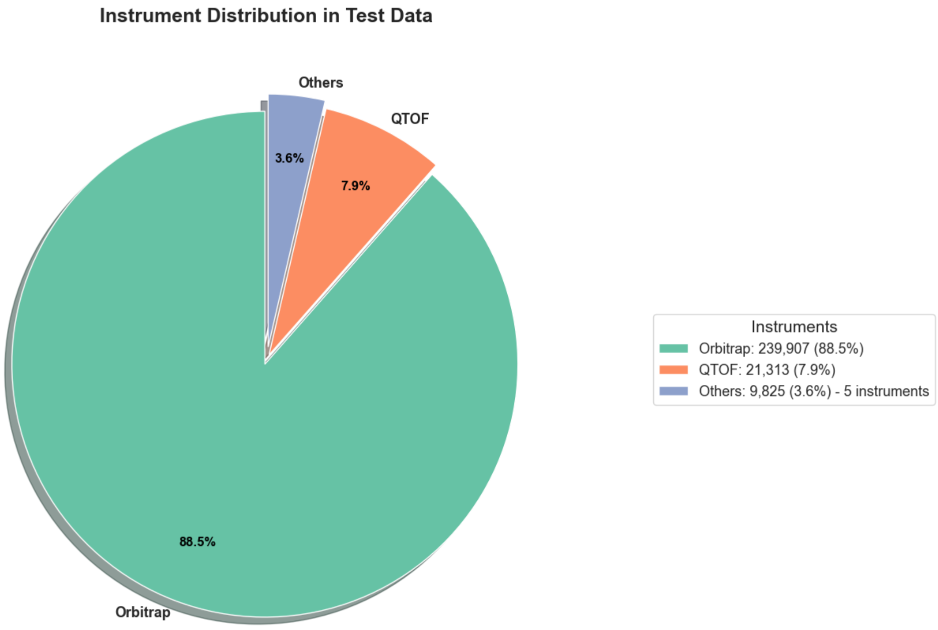


Fig. 9. Distribution of instruments in the test dataset.

A detailed analysis of these characteristics, including their implications for model behavior and dataset limitations, is provided in Chapter 6.

5.1.3 Molecular Data Processing and Search Database Construction

The molecular data processing pipeline followed the procedure defined in Subsection 4.1.2 and was executed within the unified data processing codebase. The workflow was implemented through the molecular_database_processing.ipynb script, which aggregated, transformed, and loaded molecular data into a relational database.

At the implementation level, molecular records from 24 publicly available sources (e.g., HMDB, KEGG, DrugBank, ChEBI) were ingested and mapped to a unified schema in the loading stage. Schema alignment was performed programmatically by extracting only the attributes required for downstream retrieval, especially $\mathrm{formula}$, smiles, $\mathrm{inchikey}$, and $\mathrm{monoisotopic\_mass}$. Redundant identifiers and precomputed embedding fields were explicitly excluded to reduce storage overhead and avoid inconsistencies across sources.

All records were consolidated using $\mathrm{inchikey}$ as a unique identifier, ensuring a one-to-one mapping between database entries and molecular structures. In cases of conflicting representations, priority was given to standardized SMILES derived during preprocessing.

The processed data were stored in a PostgreSQL relational database, where each entry corresponded to a unique molecular structure defined by the unified schema. This design enabled efficient integration with the retrieval pipeline and supports both structured filtering and similarity-based search.

To optimize query performance, indexing was applied at the database level. A B-tree index was defined on $\mathrm{monoisotopic\_mass}$ to support range-based candidate filtering, while an additional B-tree index on $\mathrm{formula}$ enableed efficient exact-match queries.

After model training, the database was extended with a $\mathrm{mol\_embedding}$ field, where molecular embeddings were stored as fixed-dimensional vectors. An HNSW index (cosine similarity) was constructed over this field to enable scalable approximate nearest-neighbor search during retrieval.

As a result, the implementation transformed heterogeneous molecular data into a unified, indexed, and retrieval-ready database, forming the candidate search space for the spectrum–molecule matching system.

The overall molecular database construction workflow is summarized in Pipeline 3.

PIPELINE 3
Molecular Data Processing and Database Construction

Load molecular datasets from all curated sources
Initialize empty database $\mathcal{M}$
for each record r in all sources **do**
    Extract required attributes (formula, SMILES, InChIKey, mass)
    Standardize molecular representation
    Remove redundant identifiers and unused fields
    Insert or update entry in $\mathcal{M}$ using InChIKey as unique key
end for
Create B-tree indexes on $\mathrm{monoisotopic\_mass}$ and $\mathrm{formula}$
Compute molecular embeddings using trained encoder
Store embeddings in $\mathcal{M}$ as $\mathrm{mol\_embedding}$
Build an HNSW index over the embedding space for cosine similarity-based search

5.1.4 Constructed Search Database Description

This section formally characterizes the resulting unified molecular search database. The statistical properties and composition of the database were executed using the molecular_database_eda.ipynb script.

The constructed database contained 872,935 molecular entries, each uniquely identified by an InChIKey. This guaranteed a one-to-one correspondence between database entries and molecular structures. However, 844,428 unique SMILES representations were observed, indicating that multiple entries shared identical or equivalent SMILES strings. This discrepancy arised because different InChIKeys might correspond to alternative representations of the same or closely related structures, reflecting variations in standardization or encoding.

The dataset included 147,332 unique molecular formulas. The most frequent formulas corresponded to lipid-like compounds with long carbon chains and repeated structural motifs, as summarized in Table VII.

TABLE VII
Most frequent molecular formulas in the constructed database.

Formula	Count
C77H150O17P2	3716
C76H148O17P2	3706
C75H146O17P2	3703
C78H152O17P2	3591
C74H144O17P2	3570

Additional characteristics of the constructed molecular database are presented below in Fig. 10, Fig. 11 and Fig. 12.

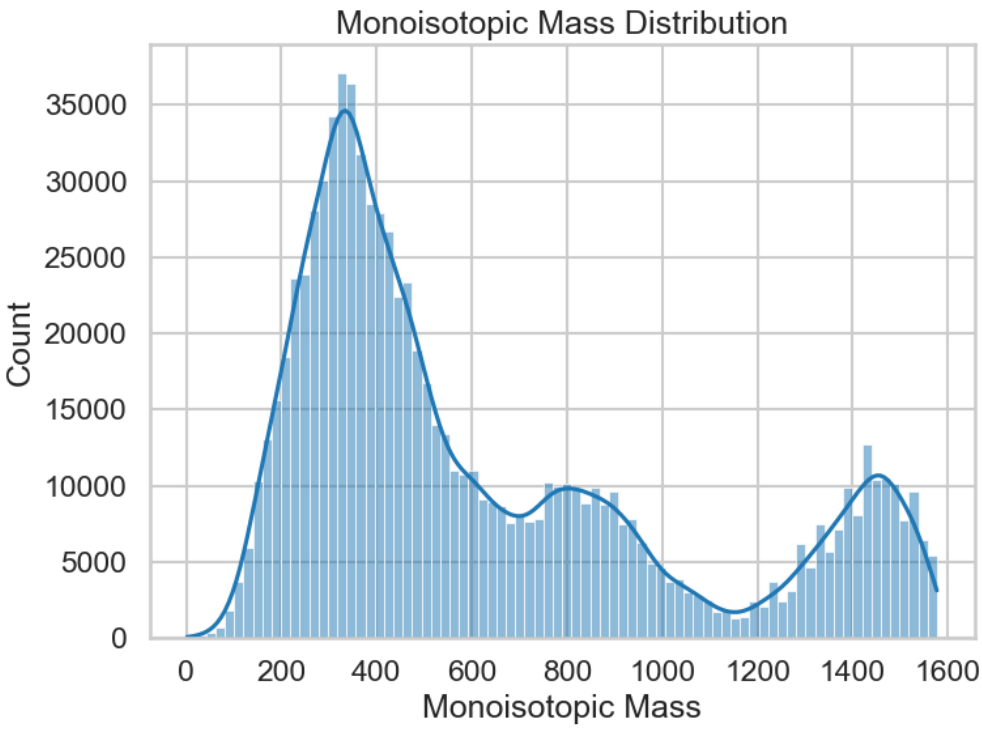


Fig. 10. Distribution of monoisotopic mass in the molecular database.

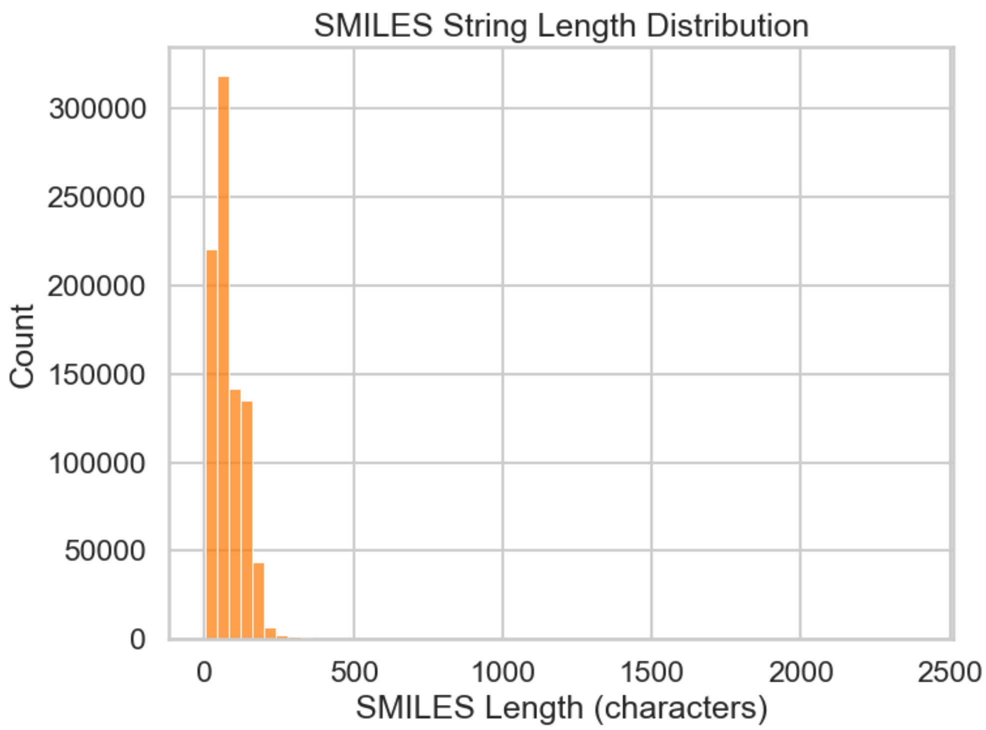


Fig. 11. Distribution of SMILES string lengths in the molecular database.

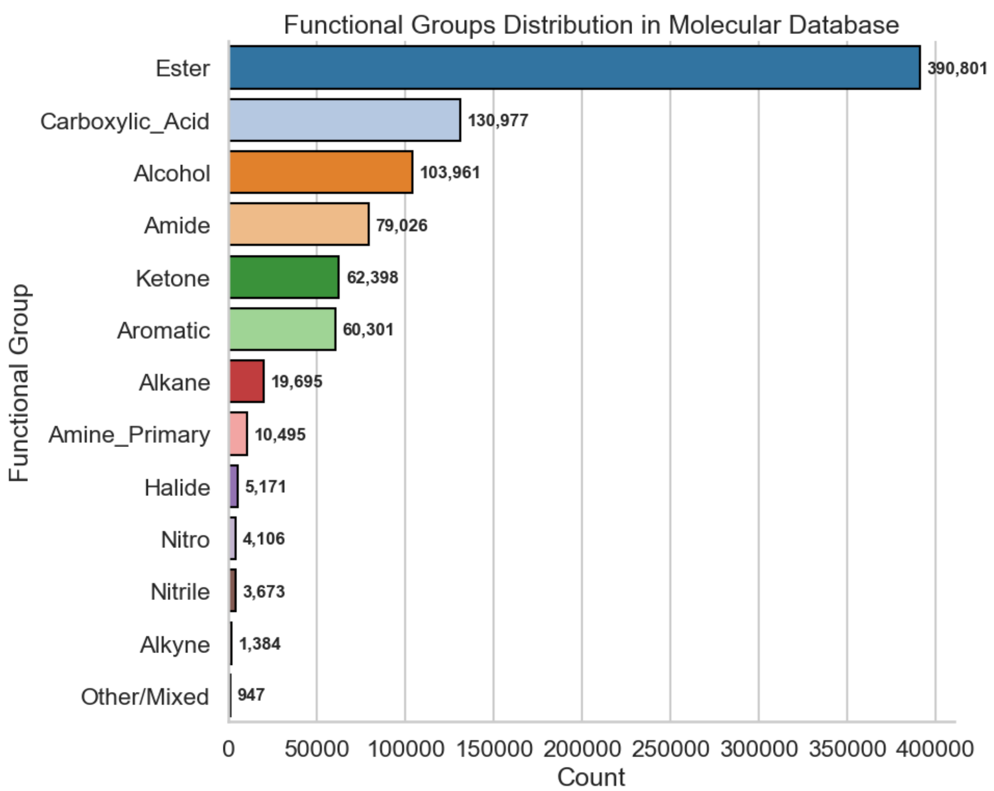


Fig. 12. Distribution of functional groups in the molecular database.

The resulting search database offered a large-scale and structurally diverse collection of molecular candidates suitable for downstream retrieval. A detailed analysis of these characteristics, including their consequences for retrieval performance and dataset limitations, is given in Chapter 6.

5.2 Model Implementation and Evaluation

This section describes how the experimental methodology defined in Subsection 4.1.3 was implemented for the JESTR and CSU-MS2 candidate models. The goal is to configure both models, train them under a single unified protocol, and evaluate them through a consistent validation procedure, so as to identify the variant most appropriate for integration into the retrieval system for spectrum–molecule matching.

The implementation preserved the original architectures and training procedures intact, and modifications are restricted to data interface alignment, training pipeline configuration, and integration into a unified evaluation workflow. As a result, the comparison reflected intrinsic model behavior rather than implementation-specific differences.

The subsections below cover the configuration and training of each model, the description of the unified evaluation pipeline, and the comparative results.

5.2.1 JESTR Model Configuration and Training

Implementation Basis and Repository Setup

The JESTR was implemented based on the official repository[2]. An adapted working repository[3] was developed to enable integration into the retrieval pipeline and controlled experimentation flow. It contained all modifications related to dataset preparation, configuration, and experiment tracking.

The software environment was configured in accordance with the original project requirements within an isolated Python environment, ensuring compatibility with the reference implementation.

The main implementation components were organized within the jestr/ module as follows:

- jestr/train.py — main training pipeline,
- jestr/params.yaml — experiment configuration,
- jestr/models/ — spectral and molecular encoder definitions,
- jestr/data/ — dataset loading and preprocessing utilities,
- jestr/eval.py and jestr/ranking_eval.py — evaluation scripts.

Pretrained checkpoints released by the authors of [10] for both the MassSpecGym and CANOPUS (NPLIB1) settings were available and considered as initialization options during model setup.

Dataset Adaptation for JESTR

Since JESTR required a benchmark-oriented tabular input format, the constructed dataset was transformed into a MassSpecGym-style TSV representation using Pipeline 4.

[2]https://github.com/HassounLab/JESTR1/tree/v2
[3]https://github.com/IVproger/JESTER

PIPELINE 4
Dataset Adaptation for JESTR Training

Load curated dataset $\mathcal{D}$
Initialize empty dataset $\mathcal{D}_{\text{tsv}}$
for each record $r \in \mathcal{D}$ **do**
    Extract spectral peaks from $\mathrm{peaks\_json}$
    Convert peaks into aligned sequences:
        $M \leftarrow$ list of m/z values
        $I \leftarrow$ list of corresponding intensities
    Serialize M and I as comma-separated strings
    Extract molecular representation:
        $s \leftarrow$ SMILES string
    Extract metadata:
        $p \leftarrow \mathrm{precursor\_mz}$
        $a \leftarrow \mathrm{adduct}$
        $k \leftarrow \mathrm{InChIKey}$
    Check validity of M, I, and s
    If any value is missing or invalid, **continue**
    Assign deterministic identifier id_r
    Assign split label (train / validation / test)
    Construct TSV record $t = (M, I, s, p, a, k, \mathrm{id}_r, \mathrm{split})$
    Add t to $\mathcal{D}_{\text{tsv}}$
end for
Export $\mathcal{D}_{\text{tsv}}$ to $\mathrm{train.tsv}$

Model Architecture

JESTR followed a dual-encoder architecture in which molecular structures and MS/MS spectra were treated as two complementary views of the same underlying object and were mapped into a shared embedding space. Rather than reconstructing intermediate objects such as fingerprints or spectra, the model learned a joint latent representation in which matching spectrum–molecule pairs were placed close to one another, while non-matching pairs were separated. Its architecture is illustrated in Fig. 13.

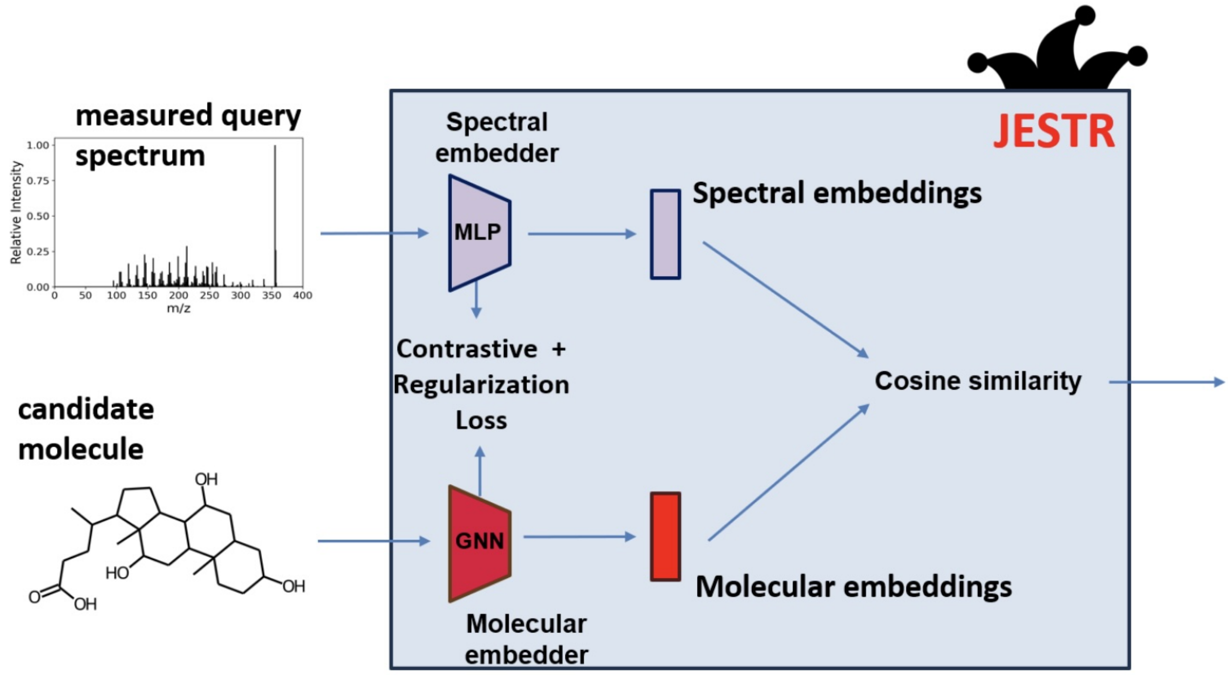


Fig. 13. Architecture of the JESTR model.

The molecular branch represented each molecule as a graph and processed it using a graph convolutional encoder. In the original formulation, atom-level descriptors such as atom type, atomic mass, valence, chirality, degree, hydrogen count, ring membership, and aromaticity were encoded at the node level, while bond-related properties were encoded at the edge level. The graph convolution layers were followed by pooling and fully connected projections to produce a fixed-dimensional molecular embedding.

The spectral branch converted each spectrum into a fixed-width binned representation. Peaks were discretized into 1 Da bins over a range of up to 1000 Da, and intensities within each bin were aggregated. The resulting vector was then normalized and log-transformed to reduce the influence of highly intense peaks. This fixed-length representation is subsequently processed by a multilayer perceptron to produce the spectral embedding.

The two embeddings were aligned into a shared latent space, where cosine similarity is used as the matching function. This design is central to the JESTR formulation, since ranking could be performed directly in the learned representation space without introducing an additional downstream ranking head.

Training Configuration

Training was configured through the $\mathrm{params.yaml}$ file of the adapted repository. In the implemented setup, GPU-based execution was used with a batch size of 1024, a learning rate of 1×10^{-4}, a maximum of 100 epochs, and validation checks performed at a relative interval of 0.5. The dataset split defined during preprocessing was preserved, approximately 80% of the data was used for training and 20% for validation, while the test subset was excluded from this stage and reserved for evaluation. The spectral encoder was configured as $\mathrm{MLP\_BIN}$, and the molecular encoder employed a three-layer GCN-based architecture with a final embedding dimension of 512.

The optimization objective adopted a contrastive formulation, where InfoNCE training promoted similarity between matching spectrum–molecule pairs than between non-matching pairs within each batch. In the implemented configuration, cosine similarity was used together with a temperature parameter of 0.05. This setup is consistent with the original design principle of learning a joint space optimized for retrieval task.

Weights Initialization

The CANOPUS (NPLIB1) pretrained checkpoint was selected as the initialization for fine-tuning, as it provides better adaptability to the constructed dataset than the MassSpecGym alternative. While the MassSpecGym model was trained on a larger and more diverse corpus, its representations were moe rigid and less responsive to distributional shifts. In contrast, the CANOPUS checkpoint, trained on a smaller-scale dataset, offered a less dominant initialization, enabling more effective adaptation to the spectral characteristics of the constructed dataset during fine-tuning. This maked model more suitable for controlled retraining within the unified experimental setting.

Training Execution and Logging

Training was carried out via the main script jestr/train.py with the configuration defined in params.yaml. The training pipeline was implemented based on the PyTorch Lightning framework, which provided structured handling of optimization, validation, checkpointing, and logging.

The observed training dynamics indicated stable convergence over the full training run. Specifically, the training loss decreased from 14.48 at the beginning of training to 7.92 at epoch 99, while the validation loss decreased from 13.69 to 10.73 over the same period. The corresponding loss curves are shown in Fig. 14. All intermediate checkpoints and training logs were stored in the experiment directory, ensuring reproducibility and enabling fair evaluation.

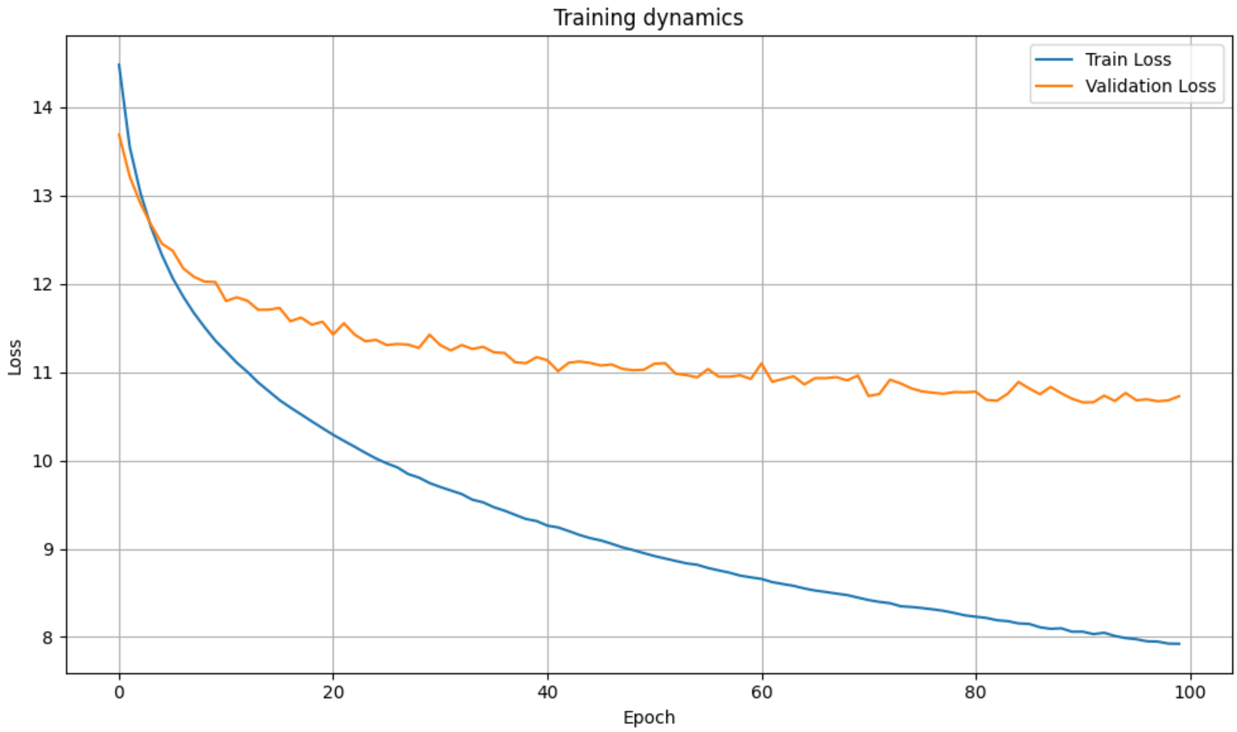


Fig. 14. Training and validation loss dynamics for the JESTR model.

5.2.2 CSU-MS2 Model Configuration and Training

Implementation Basis and Repository Setup

The implementation of CSU-MS2 in this thesis is based on the official repository[4]. To support adaptation, additional training, and integration into the developed system, the original codebase was incorporated into a dedicated project repository[5].

All dependencies were installed according to the original project instructions inside an isolated Python environment, which guarantees compatibility with the reference implementation. The codebase was organized within the CSUMS2/ module, with the main components summarized below:

[4] https://github.com/tingxiecsu/CSU-MS2
[5] https://github.com/IVproger/CSMP_thesis_project/tree/main/services/CSU-MS2/

- CSUMS2/train.py — main training pipeline,
- CSUMS2/model.py — model architecture definition,
- CSUMS2/dataloader/ — dataset processing and dataloader construction,
- CSUMS2/eval.py — embedding-level evaluation,
- CSUMS2/ranking_eval.py — retrieval and ranking evaluation.

Pretrained checkpoints released by the authors of [11], including hcd_model and qtof_model variants trained under different collision-energy regimes (low, medium, and high), were available and considered as initialization options during model setup.

Dataset Adaptation for CSU-MS2

The constructed dataset described in Subsection 5.1.2 was integrated by extending the original dataloader pipeline through dataset_wrapper.py. This custom wrapper enabled direct ingestion of the unified dataset format developed in this thesis and converted it into the input structures required by the model while preserving the original training and evaluation workflow. The dataset adaptation procedure for CSU-MS2 is summarized in Pipeline 5.

In addition to the contrastive training pipeline, the same wrapper was extended to support ranking-based evaluation. In this setting, each query spectrum was paired with a set of candidate molecules, which were converted into graph representations and assembled into batched inputs for retrieval.

PIPELINE 5
Dataset Adaptation for CSU-MS2 Training

```
Load curated dataset 𝒟
Initialize processed dataset 𝒟_proc
for each record r ∈ 𝒟 do
    Parse spectral data from peaks_json
    Convert peaks into (m/z, intensity) arrays
    Remove invalid values and sort peaks by ascending m/z
    Construct Spectrum object with metadata
    Convert SMILES to molecular graph G using RDKit
    Encode atom-level and bond-level features in G
    If spectrum or graph representation is invalid, continue to next record
    Construct spectrum–molecule pair (S, G)
    Add (S, G) to 𝒟_proc
end for
Split 𝒟_proc into training and validation sets
Serialize and cache processed datasets
Define batch collation:
  pad spectral sequences to fixed length
  aggregate molecular graphs into batched structures
```

Model Architecture

CSU-MS2 also adopted a dual-encoder design, but it relied on more expressive modality-specific encoders and an explicit cross-modal aggregation mechanism. As shown in Fig. 15, the model combined a transformer-based spectral encoder, a graph neural network-based molecular encoder, and an External Space Attention Aggregation (ESA) module for modality alignment.

On the spectral modality side, CSU-MS2 represented spectra as sequences of peak-level features rather than fixed bins. Each peak was encoded using sinusoidal m/z embeddings combined with normalized intensities, and the resulting sequence was processed by a transformer encoder without positional embeddings. This design was motivated by the set-like nature of spectra, where peaks did not follow

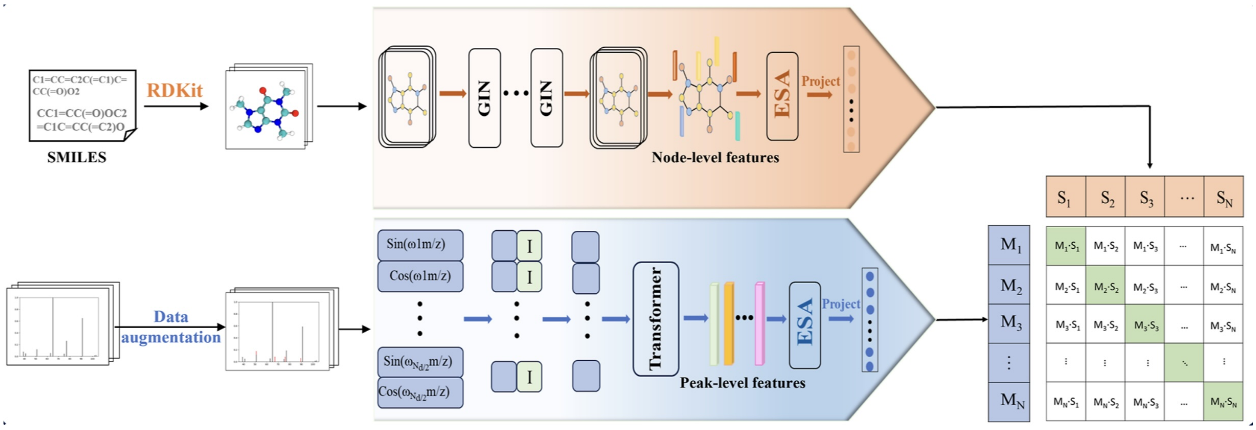


Fig. 15. Architecture of the CSU-MS2 model.

a sequential order. The spectral encoder then applied ESA to aggregate peak-level features into a compact spectral representation.

On the molecular modality side, the model constructed a graph from the SMILES representation and processed it using a GIN-based encoder. In the original formulation, node and edge attributes were derived from RDKit-generated molecular graphs, and multiple GIN layers were used to obtain node-level structural representations. These were subsequently aggregated through ESA and projected into the same embedding space as the spectral features.

The key architectural distinction of CSU-MS2 lies in this ESA-based aggregation stage. Instead of relying solely on standard pooling, ESA introduced an external learnable mapping space that performed dimension-wise attention during aggregation. This mechanism was intended to preserve discriminative feature contributions and improved the alignment between spectral and structural modalities in the shared latent space.

Training Configuration

Training was performed using the adapted CSU-MS2 pipeline under a controlled configuration. The implemented setup used a batch size of 1024 with gradient accumulation over 4 steps, a maximum of 100 epochs, a learning rate of 1×10^{-4}, warm-up over the first 5 epochs, and weight decay of 1×10^{-4}. Gradient norm clipping was applied with a maximum norm of 1.0. Mixed-precision training was enabled, including FP16 execution, to improve computational efficiency.

The dataset was split into training and validation subsets in an approximate 80/20 ratio, while the test subset was excluded from training and reserved for evaluation. The model configuration used a molecular embedding size of 300, spectral and joint embedding sizes of 256, and a feature dimension of 512. The architecture included five graph layers and three spectral encoder layers, with a graph dropout of 0.3 and a general dropout of 0.1. Mean pooling was applied for feature aggregation. Training was performed using a contrastive objective based on cosine similarity, with a temperature parameter of 0.1 and an alignment weight coefficient of 0.75.

Pretrained Initialization

The authors of [11] provided several pretrained checkpoints, including $\mathrm{hcd\_model}$ and $\mathrm{qtof\_model}$, each trained under different collision-energy regimes. In this thesis, the $\mathrm{qtof\_model}$ corresponding to medium collision energy was selected as the initialization point. This choice was motivated by the characteristics of fragmentation patterns across energy levels and their relevance to the constructed dataset. Low-energy spectra typically contained fewer fragment ions and provided limited structural information, while high-energy spectra produce more complex fragmentation patterns that might introduce additional noise and variability. The medium-energy regime offered a balance between these extremes, yielding sufficiently informative yet stable spectral representations. Furthermore, QTOF-based spectra were widely represented in public tandem mass spectrometry resources, which made this checkpoint well-suited for transfer to a heterogeneous dataset aggregated from multiple sources. As a result, this initialization supported both stable optimization and effective adaptation during fine-tuning while preserving meaningful spectral–structural relationships learned during pretraining.

Training Execution and Logging

Training was executed through CSUMS2/train.py, which implemented the full functional flow, including dataloader construction, forward and backward passes, validation scheduling, checkpointing, and metric logging. The overall dynamics are shown in Fig. 16, where the loss decreases smoothly throughout training.

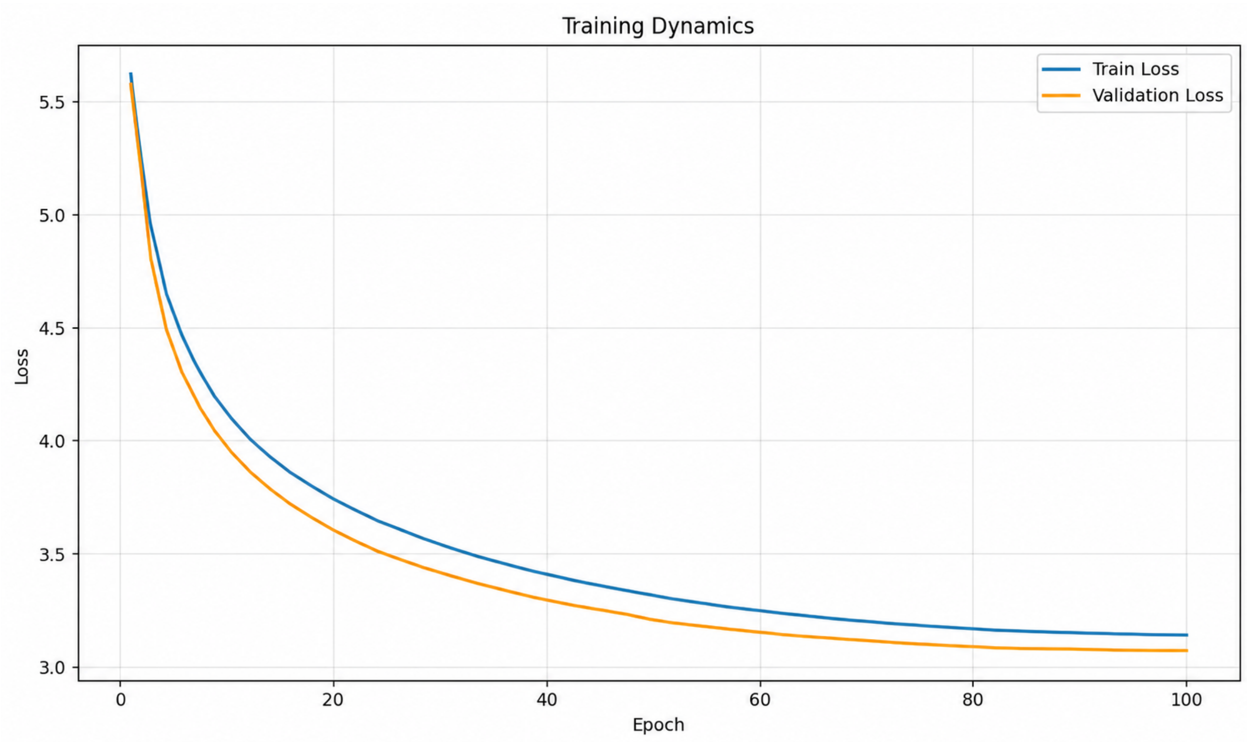


Fig. 16. Training dynamics of the CSU-MS2 model.

The Fig. 16 indicates stable convergence across the full 100-epoch run. At the beginning of training, the model exhibited a training loss of 5.62 at epoch 1 and

4.98 at epoch 2, reflecting rapid initial adaptation from the pretrained checkpoint. By the final epoch, the training loss decreased to 3.14, while the validation loss reached 3.07.

All logs, checkpoints, and intermediate evaluation artifacts were stored within the experiment workspace, ensuring reproducibility of the obtained results and enabling direct comparison with JESTR under the unified evaluation protocol.

5.2.3 Unified Evaluation Pipeline

To make sure that JESTR and CSU-MS2 were compared under identical conditions, a unified evaluation pipeline was developed and integrated into both project repositories through dedicated eval.py and ranking_eval.py scripts. All experiments were carried out on the test subset of the constructed dataset, which had been kept strictly separate from training and validation.

The evaluation procedure consisted of two complementary stages. The first one checked the quality of the learned joint embedding space. The second one evaluated retrieval performance in a candidate-ranking scenario based on molecules retrieved from the molecular search database.

Embedding space evaluation

The embedding-space analysis was set up to measure the quality of cross-modal alignment between spectra and their corresponding molecular structures within the shared latent space. The complete workflow is summarized in Pipeline 6 and consists of five stages: embedding extraction, similarity estimation, distributional analysis, inspection of local similarity structure, and assessment of distribution separation.

PIPELINE 6
Embedding-Space Evaluation

Stage E1: Embedding extraction
Load test dataset $\mathcal{D}_{test}$
Initialize spectral embeddings $\mathcal{Z}_s$ and molecular embeddings $\mathcal{Z}_m$
for each spectrum–molecule pair $(x_i, y_i) \in \mathcal{D}_{test}$ **do**
    Encode $x_i \to z_i^s$
    Encode $y_i \to z_i^m$
    Store $z_i^s \in \mathcal{Z}_s$, $z_i^m \in \mathcal{Z}_m$
end for
Stage E2: Similarity estimation
$\mathcal{S}_{self} \leftarrow \{\cos(z_i^s, z_i^m)\}_{i=1}^{N}$
Construct $\mathcal{S}_{rand}$ by pairing each z_i^s with a randomly sampled embedding from $\mathcal{Z}_m$
Stage E3: Distribution analysis
Compute statistics (mean, std, min, max, median) for $\mathcal{S}_{self}$ and $\mathcal{S}_{rand}$
Generate histogram visualizations for both distributions
Stage E4: Local structure inspection
Extract 10×10 cosine-similarity matrix block
Visualize the matrix as a heat map
Stage E5: Distribution separation assessment
Discretize $\mathcal{S}_{self}$ and $\mathcal{S}_{rand}$ into histograms
Apply smoothing to avoid zero-probability bins
Compute KL divergence between distributions

Retrieval evaluation

The retrieval evaluation assessed model performance under a realistic retrieval setting, where each query spectrum was used to rank a retrieved set of candidate molecular structures. The full procedure was summarized in Pipeline 7.

PIPELINE 7
Retrieval Evaluation

Stage R1: Query subset construction
Sample query set $\mathcal{Q} \subset \mathcal{D}_{test}$ with $\mid \mathcal{Q} \mid = 3000$
Sample with probabilities proportional to instrument property distribution
for each query spectrum $x \in \mathcal{Q}$ **do**
  Stage R2: Candidate generation
  Extract precursor m/z, adduct, and charge
  Initialize hypothesis set $\mathcal{H}$
  If adduct information is available, estimate neutral mass using adduct-specific mass shift, charge, and multiplicity
  Otherwise, generate neutral mass hypotheses using proton-based rules based on ion polarity
  Add precursor m/z as fallback hypothesis
  Round all hypotheses to fixed precision and deduplicate
  Initialize candidate set $\mathcal{C}$
  for each hypothesis $h \in \mathcal{H}$ **do**
    Retrieve candidates from database using tolerance windows:
      base tolerance $\tau = 100$ ppm
      expanded windows: 5τ and 20τ
    Add retrieved candidates to $\mathcal{C}$
  end for
  Deduplicate $\mathcal{C}$ while preserving order
  If $\mid \mathcal{C} \mid < 100$, sample additional molecules from database to ensure $\mid \mathcal{C} \mid \geq 100$
  Stage R3: Embedding-based ranking
  Encode query spectrum $x \rightarrow z_x$
  Encode all candidates $c \in \mathcal{C} \rightarrow z_c$
  Compute similarity scores $s_c = \cos(z_x, z_c)$
  Rank candidates in descending order of s_c
  Stage R4: Structure-aware evaluation
  for each ranked candidate $c \in \mathcal{C}$ **do**
    Compute Tanimoto similarity $T(c, g)$ with ground-truth molecule g
    If $T(c, g) \geq 0.8$, mark c as structurally correct
  end for
end for

This pipeline gives a unified and reproducible framework for evaluating both embedding quality and retrieval effectiveness.

5.2.4 Evaluation Results

This subsection summarizes the quantitative results obtained with the unified evaluation pipeline introduced in the previous section. It reports the embedding-space quality and retrieval performance of the two selected models, JESTR and CSU-MS2, evaluated under identical conditions.

Embedding Space Evaluation Results

The first part of the evaluation examined the quality of the learned shared embedding space. For each model, cosine similarity was computed between spectral and molecular embeddings for two scenarios: (i) ground-truth spectrum–molecule pairs (*self-similarity*) and (ii) randomly mismatched pairs (*other-similarity*). The results were summarized through descriptive statistics and the Kullback–Leibler (KL) divergence measure between the corresponding similarity distributions.

Table VIII provides a consolidated overview of the main statistical characteristics of the similarity distributions for both models.

TABLE VIII
Embedding-space evaluation results for JESTR and CSU-MS2.

Metric	JESTR	CSU-MS2
Self-similarity mean	0.5237	0.5055
Self-similarity std	0.1890	0.2130
Self-similarity min	-0.5352	-0.5070
Self-similarity max	0.9493	0.9494
Self-similarity median	0.5545	0.5351
Other-similarity mean	0.1339	0.0015
Other-similarity std	0.2284	0.1763
Other-similarity min	-0.7754	-0.5923
Other-similarity max	0.8951	0.8385
Other-similarity median	0.1354	-0.0114
KL divergence	1.6815	3.4280

To provide a more detailed view of the embedding behavior, the results for each model were examined individually.

JESTR

For the JESTR model, the self-similarity distribution exhibited a mean of 0.5237 and a median of 0.5545, indicating that matching spectrum–molecule pairs were generally mapped to relatively high similarity values. In contrast, the random-pair distribution was shifted toward lower values, with a mean of 0.1339 and a median of 0.1354. The KL divergence between the two distributions was 1.6815.

The distributional behavior is illustrated in Fig. 17, which shows the overlap between self-similarity and random-similarity values. In addition, the local structure of the embedding space is visualized in Fig. 18 through a 10×10 block of the cosine-similarity matrix.

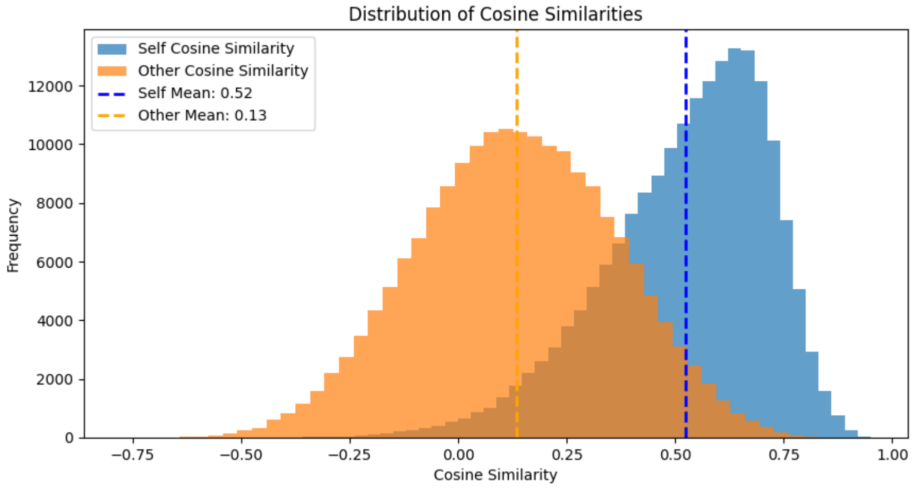


Fig. 17. Distribution of self-similarity and random-similarity values for the JESTR model.

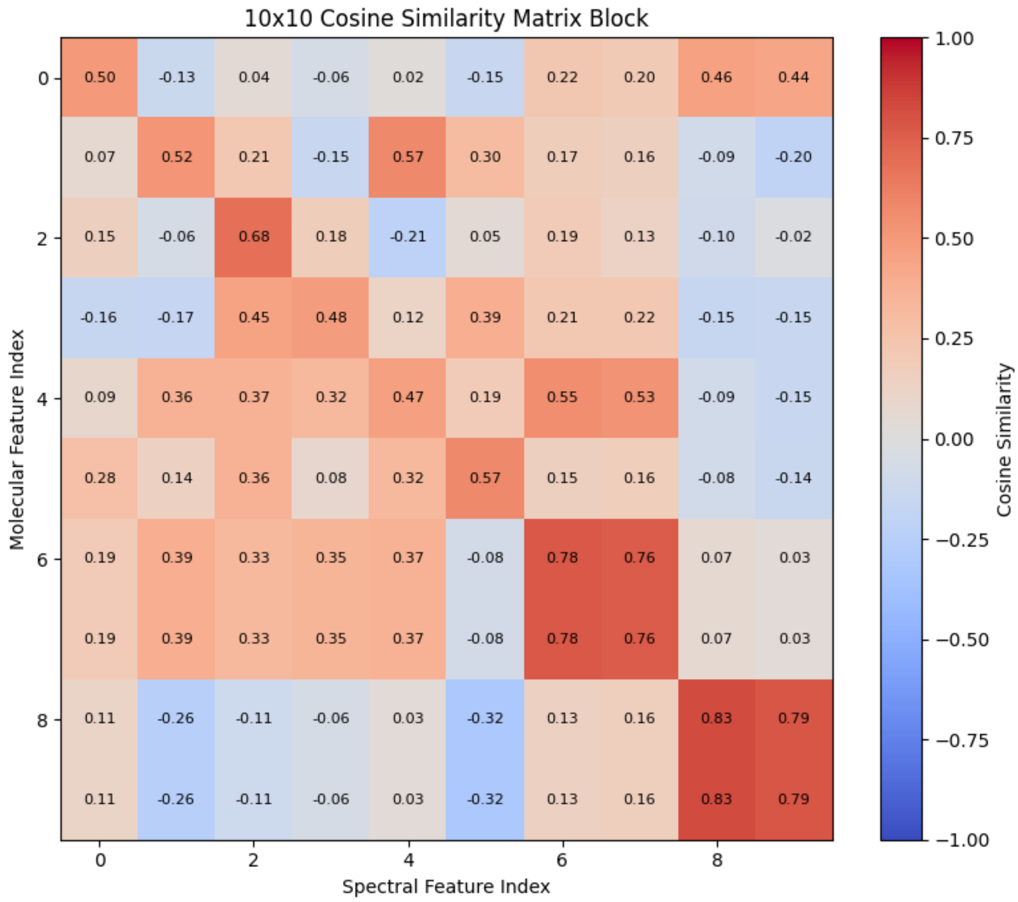


Fig. 18. Local 10×10 cosine-similarity matrix block for the JESTR model.

CSU-MS2.

For the CSU-MS2 model, the self-similarity distribution had a mean of 0.5055 and a median of 0.5351, which was comparable in magnitude to JESTR. However, the random-pair distribution weas centered substantially closer to zero, with a mean of 0.0015 and a median of -0.0114. The resulting KL divergence was 3.4280.

The corresponding similarity distributions are shown in Fig. 19, while the local similarity matrix structure appears in Fig. 20, giving a complementary view of the embedding-space organization.

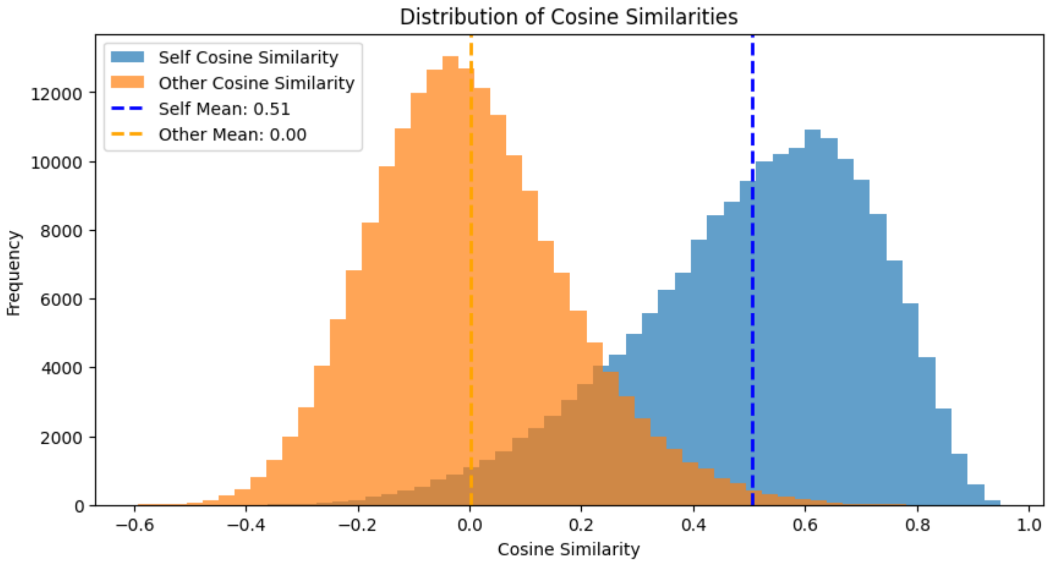


Fig. 19. Distribution of self-similarity and random-similarity values for the CSU-MS2 model.

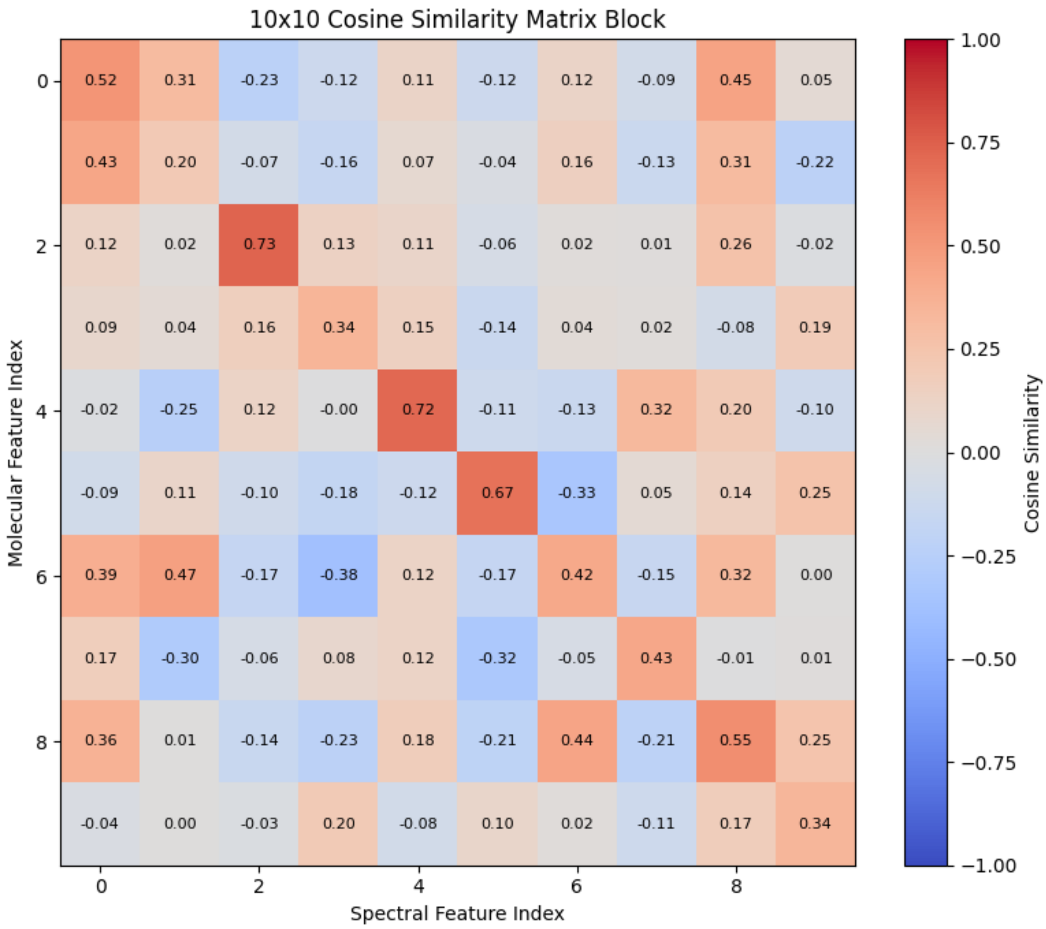


Fig. 20. Local 10×10 cosine-similarity matrix block for the CSU-MS2 model.

Both models were able to separate matching and mismatched spectrum–molecule pairs in the learned latent space, as observed in differences between self-similarity and random-similarity distributions.

Retrieval Performance Results

The second part of the evaluation assesses retrieval effectiveness in the candidate-ranking setting. Each query spectrum was matched against a candidate pool retrieved from the molecular search database, and the ranked list was evaluated using Hit Rate@K, Precision@K, Recall@K, and Tanimoto-based summary statistics for $K \in \{1, 5, 10, 100\}$.

Table IX provides a consolidated overview of the retrieval metrics for both models across different cutoff values. Complementary statistics based on structural similarity, measured using the Tanimoto coefficient, are reported in Table X.

TABLE IX
Retrieval performance of JESTR and CSU-MS2.

Metric	JESTR	CSU-MS2
Hit Rate@1	0.1137	0.2147
Precision@1	0.1137	0.2147
Recall@1	0.0988	0.1737
Hit Rate@5	0.3113	0.4633
Precision@5	0.0775	0.1204
Recall@5	0.2914	0.4172
Hit Rate@10	0.4303	0.5820
Precision@10	0.0594	0.0830
Recall@10	0.4097	0.5351
Hit Rate@100	0.8660	0.9053
Precision@100	0.0178	0.0179
Recall@100	0.8510	0.8719

TABLE X
Tanimoto similarity statistics of retrieved candidates for JESTR and CSU-MS2.

Metric	JESTR	CSU-MS2
Tanimoto@1 mean	0.2692	0.3749
Tanimoto@1 std	0.2865	0.3550
Tanimoto@1 min	0.0000	0.0000
Tanimoto@1 max	1.0000	1.0000
Tanimoto@5 mean	0.2415	0.2917
Tanimoto@5 std	0.2501	0.2971
Tanimoto@5 min	0.0000	0.0000
Tanimoto@5 max	1.0000	1.0000
Tanimoto@10 mean	0.2238	0.2513
Tanimoto@10 std	0.2277	0.2627
Tanimoto@10 min	0.0000	0.0000
Tanimoto@10 max	1.0000	1.0000
Tanimoto@100 mean	0.1572	0.1547
Tanimoto@100 std	0.1529	0.1554
Tanimoto@100 min	0.0000	0.0000
Tanimoto@100 max	1.0000	1.0000

To provide a more detailed view of model behavior, the retrieval results are described separately for each model based on the reported metrics.

JESTR

For the JESTR model, retrieval evaluation achieved a Hit Rate@1 of 0.1137, which rised to 0.3113 at $K = 5$, 0.4303 at $K = 10$, and 0.8660 at $K = 100$. The corresponding Precision@K values falled from 0.1137 at $K = 1$ to 0.0178 at $K = 100$, in line with the expected dilution of relevant candidates as the cutoff grows. Recall@K shows a consistent upward trend and reaches 0.8510 at $K = 100$. The mean Tanimoto similarity of the top-ranked candidate was 0.2692 and gradually decreased with the cutoff, dropping to 0.1572 when averaged over the top-100 candidates.

CSU-MS2

For the CSU-MS2 model, retrieval performance was consistently higher across all cutoff levels. The model reached a Hit Rate@1 of 0.2147, which growed to 0.4633 at $K = 5$, 0.5820 at $K = 10$, and 0.9053 at $K = 100$. Precision@K decreased from 0.2147 at $K = 1$ to 0.0179 at $K = 100$, following the same trend as for JESTR while staying higher at lower cutoffs. Recall@K rised steadily and reached 0.8719 at $K = 100$. The mean Tanimoto similarity of the top-ranked candidate was 0.3749, while the average similarity over the top-100 candidates was 0.1547.

To inspect ranking trends across cutoff levels, Fig. 21 shows the evolution of Hit Rate@K, Precision@K, and Recall@K for both models.

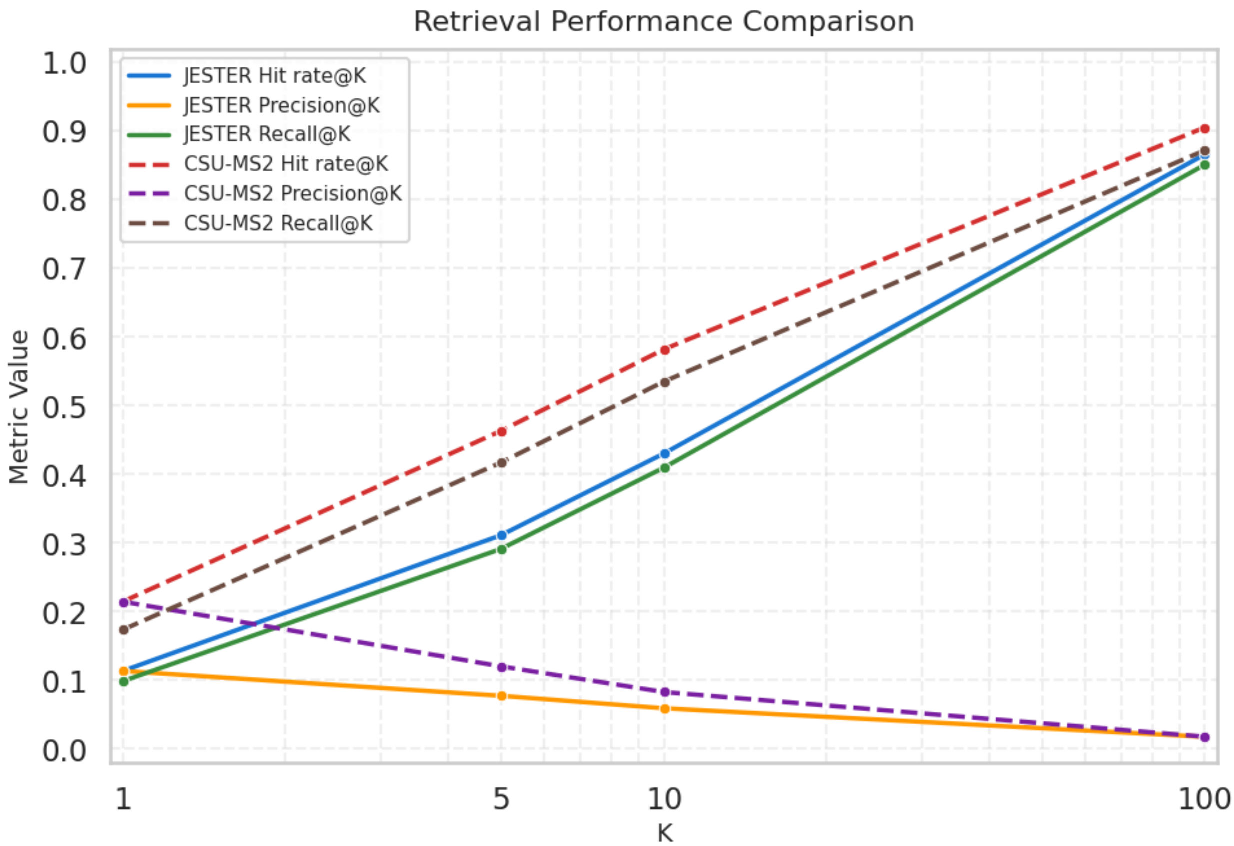


Fig. 21. Comparative retrieval performance of JESTR and CSU-MS2.

These results provided a consistent quantitative basis for evaluating retrieval effectiveness under a unified experimental setting.

5.3 System Implementation and Evaluation

This section describes how the spectrum–molecule retrieval system was put into practice and how it was evaluated as an integrated software solution. The system was developed as an independent project, with the full source code and deployment configuration available in the public repository[6].

5.3.1 Implementation of System Components

At the implementation level, the architecture introduced in Section 4.2.1 was realized as a multi-container application. Each subsystem became a separate container with its own runtime, startup procedure, and configuration file, so the boundaries between components were physical rather than purely logical. This arrangement was chosen because it allowed the retrieval pipeline to be developed, restarted, and scaled per component.

Containers were wired together through Docker Compose. The top-level csmp-search-engine compose file listed the services together with their dependency order and the way they communicated. The same file holded the remaining runtime details: exposed ports, environment variables, mounted volumes, and the internal Docker network shared by the containers. Concentrating all of this in a single configuration removed most of the differences between running the system on a developer laptop and running it on a server, since both use exactly the same compose definition.

[6] https://github.com/IVproger/csmp_search_engine.git

The components themselves were presented separately in the subsections below, each from the standpoint of how it was implemented, what it did inside the pipeline, and how it talked to its neighbors.

Client Interface

The client interface was built with the Streamlit framework, which makes it possible to develop quickly an interactive web application for spectrum analysis and visualization of retrieval results. When deployed locally, the interface was reachable in a web browser on port 8501 and acted as the primary entry point for user interaction with the system.

The web page began with a file upload interface that supported MS/MS spectral data in several formats, including .mzML, .mgf, .json, and .msp. At this stage, input validated to ensure format compatibility, and unsupported files triggered an explicit error message, enforcing correct input constraints.

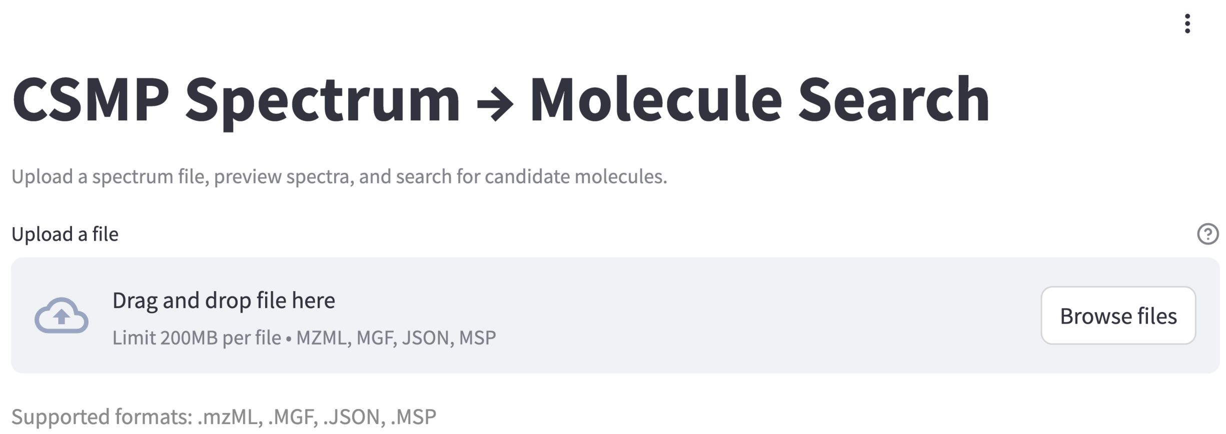


Fig. 22. File upload interface supporting multiple MS/MS spectral formats.

Following a successful upload, an optional preview mode allowed inspection of the input data. Each spectrum was visualized as a peak-intensity diagram, enabling validation and identification of potential inconsistencies prior to the processing.

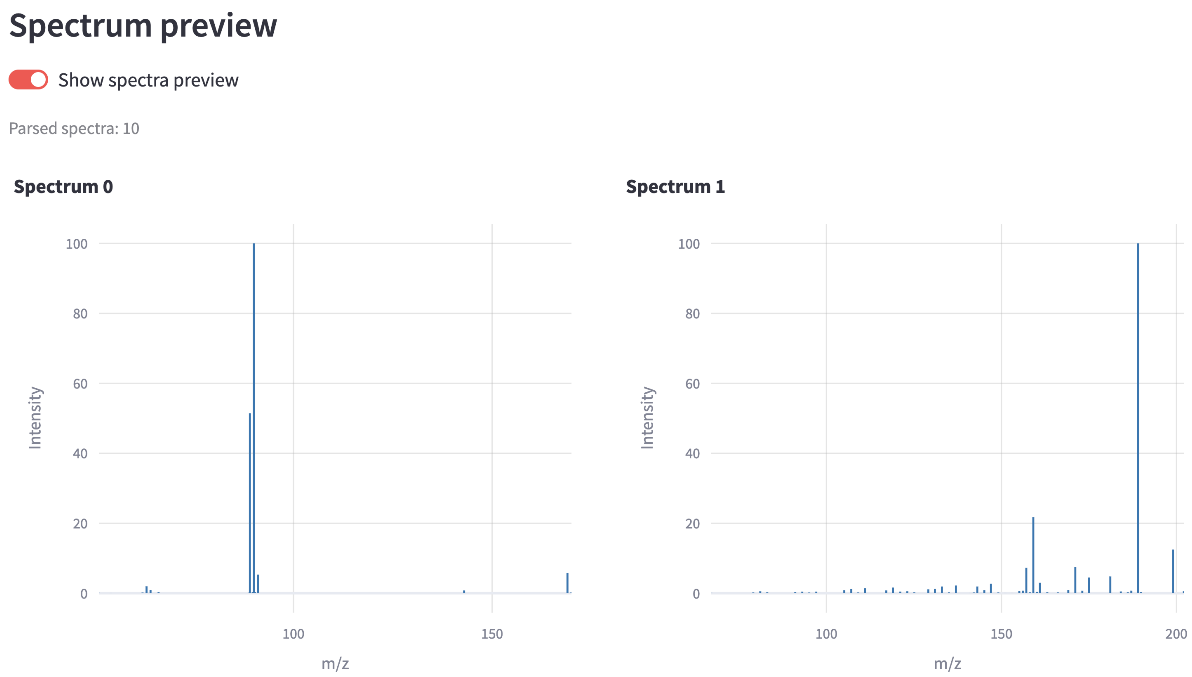


Fig. 23. Visualization of uploaded spectra in preview mode.

When the input data were verified, the retrieval process was initiated via a dedicated start annotation button. The system processed spectra sequentially and returned a ranked list of candidate molecular structures for each query.

The results were presented in two forms. The first was a tabular summary view that listed top-ranked candidates along with key attributes, including molecular mass, similarity score, and SMILES representation. This format supported efficient overview across multiple candidates.

Spectrum 0

Precursor m/z: 232.9266 | **Candidates:** 10

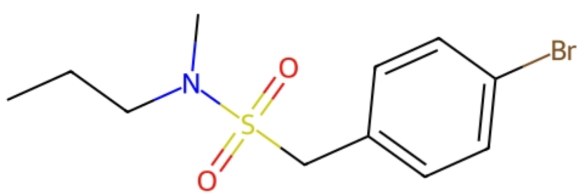
Spectrum parsed, encoded, and searched successfully.

#	Mass	Score	SMILES
1	305.0085	66.6734	CCCN(C)S(=O)(=O)Cc1ccc(Br)cc1
2	305.0051	61.008	O=C(Nc1ccc(Br)cc1)OCc1ccccc1
3	305.0085	59.5333	CCN(CC)S(=O)(=O)c1ccc(C)c(Br)c1
4	305.0085	59.3319	CCCCCNS(=O)(=O)c1cccc(Br)c1
5	305.0085	57.6188	CCN(CC)S(=O)(=O)c1cc(C)cc(Br)c1
6	305.0085	57.5677	CCCCNS(=O)(=O)c1ccc(C)c(Br)c1
7	305.0085	55.8635	CCCCNS(=O)(=O)c1cc(C)cc(Br)c1
8	305.0085	52.6058	Cc1cc(Br)cc(S(=O)(=O)NC(C)(C)C)c1
9	305.0164	50.9697	CC(=O)N(Cc1ccccc1)c1ncc(Br)cn1
10	305.0085	48.7076	CCCCCN[S@SP3](=O)(=O)c1ccccc1Br

Fig. 24. Tabular representation of retrieved molecular candidates.

The second presentation form gave a more detailed view of individual candidates. Each candidate appeared as a separate card containing a 2D molecular structure visualization and the associated metadata, which made qualitative inspection of retrieval results possible.

Candidate 1



Monoisotopic mass
305.008511852

Similarity score
66.6734

SMILES:

```
CCCN(C)S(=O)(=O)Cc1ccc(Br)cc1
```

Fig. 25. Detailed visualization of candidate molecules.

To accommodate different usage scenarios, the interface could hide the detailed visualization in favor of a tabular-only mode, which it more efficient for large-scale analysis.

The client interface offered a streamlined interaction layer that balanced usability with analytical functionality, supporting both rapid inspection and detailed examination of spectrum–molecule matching results.

API Layer with Spectrum Processing Module

The API layer was built on the FastAPI framework and acted as the central orchestration component of the system. After local deployment, the service listened on port 8000 and exposed HTTP endpoints for receiving input data, coordinating the processing steps, and returning structured retrieval results. In addition to request handling, this layer integrated the spectrum processing module, which is responsible for parsing and standardizing the incoming spectral data.

The API exposed the following two primary endpoints:

- GET /health — returns the service status and is used for monitoring and container readiness verification;
- POST /annotate-spectrum — accepts an uploaded spectral file, initiates the annotation process, and returns structured results defined using Pydantic schemas to ensure type safety and consistency.

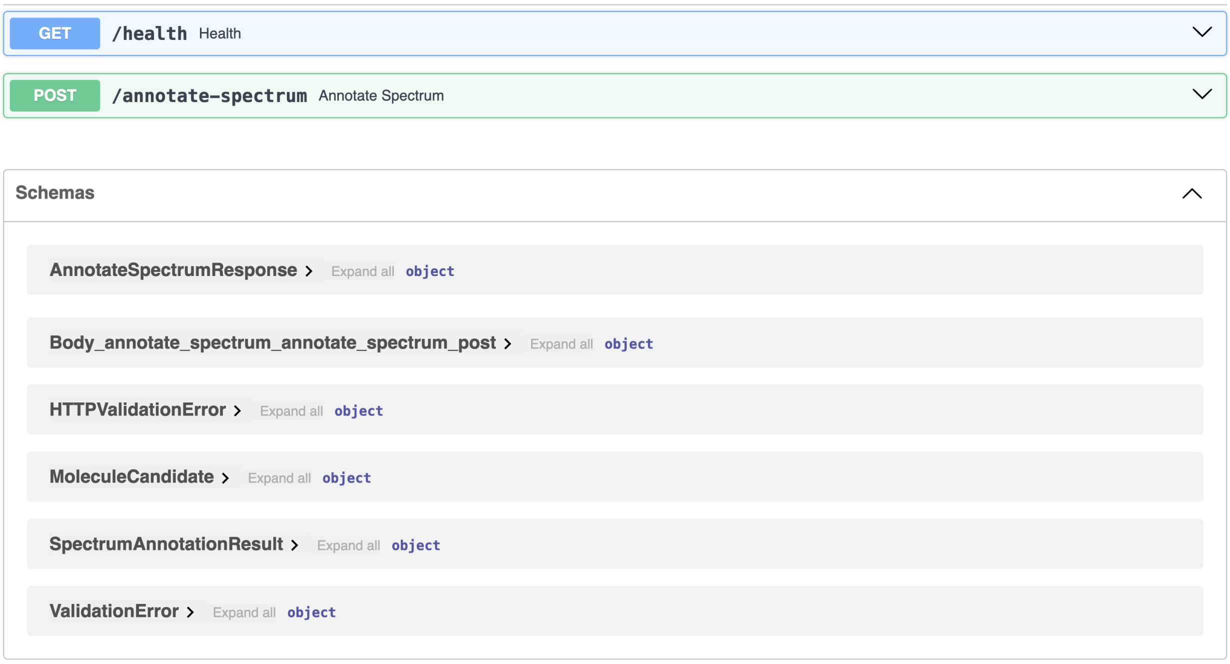


Fig. 26. FastAPI interface with available endpoints and response schemas.

The schema layer was built around five entities. The smallest of them was Peak, which simply holded an $(m/z, \text{intensity})$ pair for one spectral peak. A complete spectrum was expressed by ParsedSpectrum, which keeped the spectrum identifier, the precursor data, the list of peaks, any optional chemical attributes, and a status field describing the outcome of parsing.

On the molecular side, every entry returned by the retrieval engine was described by MoleculeCandidate. This object carried the SMILES string of the molecule together with its mass and the similarity score assigned during ranking. For one input spectrum the system collected such candidates inside SpectrumAnnotationResult, which added the spectrum identifier so that results could be matched back to the input.

The outermost wrapper was AnnotateSpectrumResponse. Its role was to act as the HTTP-level response payload: it reported the overall processing status, the metadata of the uploaded file, and the per-spectrum annotation results assembled inside it.

The spectrum processing module was integrated within the API layer and operated on raw input file bytes. It routed the input to format-specific parsers implemented using external libraries such as matchms and pymzML. The module validated and extracted spectral peaks and metadata, transforming heterogeneous inputs into a unified ParsedSpectrum representation.

The API layer coordinated the execution of the retrieval pipeline. Upon receiving a request, it validated the input, invoked the parsing module, and managed subsequent interactions with downstream services. In particular, it dispatched batched spectrum data to the inference service for embedding generation and initiated retrieval queries against the molecular database.

Finally, the API aggregated the retrieved candidates, formats the results according to the defined schemas, and returned a structured response to the client interface.

Inference Service

The inference service was implemented using the NVIDIA Triton Inference Server container, which provided a scalable and high-performance environment for serving machine learning models. After deployment, the service exposed both HTTP and gRPC protocols and was responsible for generating fixed-dimensional embedding vectors for input spectra.

The deployed model configuration is summarized below:

- model_name = spectrum_encoder
- backend = onnxruntime
- model_file = csu-ms-2-spec-encoder.onnx
- max_batch_size = 64

- $\text{instance\_group} = \{\text{KIND\_CPU}, \text{count} = 1\}$

The Triton enabled flexible deployment across heterogeneous hardware environments. In particular, the execution mode (CPU or GPU) could be configured at the service level, allowing the system to dynamically adapt to available hardware resources without changes to the application logic. This capability ensured portability and simplified deployment in different hardware infrastructures.

To enable compatibility with Triton, the spectral encoder was converted to the ONNX format and executed via the onnxruntime backend. This representation allowed efficient inference and supported hardware-level optimizations.

During execution, the inference service received batched spectral data from the API layer via gRPC calls. Each batch was processed by the encoder to produce embedding vectors in a shared latent space, which were used for similarity-based retrieval.

By isolating model inference as a dedicated service, the system gained better scalability, more efficient use of resources, and the ability to update or replace the model independently from the rest of the workflow.

Retrieval Engine and Molecular Search Database

The retrieval engine was responsible for identifying and ranking candidate molecular structures for a given query spectrum. At runtime, the API layer forwarded parsed spectra and embedding vectors to the db_search_client module, which carried out the retrieval procedure. The process combined mass-based filtering with embedding-based ranking and was summarized in Pipeline 8.

PIPELINE 8
Retrieval Engine

Input: query spectrum x, molecular database $\mathcal{M}$
Extract precursor m/z, adduct, and charge from x
Stage 1: Candidate generation via mass constraints
Estimate neutral mass m_n using precursor m/z, adduct, and charge
Define tolerance window τ in ppm around m_n
Retrieve candidate set $\mathcal{C}$ such that $\mid m_{\text{mono}} - m_n \mid \leq \tau$
Stage 2: Mass-constrained vector retrieval
If $\mathcal{C} \neq \emptyset$, encode query spectrum $x \to z_x$
If $\mathcal{C} \neq \emptyset$, process each candidate molecule $c \in \mathcal{C}$:
  retrieve embedding z_c
  compute similarity score $s_c = \cos(z_x, z_c)$
If $\mathcal{C} \neq \emptyset$, rank candidates in descending order of s_c
Stage 3: Fallback retrieval
If $\mathcal{C} = \emptyset$, retrieve all molecules from $\mathcal{M}$
If $\mathcal{C} = \emptyset$, encode $x \to z_x$
If $\mathcal{C} = \emptyset$, process each molecule $c \in \mathcal{M}$:
  retrieve embedding z_c
  compute $s_c = \cos(z_x, z_c)$
If $\mathcal{C} = \emptyset$, rank all candidates by s_c
Stage 4: Result construction
for each ranked candidate c **do**
  Convert cosine distance to normalized similarity score
  Construct $\mathrm{MoleculeCandidate}$(SMILES, mass, similarity)
end for
Output: ranked candidate list

5.3.2 System Evaluation Implementation and Results

Experimental Setup and Evaluation Procedure

The system-level evaluation assessed the efficiency, scalability, and robustness of the deployed retrieval pipeline under realistic operating conditions. The evaluation protocol followed the methodology defined in Section 4.2.2, which treated the system as a file-based inference service that processed multiple MS/MS spectra per request.

To keep all experiments comparable, a unified request timeout was used. Each request had to complete within a maximum of 300 seconds. The same constraint applied both to single-file and to parallel processing scenarios. In the case of concurrent execution, all requests within a batch had to satisfy this condition for the run to be considered successful. This convention defined a consistent notion of system stability and made evaluation across different settings comparable.

The experiments were run on a local machine with a 10-core CPU and 16 GB of RAM. Within this setup, the evaluation inputs were drawn from the test subset of the constructed dataset. The spectra were saved into input files and arranged into scenario-specific groups designed to reflect different workload conditions.

Scenario 1: Per-spectrum processing time assessment

This scenario relied on a single input file containing 1000 spectra. The total processing time of the file was measured and then normalized to obtain the average processing time per spectrum. To ensure that the result is stable, the experiment was repeated several times under identical conditions, and the reported value corresponds to the mean across runs.

Scenario 2: Parallel file processing assessment

This scenario examined concurrent processing on a heterogeneous set of input files. Two groups of files were prepared:

- 100 small files, each containing between 5 and 10 spectra;
- 10 large files, each containing 500 spectra.

These files were combined into mixed batches in order to simulate realistic workloads with varying input sizes. For each concurrency level F, a batch of F distinct files was processed simultaneously. The system was considered stable if all requests completed successfully. The maximum sustainable concurrency F_{parallel}

was determined by increasing F from 1 to 12.

Scenario 3: Maximum supported file size assessment

To evaluate system capacity with respect to input size, a series of files with increasing numbers of spectra was generated. The evaluated file sizes included 100, 1000, 2000, 3000, 4000, 5000, 7000, and 10000 spectrum records. Each file was processed sequentially, and the maximum supported size $N_{\max}$ was defined as the largest file processed without failure.

Scenario 4: Robustness to invalid input assessment

This scenario uses corrupted input files obtained by introducing controlled errors into valid spectra records. The following three types of corrupted inputs were constructed:

- file with 50 spectra having invalid precursor information;
- file with malformed XML structure;
- truncated file with incomplete obligatory data.

These inputs were used to verify that the system handles incorrect data safely, without crashes, and returns consistent error responses. In addition, record-level failures were tracked to compute the overall error rate E.

Across all scenarios, experiments were conducted through automated request execution on the API-level. For each test case, the following metrics were recorded: request success status, processing time, number of processed records, and the presence of degraded or failed outputs.

The quantitative results obtained under this experimental setup are presented in the following section.

System Evaluation Results

Scenario 1: Per-spectrum processing time assessment

In this scenario, all executions completed successfully, with consistent HTTP status 202, payload status $\mathrm{accepted}$, and full preservation of ranking results for all 1000 spectrum records in each run. The measured processing times are summarized in Table XI, which reports both the total processing time per file and the corresponding normalized processing time per spectrum.

TABLE XI
Per-spectrum processing time results.

Run	T_{file} (s)	T_{avg} (s/record)
1	81.2019	0.081202
2	92.3456	0.092346
3	88.5678	0.088568
4	85.4321	0.085432
5	102.1234	0.102123
Mean	—	0.089934

The results demonstrated stable processing behavior across repeated runs, with the average per-spectrum processing time equal to $T_{\text{avg}} = 0.089934$ s/record.

Scenario 2: Parallel file processing assessment

The results of concurrent execution across all tested concurrency levels are summarized in Table XII. For each level F, the table reports batch composition, total processing time, average per-file latency, and execution status. Large files corresponded to inputs containing 500 spectra, whereas small files contained 5–10 spectra.

TABLE XII
Parallel processing results across concurrency levels.

F	Large	Small	Batch time (s)	Avg. latency (s)	All OK
1	1	0	34.750	34.706	Yes
2	1	1	36.648	19.218	Yes
3	1	2	45.179	16.822	Yes
4	2	2	101.953	51.831	Yes
5	2	3	92.031	29.227	Yes
6	3	3	162.377	53.594	Yes
7	3	4	178.938	85.008	Yes
8	4	4	202.062	66.347	Yes
9	4	5	204.355	59.402	Yes
10	5	5	256.920	79.273	Yes
11	5	6	281.186	84.854	Yes
12	6	6	28.222	17.916	No

As shown in Table XII, all batches up to $F = 11$ completed successfully. The corresponding maximum concurrency level was therefore $F_{\text{parallel}} = 11$.

At $F = 12$, the system no longer satisfied the success criterion. Although all requests were accepted at the HTTP level, logs indicate that the Triton inference service became unresponsive under increased load, preventing processing of incoming batches and resulting in degraded outputs for most requests.

Scenario 3: Maximum supported file size assessment

The results for increasing input file sizes are presented in Table XIII, which reports the execution status, processing time, and number of successfully returned records.

TABLE XIII
Maximum supported file size results.

File	Records	Status	Time (s)	Returned
capacity_100.mzML	100	OK	9.874	100
capacity_1000.mzML	1000	OK	74.601	1000
capacity_2000.mzML	2000	OK	171.592	2000
capacity_3000.mzML	3000	OK	292.493	3000
capacity_4000.mzML	4000	FAIL	300.385	0

The evaluation was performed sequentially, processing input files of increasing size. Execution was terminated at the first failure, and larger inputs were not evaluated.

As shown in Table XIII, the file containing 4000 records exceeded the predefined request timeout and did not satisfy the success criterion. The processing time of 300.385 s vs. 300 s indicates that the failure was caused by the timeout constraint rather than an intrinsic system limitation. The practical upper bound lied between 3000 and 4000 records.

Scenario 4: Robustness to invalid input assessment

The behavior of the system under corrupted input conditions is summarized in Table XIV. The table reports the HTTP response, execution status, processing time, and observed effect for each test case.

TABLE XIV
Robustness evaluation results.

File	HTTP	Status	Time (s)	Effect
corrupted_invalid_precursor.mzML	202	PASS	0.070	50 degraded records
corrupted_malformed_tags.mzML	400	PASS	0.013	parsing error
corrupted_truncated.mzML	400	PASS	0.008	parsing error

Across all test cases, the system operated without crashes or uncontrolled behavior. At the request level, malformed and truncated inputs produce controlled

HTTP 400 responses, while the file with invalid precursor metadata was accepted with HTTP 202 response. Consequently, no unexpected failures were observed, and the request-level failure rate is

$$F = 0, \tag{33}$$

as all responses corresponded to controlled and correctly handled outcomes.

At the record level, the file with invalid precursor information produced 50 degraded records out of 50 processed spectra, resulting in

$$D = \frac{50}{50} = 1.0, \tag{34}$$

indicating complete degradation due to missing required metadata.

Overall, the system satisfied the robustness criteria by correctly distinguishing between recoverable and non-recoverable input errors and maintaining safe execution with consistent error reporting.

The presented results provide a structured quantitative characterization of the system with respect to processing efficiency, concurrency limits, input size capacity, and robustness to invalid data. A detailed interpretation of these findings and their implications for system design and practical deployment is provided in Chapter 6.

Chapter 6

Analysis and Discussion

This chapter analyzes the scientific validity, practical applicability, and broader implications of the developed spectrum–molecule retrieval framework. The discussion focuses on how the constructed data foundation, evaluated models, and implemented retrieval system collectively contribute to the original objective of establishing a scalable, reproducible, and deployable approach for tandem mass spectrum–molecular structure matching. Particular attention is given to methodological strengths, observed limitations, alignment with existing literature, and the extent to which the proposed approach contributes to both academic novelities and practical application.

6.1 Analysis of Data Curation Results

The data curation stage establishes the foundational conditions under which all subsequent model benchmarking and retrieval system evaluation become scientifically meaningful. Within the context of this thesis, the primary objective of data preparation is not merely large-scale aggregation, but the construction of a standardized, chemically credible, and engineering-reliable spectrum–molecule resource capable of supporting reproducible retrieval experiments. The resulting dataset and molecular search database indicate that this objective is largely achieved, while also revealing several practical and scientific limitations.

6.1.1 Analysis of the Curated MS/MS – Molecule Dataset

From a scientific perspective, the curated MS/MS dataset demonstrates substantial structural and spectral breadth while preserving methodological consistency. The final training split contains 1,111,380 validated spectrum–molecule pairs across 64,234 unique molecular structures, while the test split contains 271,045 pairs across 13,523 unique structures, with explicit molecule-level separation enforced during splitting. This separation matters because it directly limits structural leakage between training and evaluation, and as a result the downstream retrieval metrics describe genuine generalization rather than memorization. In practical terms, the design comes much closer to open-world molecular retrieval, where previously unseen structures must be matched from spectral evidence. At the same time, comparable train/test spectral statistics, including peak counts, TIC distributions, and precursor mass ranges, suggest that evaluation conditions remain controlled and methodologically coherent rather than artificially divergent. Maintaining comparable spectral statistics between the training and test partitions preserves benchmark fairness while maintaining structural novelty.

The diversity of adduct types, precursor masses, functional groups, and instrument distributions further indicates that the dataset captures a broad, though not chemically uniform, representation of tandem MS/MS conditions. The training set includes 83 adduct types, while the test set includes 45, with both primarily dominated by common ionization states such as [M+H]+ and [M+Na]+. This suggests that the dataset is sufficiently heterogeneous to expose models to multiple ionization behaviors while still reflecting realistic concentrations around dominant laboratory acquisition modes. Functional group distributions similarly demonstrate substantial chemical breadth, although they remain concentrated around common metabolomics-relevant classes rather than chemically balanced universality. Consequently, the dataset should be interpreted not as an exhaustive representation of molecular space, but as a practically broad and operationally relevant resource that reflects real-world chemical prevalence more closely than artificially balanced benchmarks.

From an engineering perspective, one of the most important outcomes is the successful transformation of heterogeneous public sources into a unified schema with normalized metadata, standardized molecular annotations, deduplicated spectrum – molecule pairs, and reproducible train/test splitting logic. This infrastructure-level contribution is particularly significant because prior spectrum–molecule research frequently prioritizes model architecture while relying on fragmented, source-specific, or inconsistently processed datasets. In contrast, this thesis places reproducibility, schema consistency, and retrieval-system readiness at the center of design. Hash-based duplicate removal and molecule-level splitting introduce methodological discipline that reduces raw data volume, but this trade-off is justified because downstream benchmarking validity depends more heavily on structural integrity and fair evaluation than on maximal dataset size alone.

6.1.2 Analysis of the Constructed Molecular Search Database

The constructed molecular database further strengthens this foundation by extending beyond model-development requirements toward deployment-oriented retrieval infrastructure. With 872,935 molecular entries, 147,332 unique molecular formulas, B-tree indexing for exact structural filtering, and HNSW indexing for scalable embedding search, the database functions not simply as a static repository but as an operational retrieval space. The broad monoisotopic mass range and substantial formula diversity suggest that candidate generation operates across chemically diverse search spaces, while the integration of exact indexing and approximate nearest-neighbor retrieval reflects an explicitly engineering-driven design philosophy. This distinction is important because many prior studies evaluate retrieval algorithms without constructing scalable retrieval-ready search spaces. In the proposed framework, the molecular database architecture itself constitutes part of both the scientific and practical contributions.

6.1.3 Relation to Existing Benchmarks and Datasets

A major distinguishing advantage of this work emerges when considered relative to the current benchmarks landscape. Existing public resources such as NPLIB1/CANOPUS and MassSpecGym represent important milestones for standardized MS/MS machine learning, but they primarily function as individual benchmark environments rather than unified retrieval infrastructures. NPLIB1 provides approximately 19,687 spectra for 8,553 molecules, while MassSpecGym substantially expands the benchmark scale to roughly 231,000 spectra across approximately 29,000 molecules. However, these resources remain comparatively specialized in scope and are generally optimized for benchmark standardization rather than comprehensive deployment-oriented integration. By systematically aggregating and harmonizing multiple large public sources, including GNPS-derived datasets, CANOPUS/NPLIB1-associated resources, and MassSpecGym-scale collections, this thesis constructs an integrated corpus of approximately 1.38 million validated spectrum–molecule pairs alongside a retrieval database containing

872,935 indexed molecular candidates. The primary significance of this integration is therefore not solely quantitative scale, but structural unification: previously fragmented benchmark solutions are transformed into a single standardized, retrieval-compatible framework.

This benchmark-unifying strategy offers several practical advantages:

1. **Reduced benchmark dependence:** Combining multiple benchmark paradigms decreases overreliance on the assumptions, biases, or domain constraints of any single source.
2. **Improved deployment realism:** Multi-source integration matches real laboratory conditions more closely, since retrieval systems must operate across heterogeneous instruments, adduct types, and chemical domains rather than within isolated benchmark settings.
3. **Unified benchmarking and deployment support:** The resulting approach simultaneously supports fair model benchmarking and deployable retrieval-oriented data design, which current public datasets rarely achieve within a single framework.

In this sense, the contribution is not merely "larger than existing benchmarks." It is functionally broader because it establishes a comprehensive and diversified retrieval foundation.

6.1.4 Limitations and Conclusions

Despite these strengths, several important limitations remain and should be explicitly defined in the interpretive boundaries of the proposed data curation strategy.

- **Structural simplification:** Standardization through SMILES and InChIKey harmonization may reduce stereochemical specificity or simplify representational nuances that can be chemically meaningful in certain retrieval contexts.

- **Source heterogeneity:** Aggregation across multiple public datasets introduces unavoidable variability in annotation quality, acquisition conditions, and domain composition, which may affect consistency across chemical subspaces.
- **Preprocessing trade-offs:** Data preparation places standardization, interoperability, and reproducibility above maximal chemical nuance, which implies that some rare, ambiguous, or technically complex records may be left out.

Taken together, these constraints make it clear that the data curation pipeline should not be treated as producing a chemically complete representation of molecular space. Its main contribution lies in giving a practically robust, methodologically defensible, and engineering-ready retrieval foundation.

Overall, the principal significance of this stage lies not in claiming absolute chemical comprehensiveness, but in demonstrating that the following can be systematically engineered within a single framework:

1. **Large-scale integration** of heterogeneous spectrum–molecule resources,
2. **Reproducible and fair benchmarking** through disciplined curation and splitting strategies,
3. **Benchmark unification** across previously fragmented public ecosystems,
4. **Deployment-aware data architecture** that supports practical retrieval system design.

Whereas many prior studies focus primarily on model-centric innovation within narrower benchmark settings, this thesis demonstrates that retrieval quality is fundamentally bounded not only by model architecture but also by data architecture, candidate space design, and infrastructure validity. Consequently, the data curation stage does not merely assemble a large dataset. It establishes the trustworthy experimental and operational substrate required for scientifically credible benchmarking and practical spectrum–molecule retrieval.

6.2 Analysis of Model Evaluation Results

The model evaluation stage was designed not to develop a novel spectrum–molecule architecture, but to determine how effectively current state-of-the-art dual-encoder solutions can be reproduced, fairly benchmarked, and adapted for integration into the proposed retrieval framework. Within this context, the principal objective was practical and methodological rather than architectural innovation. The comparative analysis focuses on how existing models perform under unified data conditions, standardized retraining, and retrieval-oriented evaluation, rather than relying exclusively on originally reported performance metrics from prior publications.

A critical practical finding during model evaluation of this stage is that publicly released pretrained checkpoints did not provide sufficiently reliable or transferable performance for direct deployment within the constructed large-scale heterogeneous retrieval environment. As a result, additional model training under the unified protocol described in the methodology became necessary to obtain predictable and scientifically interpretable behavior. This outcome is methodologically important because it demonstrates that published pretrained weights may not automatically generalize across broader aggregated datasets, particularly when candidate spaces, source heterogeneity, and retrieval constraints differ from original benchmark assumptions. Consequently, reproducible adaptation proved more important than nominal model availability alone.

6.2.1 Embedding Space Quality and Representation Reliability

The first major criterion for model success was the quality of the learned shared latent space, since retrieval effectiveness fundamentally depends on whether matching spectrum–molecule pairs are separated from mismatched pairs in a discriminative and operationally useful manner. Both JESTR and CSU-MS2 demonstrate meaningful positive-versus-random separation, confirming that each architecture successfully learns chemically relevant cross-modal structure under retraining conditions.

However, unified evaluation reveals important differences in representation behavior:

- **JESTR:** Higher raw self-similarity mean (0.5237), but substantially higher random-pair similarity mean (0.1339), indicating weaker negative-pair suppression.
- **CSU-MS2:** Comparable self-similarity mean (0.5055), but near-zero random-pair similarity mean (0.0015), indicating stronger embedding discrimination.
- **KL divergence:** CSU-MS2 achieves substantially stronger distributional separation (3.4280 vs. 1.6815), suggesting a more retrieval-suitable embedding space.

The findings indicate that good retrieval representation is not determined only by maximizing similarity on correct pairs. Practical retrieval quality also depends on suppressing similarity for incorrect candidates. From an engineering standpoint, CSU-MS2's stronger negative-space separation appears more useful than a marginally higher positive-pair similarity, because retrieval systems operate within large candidate pools where suppressing false positives is critical.

6.2.2 Retrieval Performance Under Unified Candidate Ranking

The second criterion, more important from an operational point of view, was retrieval effectiveness under realistic candidate-ranking conditions. Across all major retrieval metrics, CSU-MS2 consistently outperformed JESTR.

Several comparative observations stand out:

1. **Top-ranked precision:** CSU-MS2 reaches almost double the Hit Rate@1 of JESTR (0.2147 vs. 0.1137), pointing to substantially stronger first-choice retrieval reliability.
2. **Early ranking superiority:** CSU-MS2 keeps a clear advantage at Hit Rate@5 and Hit Rate@10, which suggests more effective prioritization of relevant candidates in practical shortlist scenarios.
3. **Candidate pool saturation:** At Hit Rate@100, performance differences narrow (0.9053 vs. 0.8660), indicating that both models often retrieve relevant structures eventually, but CSU-MS2 ranks them more effectively.
4. **Structural relevance:** Higher Tanimoto@1 for CSU-MS2 (0.3749 vs. 0.2692) suggests better top-ranked structural similarity, which is particularly important for practical annotation workflows.

These results suggest that CSU-MS2's principal advantage lies less in absolute retrieval possibility and more in ranking quality under constrained cutoffs. This distinction is operationally significant because real-world laboratory systems derive substantially more value from highly accurate top-ranked candidates than from broader low-priority retrieval coverage.

6.2.3 Interpretation Relative to Model Architecture

The difference between JESTR and CSU-MS2 can be interpreted through the way each model represents spectra and molecules before computing cross-modal similarity. JESTR uses a comparatively lightweight design, where the spectrum encoder is based on an MLP and the molecular encoder is based on a graph neural network. This architecture is sufficient to learn a general alignment between matching spectra and molecular structures, which is reflected in the separation between self-similarity and random-pair similarity. However, the relatively high random-pair similarity mean indicates that JESTR does not suppress incorrect candidates as effectively as CSU-MS2. In retrieval terms, this leads to weaker discrimination among chemically plausible but incorrect candidates.

CSU-MS2 demonstrates stronger retrieval behavior because its architecture is more directly aligned with the direct matching task. The transformer-based spectral encoder is better suited to modeling fragment-level relationships within MS/MS spectra, while the GIN-based molecular encoder provides a stronger graph representation of molecular structure. In addition, the ESA module is designed to improve cross-modal alignment between spectral and molecular representations. This architectural combination helps explain why CSU-MS2 does not mainly outperform JESTR by increasing self-similarity for correct pairs, since the self-similarity means are comparable. Its main advantage is the stronger reduction of similarity for random pairs, which produces higher KL divergence and improves candidate ranking.

This interpretation is also consistent with the retrieval results. CSU-MS2 achieves better Hit Rate@1, Hit Rate@5, and Hit Rate@10 because it ranks correct or structurally closer candidates earlier in the list. At larger cutoffs, such as Hit Rate@100, the gap between the models becomes smaller, which suggests that both models can often place the correct molecule somewhere in the candidate list. The practical difference is therefore not only whether the correct molecule can be retrieved, but also how early it appears in the ranking. For a real retrieval system, this early-ranking behavior is more important because users typically inspect only a limited number of top candidates.

Thus, the results do not prove that one architectural family is universally superior. They show that, under the unified dataset and retrieval protocol used in this thesis, CSU-MS2 provides a more suitable balance between spectral representation, molecular graph encoding, and cross-modal alignment. This makes it a more appropriate choice for integration into the final retrieval system.

6.2.4 Failure Boundaries and Performance Constraints

Despite CSU-MS2's superior performance, neither model approaches perfect retrieval accuracy. Even the best-performing model achieves Hit Rate@1 of 0.2147, indicating that exact top-ranked identification remains limited in the majority of real retrieval cases. This limitation is scientifically important because it reinforces a broader constraint already recognized in spectrum–molecule research: tandem mass spectra alone often underdetermine full molecular structure.

Several factors likely contribute to this ceiling:

- Structural isomerism and chemically similar alternatives,
- Incomplete fragmentation information,
- Candidate-space complexity,
- Representation limitations of current dual-encoder architectures.

Thus, model superiority should not be interpreted as solving the spectrum–molecule problem completely. Instead, the benchmarking results suggest that current retrieval-oriented representation learning significantly improves candidate prioritization, while still operating within fundamental chemical ambiguity constraints.

From a broader methodological perspective, these limitations indicate that further progress in spectrum–molecule matching will require retrieval architectures capable of integrating additional chemically informative modalities beyond MS/MS data alone in order to reduce structural ambiguity and improve molecular identification accuracy.

6.2.5 Limitations and Conclusions

Several limitations define the scope of interpretation for this benchmarking stage:

- Only two candidate architectures were evaluated, reflecting engineering feasibility rather than exhaustive architectural comparison.
- Model adaptation required retraining because the original pretrained checkpoints were insufficiently transferable.
- Results are bounded by the constructed dataset, molecular database, and unified training protocol.

Overall, the principal contribution of this stage lies not in proposing a new representational learning framework, but in demonstrating that systematic engineering evaluation can meaningfully identify which current architectures are most suitable for reproducible, deployment-oriented spectrum–molecule retrieval. Under the conditions tested in this thesis, CSU-MS2 provides the strongest balance of embedding discrimination, retrieval ranking quality, and system integration suitability.

6.3 Analysis of Retrieval System Performance

The retrieval system evaluation assesses whether the proposed framework can operate not only as an experimental model pipeline, but also as a practical software solution for spectrum–molecule search. In contrast to model evaluation, which focuses on ranking quality, this stage analyzes system-level properties: processing efficiency, concurrency, file-size capacity, and robustness to invalid input. These metrics are directly relevant to the practical niche of the proposed solution, because deployable retrieval systems must provide not only accurate candidate ranking, but also predictable execution, controlled error handling, and usability under realistic laboratory workloads.

6.3.1 Efficiency and Scalability of the Retrieval Pipeline

The obtained results indicate that the implemented retrieval pipeline achieves stable and practically effective processing speed under the tested configuration. In the per-spectrum processing experiment, all runs completed successfully, with an average processing time of 0.089934 seconds per spectrum. This performance is particularly meaningful when interpreted through established usability principles. According to Nielsen's usability engineering framework, approximately 0.1 seconds represents the threshold at which system response is perceived as effectively instantaneous by users [98]. Under the tested local setup with a 10-core CPU and 16 GB of RAM, the proposed system therefore operates within the approximate boundary of immediate human-perceived responsiveness on a per-spectrum basis while consistently processing a file containing 1000 spectra with full preservation of ranking outputs. From both practical and engineering perspectives, this suggests that the proposed retrieval architecture achieves a level of responsiveness compatible with interactive laboratory workflows. More broadly, these results demonstrate that representation-driven spectrum–molecule retrieval can be implemented not only as an experimental computational framework, but also as a deployable system with operational performance approaching real-time user expectations.

The parallel processing experiment further clarifies the operational capacity of the system. The system completed all mixed-file batches successfully up to $F = 11$ concurrent requests, while degradation appeared at $F = 12$ due to the Triton inference service becoming unresponsive under increased load.

The maximum file-size experiment shows that files containing up to 3000 spectra are processed successfully within the predefined 300-second timeout. The 4000-spectrum file exceeds the timeout slightly, with a measured processing time of 300.385 seconds, and therefore fails according to the evaluation criterion. This result places the practical upper bound between 3000 and 4000 spectra per request under the current deployment configuration. Importantly, the failure is caused by the timeout constraint rather than by uncontrolled system behavior or data-processing collapse, which makes the limitation measurable and operationally manageable.

From an engineering perspective, these results indicate that the system supports stable moderate-scale concurrent usage under local resource constraints while preserving predictable operational boundaries. Within the context of practical laboratory deployment, this behavior is fully consistent with realistic usage requirements. Most laboratory environments typically involve relatively small numbers of analysts or researchers interacting with molecular identification systems simultaneously, rather than enterprise-scale mass concurrency. Accordingly, the observed concurrency ceiling and large-file capacity suggest that the proposed system is appropriately aligned with its intended operational niche as a deployable research and laboratory support platform. Rather than prioritizing extreme throughput at the expense of system transparency or methodological flexibility, the implementation successfully satisfies the practical workload conditions most relevant to small- and medium-scale analytical settings.

6.3.2 Robustness and Reliability Under Invalid Input

Robustness evaluation demonstrates that the system handles invalid input in a controlled and operationally safe manner across different categories of corrupted data. Malformed and truncated files consistently return HTTP 400 parsing errors, while files containing invalid precursor metadata are accepted for processing but produce degraded analytical records rather than uncontrolled failure. Across all corrupted-input scenarios, no crashes or unstable execution behavior are observed, resulting in a request-level failure rate of 0.0%. At the record level, the invalid precursor test produces degradation for all 50 processed spectra, corresponding to complete record-level degradation ($D = 1.0$). This behavior is analytically significant because it demonstrates that the system clearly distinguishes between structurally invalid files, which cannot be processed, and structurally valid inputs that remain computationally processable but lack the metadata required for successful retrieval. Since the implemented retrieval strategy depends directly on precursor mass and adduct metadata for candidate generation, missing or invalid precursor information prevents meaningful retrieval even when file-level execution remains stable.

6.3.3 Position Relative to Existing Retrieval Systems

The proposed system occupies a distinct niche between commercial laboratory ecosystems and open academic tools. Commercial systems such as Compound Discoverer, MassHunter, MetaboScape, NIST MS Search, and METLIN remain stronger in workflow maturity, vendor integration, curated spectral libraries, and enterprise support. However, these platforms are typically proprietary, license-dependent, and primarily focused on in-library identification. Open-source systems such as SIRIUS, MS-DIAL, MZmine, GNPS/MASST, MetFrag, and CFM-ID provide greater transparency and scientific flexibility, but most remain centered on spectral libraries, fragmentation heuristics, or candidate-specific scoring rather than deployable representation-driven retrieval.

The proposed system is positioned differently as an open, service-oriented, representation-driven retrieval framework that combines:

- **Curated spectrum–molecule dataset:** Structured for unified machine learning adaptation and evaluation;
- **Indexed molecular candidate database:** Large-scale retrieval-ready molecular search space;
- **Mass-filtered embedding retrieval:** Candidate generation with embedding-based molecular ranking;
- **Dedicated user interface:** Accessible spectrum upload, analysis, and result interpretation;
- **Controlled robustness:** Predictable behavior under malformed or incomplete input conditions.

This positioning makes the system competitive not as a full replacement for mature proprietary ecosystems, but as a practical research-to-deployment alternative for modern machine-learning-based retrieval.

From an economic perspective, the system is also viable. Under the tested configuration, deployment is achieved on standard workstation-class hardware (10-core CPU, 16 GB RAM), which typically corresponds to approximately \$1,500–\$3,000 USD in initial infrastructure cost. In contrast, commercial laboratory ecosystems frequently require recurring licensing and support costs that commonly exceed \$5,000–\$20,000 annually, depending on platform scale. Over a three-year period, this produces an approximate comparison:

$$C_{\text{proposed}} \approx 3{,}000\text{–}5{,}000, \tag{35}$$

$$C_{\text{commercial}} \approx 15{,}000\text{–}60{,}000+. \tag{36}$$

While these estimates depend on deployment context, they demonstrate that the proposed framework can offer substantially lower long-term operational costs, particularly for academic laboratories, smaller research groups, or institutions seeking scalable retrieval without vendor lock-in.

Overall, the system's niche is defined by three factors: methodological transparency, deployable representation-driven retrieval, and lower economic barriers. The obtained system-level metrics indicate that this can be achieved while maintaining practical processing speed, moderate concurrency, robust behavior, and realistic deployment feasibility.

6.3.4 Limitations and Conclusions

The system-level results are satisfactory within the engineering scope of this thesis, but they also define clear limitations.

- **Hardware-bound scalability:** The concurrency limit of $F_{\text{parallel}} = 11$ reflects the current deployment configuration and available local resources.

- **Timeout-bound file capacity:** The practical file-size limit lies between 3000 and 4000 spectra per request under the 300-second timeout constraint.
- **Metadata dependency:** Invalid precursor information leads to complete record-level degradation, showing that the current retrieval workflow depends strongly on correct precursor and adduct metadata.
- **Limited production validation:** The system was evaluated under controlled experimental workloads rather than long-term deployment in an active laboratory environment.

Overall, the retrieval system evaluation shows that the proposed framework is practically viable under the tested conditions. It does not yet match the operational maturity of commercial vendor ecosystems, but it provides a transparent and extensible alternative focused on representation-driven retrieval. The main contribution of this stage is therefore the demonstration that modern spectrum–molecule embedding models can be embedded into a functioning retrieval service with measurable efficiency, scalability, and robustness.

6.4 Implications and Future Work

6.4.1 Implications for Practice

The practical implication of this thesis is that the proposed framework is sufficiently mature to be tested outside the controlled experimental setting. The obtained results show that the system can process realistic MS/MS files, support moderate concurrent usage, handle invalid inputs safely, and provide an accessible user interface for non-engineering users. Therefore, the next practical step is not only additional offline benchmarking, but validation in an active laboratory workflow.

This direction has already received external scientific interest. On November 17–21, 2025, the proposed approach was presented at the II Scientific Conference “Artificial Intelligence in Chemistry and Materials Science,” organized by the Zelinsky Institute of Organic Chemistry of the Russian Academy of Sciences[1]. The discussion initiated at this conference led to preliminary communication about a possible pilot in a real laboratory environment. Such validation is important because it can evaluate aspects that cannot be fully measured through benchmark metrics alone, including usability, trust in ranked candidates, compatibility with existing analytical routines, and the practical value of the system for unknown-compound interpretation.

In practical terms, the most relevant near-term application is the use of the system as an auxiliary decision-support tool for molecular identification. It should not be treated as a replacement for established vendor platforms or expert interpretation. Instead, it can provide an additional transparent and extensible candidate-ranking layer, especially in cases where conventional spectral-library search is limited by database coverage. This makes the framework particularly relevant for academic laboratories, research groups, and early-stage analytical workflows where openness, adaptability, and lower deployment cost are important.

Thus, the main practical outcome of the thesis is the transition from a model-centered research idea to a system that can be realistically piloted, evaluated by domain specialists, and further adapted to laboratory requirements.

[1] https://ai2025.zioc.ru/

6.4.2 Directions for Future Research

The results obtained in this thesis demonstrate that scalable and deployable spectrum–molecule retrieval can be practically realized through the combination of unified data engineering and modern cross-modal retrieval approaches. At the same time, the conducted experiments indicate that the principal remaining limitation is now associated with the retrieval capability of current machine learning models rather than with system construction itself. Although the developed framework successfully supports large-scale molecular search, the achieved retrieval accuracy shows that existing models still remain insufficient for consistently reliable molecular identification.

The primary direction for future research is therefore the development of more effective retrieval architectures capable of operating under the large-scale conditions established in this work. CSU-MS2 demonstrates clear improvements over JESTR; however, the observed Hit Rate@1 ceiling suggests that current dual-encoder approaches still have limited ability to reliably separate structurally similar molecular candidates. Further progress will therefore require improvements both in representation learning quality and in the chemical information available to the retrieval process.

Particularly important directions for future research include:

1. **Advanced retrieval architectures:** Development of models trained directly on unified datasets and optimized for both retrieval accuracy and deployment-oriented operation.
2. **Improved molecular representations:** Integration of stereo-aware, three-dimensional, and physicochemical molecular features to improve structural representation quality.
3. **Multimodal retrieval approaches:** Incorporation of additional chemically

informative modalities beyond MS/MS spectra to reduce structural ambiguity and improve candidate discrimination.

4. **Expanded candidate-space quality:** Construction of broader and more chemically representative molecular search databases.

Overall, the infrastructure established in this thesis provides a foundation for subsequent research on scalable molecular retrieval. Future improvements in identification quality will depend primarily on advances in retrieval modeling and multimodal molecular representation.

Chapter 7

Conclusion

This thesis has addressed the problem of scalable spectrum–molecule matching for tandem mass spectrometry by developing and evaluating an end-to-end retrieval framework for matching MS/MS spectra with candidate molecular structures. The main contributions and obtained results are summarized below.

The first major contribution of this thesis is the development of a unified data foundation for spectrum–molecule retrieval[1]. Heterogeneous public MS/MS resources were systematically curated into a reproducible dataset containing **1,382,425 validated spectrum–molecule pairs**, including **1,111,380 training pairs** and **271,045 test pairs**, while molecule-level separation between the splits was enforced to reduce structural leakage and support more reliable evaluation of model generalization. In addition, a molecular search database containing **872,935 candidate structures** was constructed and indexed for both mass-based candidate generation and embedding-based ranking. These results directly address **RQ1**, demonstrating that heterogeneous and partially inconsistent public MS/MS resources can be transformed into a structurally consistent and retrieval-

[1] https://github.com/IVproger/csmp-data-pipeline

ready benchmark through systematic preprocessing, molecular standardization, deduplication, metadata validation, and leakage-aware dataset construction. The study therefore shows that reliable spectrum–molecule retrieval depends not only on model design, but also on disciplined large-scale data engineering capable of producing reproducible training and evaluation conditions.

The second contribution is the unified evaluation of two modern dual-encoder spectrum–molecule models, JESTR and CSU-MS2. Under identical training and retrieval conditions, CSU-MS2 demonstrated consistently stronger embedding-space separation and retrieval effectiveness, achieving a **KL divergence of 3.4280** compared with **1.6815** for JESTR, **Hit Rate@1 of 0.2147** compared with **0.1137**, and **Hit Rate@100 of 0.9053** compared with **0.8660**. These findings directly address **RQ2** by showing that modern cross-modal contrastive learning architectures are capable of learning meaningful shared representations between MS/MS spectra and molecular structures under standardized experimental conditions. At the same time, the evaluation results indicate that current retrieval quality remains insufficient for reliable one-to-one molecular identification in large open search spaces, particularly at top-ranked positions. Consequently, the obtained results suggest that representation-driven spectrum–molecule retrieval should presently be viewed primarily as a hypothesis-generation and candidate-ranking approach capable of narrowing the search space for downstream expert interpretation, rather than as a fully autonomous identification solution.

The third contribution is the implementation of CSU-MS2 within a deployable retrieval system[2]. The developed system combines a web interface, API layer, spectrum preprocessing module, inference service, and indexed molecular database into a unified retrieval workflow in which candidate molecules are first

[2]https://github.com/IVproger/csmp_search_engine.git

filtered using precursor-mass constraints and subsequently ranked through embedding similarity. System evaluation demonstrated an average processing time of **0.089934 seconds per spectrum**, stable execution under up to **11 concurrent file requests**, successful processing of files containing up to **3000 spectra** within defined timeout constraints, and a request-level failure rate of $F = 0$ under malformed input scenarios. These results directly address **RQ3**, demonstrating that representation-driven spectrum–molecule matching can be integrated into a deployable and computationally efficient software environment capable of processing realistic analytical workloads while maintaining operational robustness. This finding is particularly important because many recent spectrum–molecule learning studies remain limited to isolated experimental prototypes without demonstrating practical deployment feasibility.

The significance of this thesis lies in demonstrating that scalable AI-assisted molecular annotation constitutes a complex multidisciplinary scientific and engineering challenge that extends far beyond isolated model development. The obtained results show that practical spectrum–molecule retrieval requires the coordinated integration of **large-scale data engineering**, **standardized benchmarking methodology**, **cross-modal machine learning**, **retrieval-oriented database infrastructure**, and **deployable software-system design**. Such integrated retrieval-system design is particularly relevant for open academic and research environments, where transparent, reproducible, and extensible alternatives to proprietary spectral-library-dependent systems remain limited.

At the same time, the conducted study highlights that practical molecular annotation from MS/MS spectra remains an open problem with substantial scientific and technical limitations. Although the developed framework demonstrated stable large-scale retrieval capability, the achieved top-ranked retrieval accuracy remains insufficient for fully autonomous molecular identification in realistic open-world chemical search spaces. In addition, the available public MS/MS resources remain inherently heterogeneous and incomplete, limiting both dataset diversity and chemical-space coverage despite large-scale curation efforts. Furthermore, spectrum–molecule matching itself represents a fundamentally difficult multimodal learning problem because fragmentation behavior depends not only on molecular structure, but also on experimental conditions, ionization modes, collision energies, instrument characteristics, and other partially observed analytical factors that are not fully captured within current retrieval formulations.

Therefore, future research should focus on both methodological and system-level advancements aimed at improving retrieval quality and practical applicability. One of the most important directions is the development of more advanced **multimodal retrieval architectures** capable of integrating complementary analytical evidence beyond MS/MS spectra alone. In particular, future systems may benefit from incorporating **retention time**, **isotope distributions**, **ion mobility measurements**, **collision-energy-aware representations**, and other orthogonal modalities into unified retrieval frameworks. Additional improvements may also arise from richer molecular encodings, including **stereo-aware**, **conformational**, and **three-dimensional representations**, as well as from larger and more chemically diverse training datasets. From an engineering perspective, future work should additionally include **laboratory pilot studies, long-term deployment validation, and evaluation of expert interaction with ranked candidate lists during real unknown-compound interpretation scenarios**.

Overall, this thesis demonstrates that the developed system and experimental results establish a reproducible foundation for future research toward more accurate, robust, and operationally practical AI-assisted molecular annotation platforms.

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